How Much Sustainability is Really in Stock Prices?*

Enrica Archetti[1] Eva Lütkebohmert[2] Markus Pelger[3] Marcus Rockel[4]

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Abstract

We find no robust evidence that sustainability information predicts stock returns. Machine learning on 22 environmental characteristics from four providers predicts neither returns nor their component orthogonal to 45 conventional firm characteristics. The same protocol recovers the known predictability of conventional characteristics, and the sustainability models' confidence intervals exclude Sharpe ratios above half of that benchmark. Replicating the green-minus-brown evidence of Pástor, Stambaugh, and Taylor (2022), we show that its significance as a return predictor hinges on sample period, universe, sort granularity, and prediction target. Across 132 models that vary variables, sample period, and prediction target, none survives multiple-testing corrections.

Keywords: climate finance, ESG, return predictability, machine learning, multiple testing, replication

JEL classification: G12, G14, C45, C58, Q54

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[1]Stanford University, Department of Management Science & Engineering. Email: archetti@stanford.edu.

[2]University of Freiburg, Department of Economics. Email: eva.luetkebohmert@finance.uni-freiburg.de.

[3]Stanford University, Department of Management Science & Engineering and NBER. Email: mpelger@stanford.edu.

[4]University of Freiburg, Department of Economics. Email: marcus.rockel@finance.uni-freiburg.de.

1 Introduction

Climate change has triggered one of the largest reallocations of capital in modern financial history. As of mid-2024, more than 100 countries, responsible for roughly 82% of global greenhouse gas emissions, had announced net-zero pledges, complemented by over 9,000 corporations and 600 financial institutions.[5] Trillions of dollars of assets under management are tied to mandates that incorporate environmental, social, and governance (ESG) information, and an entire industry of data providers sells sustainability scores to investors who use them in portfolio construction. Whether this information is priced, that is, whether sustainability characteristics predict stock returns beyond what conventional firm characteristics already capture, is one of the central questions of climate finance. The empirical literature offers strikingly contradictory answers. Bolton and Kacperczyk (2021) document a carbon premium, with high-emission firms earning higher returns. Pástor, Stambaugh, and Taylor (2022) document the opposite pattern in realized returns. Green stocks outperformed brown stocks over the 2010s, and a long–short "green-minus-brown" portfolio earned an annualized Sharpe ratio above one. Pedersen, Fitzgibbons, and Pomorski (2021) and Aswani, Raghunandan, and Rajgopal (2024) argue that much of the apparent ESG–return relation reflects correlation with firm fundamentals or vendor estimation procedures rather than sustainability itself.

Three features of the setting make this a hard inference problem and help explain the disagreement. First, the data are young, the most widely used scores exist for little more than a decade, and the panel is extraordinarily sparse. Second, the information is contested, since the major providers disagree with each other, and their scores are built from the same firm fundamentals that already predict returns. Third, the researcher chooses the provider, the variables, the stocks, the start date, the portfolio sorts, and the return target. Each choice is defensible on its own, and together they create a large space in which significant results can be found by chance. This paper brings to the question the discipline that the machine learning and anomalies literatures have developed for situations exactly like this one, a flexible prediction protocol whose power is verified rather than assumed, a nonparametric control for conventional information, and explicit corrections for the search over designs. Our answer is that essentially no return predictability from sustainability information can be established credibly.

Our first contribution is a comprehensive out-of-sample prediction test. We assemble 22 environmental characteristics from the four providers most used in research and practice

[5] United Nations Net-Zero Coalition, https://www.un.org/en/climatechange/net-zero-coalition, accessed June 2024.

(MSCI, S&P Global, Refinitiv, and Trucost) for U.S. stocks over the 2002–2024 period, drawn from a universe of 513 sustainability-related variables, and we predict next-month abnormal returns with gradient-boosted trees, ensembled over random seeds, with hyperparameters chosen on an economic objective, the Sharpe ratio of a long–short decile portfolio. Because the panel is short, we adopt the cross-out-of-sample evaluation of Kaniel, Lin, Pelger, and Van Nieuwerburgh (2023), which delivers an out-of-sample prediction for every observation. Applied to the sustainability characteristics, the protocol finds nothing. The long–short Sharpe ratio and the information coefficient are both slightly negative and statistically indistinguishable from zero, the deciles do not separate, and the cumulative-return paths of the top and bottom deciles cross repeatedly.

A null result is only as informative as the power of the test, and we measure that power in three ways rather than assume it. Applied to 45 conventional firm characteristics, the same protocol produces a value-weighted long–short decile portfolio with an annualized Sharpe ratio of 0.72, in line with the machine learning asset-pricing literature, and a Ledoit and Wolf (2008) test confirms that this Sharpe ratio is significantly higher than those of the sustainability models. The 95% confidence intervals of the sustainability models exclude Sharpe ratios above half of what the characteristics deliver. When applied to a placebo panel in which a synthetic signal of known strength is added to the sustainability variables, the protocol detects a signal as strong as that of the characteristics in every simulated sample, and a signal about a third as strong in 92% of them. The comparison does not handicap the sustainability variables, whose models are evaluated with a scheme that lets them learn from data a real-time investor would not yet have, while the characteristics model is evaluated on a strict chronological split.

A natural objection is that sustainability premia could be masked by correlated exposures to conventional characteristics, or, conversely, that any predictability from sustainability data is fundamentals in disguise. We address both with a two-step orthogonalization. In the first step we estimate the conditional mean of abnormal returns given the 45 conventional characteristics with the same carefully tuned machine learning protocol and compute out-of-sample residuals. A residual relative to the conditional mean is orthogonal to every function of the characteristics, not only to the linear combinations that a regression removes. Our residuals have this property up to the estimation error of the conditional mean, and the out-of-sample performance of the step-1 model indicates that this error is small. A direct check confirms it, as a model that predicts the step-1 residuals with the 45 characteristics themselves finds no exploitable structure in them. In the second step, we ask whether sustainability information predicts these residuals. It does not. The long–short Sharpe ratio is again slightly negative and insignificant, and the information coefficient is essentially zero.

Sustainability characteristics neither provide standalone return predictability nor predict the return variation left unexplained by conventional firm characteristics. There is nothing for the controls to subsume.

The null result is not an artifact of our particular choices. We repeat the analysis with each provider's variables separately, with emissions only, with ESG scores only, with all 513 variables, with rank-transformed inputs, with Elastic Net and a feedforward neural network in place of gradient boosting, with equal-weighted, prediction-weighted, tercile and quintile portfolios, and without any imputation of the sustainability panel. This family of 16 models and 32 tests contains three nominally significant positive results and two significantly negative ones at the 5% level, and none survives a multiple-testing correction, whether the tests are one-sided or two-sided. Emissions alone, the variables behind the carbon premium of Bolton and Kacperczyk (2021), predict neither returns nor residuals.

Our second contribution explains why parts of the literature found otherwise. We take the most prominent evidence of green outperformance, the green-minus-brown (GMB) portfolio of Pástor, Stambaugh, and Taylor (2022) (henceforth PST), as a laboratory. PST sort stocks on a greenness score built from MSCI environmental data and report an annualized Sharpe ratio above one for November 2012 to December 2020. They are careful to attribute this outperformance to unanticipated increases in climate concerns rather than to high expected returns. Much of the subsequent literature and of investment practice has nonetheless read green outperformance as evidence of a return premium or an investable signal. We ask how robust that reading is. We reproduce PST's result on their sample period, their restricted universe, and their tercile sorts (Sharpe ratio 1.02 against their 1.14), and then perturb each design choice toward an equally standard alternative. Extending the sample back to 2007, the first year in which the underlying MSCI data exist, roughly halves the Sharpe ratio, and so does extending it forward to 2024. A bootstrap shows that the usual justification for discarding the early years, that the cross-section is too thin, does not hold, because sampling the median pre-2012 number of firms within the post-2012 period reproduces the post-2012 Sharpe ratio. Moving from terciles to deciles, the standard granularity of the anomalies literature, removes the significance of the score-sorted portfolio in every sample that extends beyond PST's window, because the score assigns stocks to the extreme buckets no better than random assignment. Replacing realized returns with characteristic-orthogonal residuals halves the Sharpe ratio inside PST's window and removes the significance of the score-sorted portfolio in every sample outside it, and moving to the full imputed universe weakens the score-sorted results in the samples that start in 2007.

Finally, we make the implicit search over specifications explicit by estimating models on $1, 2, \ldots, 22$ sustainability variables for three sample starts and two targets. The resulting

family of 132 long–short portfolios contains cells with t-statistics above two and cells with t-statistics below minus two, and the same nine variables predict characteristic-orthogonal residuals significantly when the sample starts in 2002 and earn negative spreads when it starts in 2007 or in 2012. Not a single specification survives a Holm (1979) correction, none survives Benjamini and Hochberg (1995) at the 5% false-discovery rate, and Diebold and Mariano (1995) and Ledoit and Wolf (2008) tests cannot distinguish any multivariate sustainability model from the univariate greenness score.

Taken together, the evidence supports a simple conclusion. Sustainability characteristics, as sold by the major data providers and used in academic studies, contain no detectable information about future stock returns beyond conventional characteristics, and little detectable information at all. Where green outperformance appears, it lives in one particular window, one particular universe, and one particular sort coarseness, and it does not survive the inferential discipline that the anomalies literature now considers standard (Harvey, Liu, and Zhu, 2016; Hou, Xue, and Zhang, 2020; Jensen, Kelly, and Pedersen, 2023). We emphasize what our results do and do not say. They do not say that climate risk is unimportant, that it should not be priced, or that investors' nonpecuniary preferences for green assets (Pástor, Stambaugh, and Taylor, 2021; Zerbib, 2022; Hong and Kacperczyk, 2009) are irrational. Equilibrium models predict that green assets can earn *lower* expected returns precisely because investors value them, and that realized green outperformance can reflect unexpected shifts in tastes or climate concerns (Pástor, Stambaugh, and Taylor, 2021; Ardia, Bluteau, Boudt, and Inghelbrecht, 2023). Our results concern a sharper question, whether publicly available sustainability information predicts returns out of sample in a way an investor or researcher could credibly exploit or even detect. The answer is no. This matters for three audiences. For investors, strategies marketed on the premise that ESG information improves return forecasts lack empirical support. For researchers, the sustainability–returns literature provides a cautionary case study in which economically motivated but discretionary design choices generate publishable significance. For policymakers and standard setters, the result that commercial sustainability scores carry essentially no return-relevant information beyond fundamentals speaks to the informational value of current disclosure regimes.

1.1 Related literature

Our paper contributes to four strands of literature. First, a large literature studies whether sustainability characteristics are priced. Bolton and Kacperczyk (2021, 2023) find that high-emission firms earn higher returns, interpreting this as compensation for carbon-transition risk, and Hsu, Li, and Tsou (2023) document a premium for toxic emissions, and Zhang

(2025) qualifies the international evidence. Berk and van Binsbergen (2025) argue that the price impact of sustainable investing is small. Pástor, Stambaugh, and Taylor (2022) show that German green bonds and U.S. green stocks outperformed in the 2010s and attribute the outperformance to unexpected increases in environmental concerns, building on the equilibrium framework of Pástor, Stambaugh, and Taylor (2021). Hong and Kacperczyk (2009) document a sin-stock premium driven by norm-constrained investors. Ardia, Bluteau, Boudt, and Inghelbrecht (2023) link green-minus-brown returns to climate-concern shocks, and Engle, Giglio, Kelly, Lee, and Stroebel (2020) construct portfolios that hedge climate news. Krueger, Sautner, and Starks (2020) and Giglio, Maggiori, Stroebel, Tan, Utkus, and Xu (2025) document how institutional and retail investors perceive climate risks and ESG investing. Relative to this literature, our contribution is a comprehensive assessment of return predictability from sustainability information under a protocol with verified power, which fails to detect any robust signal.

Second, a growing literature studies how sustainability is measured. Berg, Kölbel, and Rigobon (2022) document large disagreement across ESG raters, Christensen, Serafeim, and Sikochi (2022) show that disagreement increases with disclosure, and Avramov, Cheng, Lioui, and Tarelli (2022) show that rating uncertainty weakens the link between ESG and expected returns. Aswani, Raghunandan, and Rajgopal (2024) show that the carbon–return relation is driven by vendor-estimated emissions and disappears for firm-reported emissions, suggesting that vendor methodologies embed fundamentals. Duan and Pelger (2025) reach the same conclusion with a method that delivers valid inference when covariates are missing. Treating vendor estimates as missing rather than observed, they find no relation between returns and emissions from firm operations and a negative relation for value-chain emissions. Pedersen, Fitzgibbons, and Pomorski (2021) make the related theoretical and empirical point that ESG scores correlate with characteristics that already predict returns. Our two-step design generalizes these concerns nonparametrically. Instead of controlling for a handful of fundamentals in a linear regression, we remove the predictability that a flexible model extracts from 45 characteristics, and ask what sustainability information adds.

Third, we build on the use of machine learning in asset pricing. Gu, Kelly, and Xiu (2020), Chen, Pelger, and Zhu (2024), and Kelly, Pruitt, and Su (2019) establish that flexible machine learning methods extract substantial out-of-sample return predictability from firm characteristics. Kaniel, Lin, Pelger, and Van Nieuwerburgh (2023) apply these tools to mutual fund selection and develop the cross-out-of-sample evaluation we adopt. Bryzgalova, Lerner, Lettau, and Pelger (2025) provide the imputation method we use for missing characteristics, and Duan and Pelger (2025) develop inference with imputed covariates. We use machine learning not to find new predictability but to bound it. Because tree ensembles

capture nonlinearities and interactions, a null result is informative about a large class of functions of the inputs, not only about linear combinations. Our positive control replicates the consensus finding that characteristics predict returns, which disciplines the interpretation of the sustainability null.

Fourth, our paper relates to the literature on replication, data snooping, and multiple testing. Harvey, Liu, and Zhu (2016) argue that the threshold for declaring new factors should account for the multiplicity of tests. Hou, Xue, and Zhang (2020) show that most anomalies fail replication under standard methodological choices. Linnainmaa and Roberts (2018) and McLean and Pontiff (2016) document out-of-sample and post-publication decay, and Chordia, Goyal, and Saretto (2020) quantify the rate of false rejections in the anomalies universe, while Jensen, Kelly, and Pedersen (2023) provide a more optimistic Bayesian reading. Closest to our design dimension, Lioui and Tarelli (2022) show that the estimated ESG factor premium is highly sensitive to how the factor is constructed. We bring this agenda to climate finance. We treat the design choices of the sustainability literature as a search space, replicate the headline result, and show that it does not survive the discipline this literature prescribes. Methodologically, the combination of within-study design perturbation, bootstrap calibration of subsample claims, and multiple-testing corrections over the design space provides a template for evaluating contested predictability claims more generally (Welch and Goyal, 2008).

The remainder of the paper proceeds as follows. Section 2 describes the data. Section 3 presents the two-step design, the estimation protocol, and the inference framework. Section 4 reports the main prediction results and measures the power of the protocol. Section 5 dissects the green-minus-brown replication and quantifies the contribution of each research-design choice. Section 6 collects robustness checks. Section 7 concludes. Variable definitions, implementation details, the data appendix, and additional results appear in the Appendix and the Internet Appendix.

2 Data

2.1 Returns and conventional firm characteristics

Our prediction target is the abnormal stock return. Predicting the level of stock returns is dominated by the common market component, which makes total returns a statistically poor target for cross-sectional prediction, whereas abnormal returns remove the level effect and turn prediction into a relative, cross-sectional objective (Kaniel, Lin, Pelger, and

Van Nieuwerburgh, 2023). For each stock i and month t, we define the abnormal return as

$$r_{i,t}^{\text{abn}} = \underbrace{r_{i,t} - r_{f,t}}_{r_{i,t}^{\text{excess}}} - r_{m,t}, \quad \text{where } r_{m,t} = \sum_{i=1}^{N_t} w_{i,t-1}\, r_{i,t}, \quad \text{and } w_{i,t} = \frac{\text{MC}_{i,t}}{\sum_{j=1}^{N_t} \text{MC}_{j,t}}, \tag{1}$$

where $r_{f,t}$ is the one-month Treasury rate from the Kenneth R. French data library and $r_{m,t}$ is the return of the N_t stocks in our universe weighted by their market capitalization $\text{MC}_{i,t-1}$ at the end of the previous month, which is a close proxy for the CRSP value-weighted market given the breadth of the sample. All models predict abnormal returns. Portfolio returns are reported in excess of the risk-free rate. For a long–short portfolio the market component cancels, so the two conventions coincide.

The investment universe and the conventional characteristics follow Bryzgalova, Lerner, Lettau, and Pelger (2025). The universe is liquid and excludes penny and micro-cap stocks, so that our results are not driven by stocks that are expensive or impossible to trade. We use 45 characteristics spanning the standard categories (value, investment, profitability, intangibles, past returns, and trading frictions), listed in Appendix Table A.1. Missing characteristics are imputed following Bryzgalova, Lerner, Lettau, and Pelger (2025), with the backward cross-sectional (B-XS) model when past observations are available and the cross-sectional (XS) model otherwise, both free of look-ahead bias. The characteristics panel begins in January 1977 and extends through November 2024, and includes 21,475 stocks for a total of 2,599,933 stock-month observations.

2.2 Sustainability characteristics

We assemble stock-level sustainability information from the four providers most widely used in the literature and in industry, S&P Global (298 raw variables), Refinitiv (7), Trucost (148), and MSCI (60), for a total of 513 sustainability-related variables. From this universe we select 22 variables that are the most widely used in academic studies and have the most direct economic interpretation. They are absolute greenhouse-gas emissions at all scopes (Trucost and S&P Global), estimated total CO_2-equivalent emissions, provider-level ESG and environmental-pillar scores (Refinitiv, S&P Global, MSCI), resource-use scores, environmental innovation scores, and the MSCI environmental pillar score and weight that underlie the greenness measure of Pástor, Stambaugh, and Taylor (2022). The 22 variables are listed in Appendix Table A.2. Following Bolton and Kacperczyk (2021) and Aswani, Raghunandan, and Rajgopal (2024), raw emissions enter in logs, and the other variables are left unchanged. The emission variables combine firm-reported values with the providers' own estimates for firms that do not report. Aswani, Raghunandan, and Rajgopal (2024) and

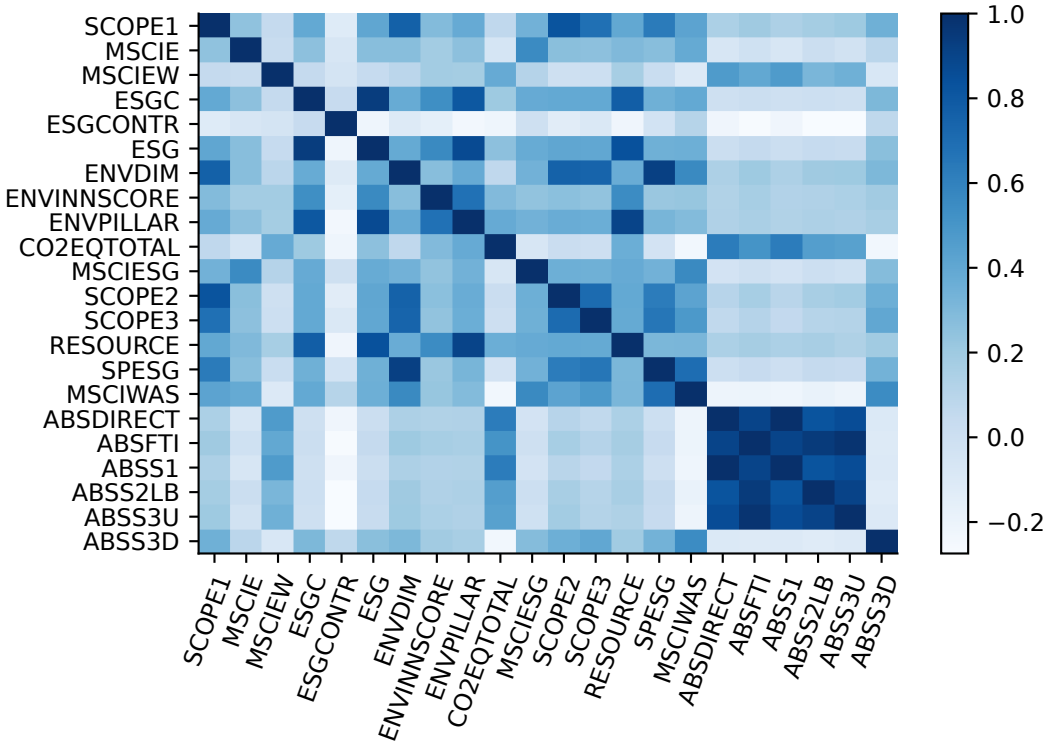


Figure 1: Correlation structure of sustainability characteristics. The figure shows the correlation matrix of the 22 sustainability variables listed in Appendix Table A.2.

Duan and Pelger (2025) show that such estimates are functions of firm fundamentals and that this is what produces a carbon–return relation in regressions. Our residual test of step 2 removes exactly this channel, because it conditions on the fundamentals nonparametrically. Section 6 repeats the analysis with the full set of 513 variables, with each provider's variables separately, with emissions only, with ESG scores only, and with rank-transformed inputs.

Figure 1 shows the correlation matrix of the 22 variables. Two patterns stand out. First, variables that measure the same quantity from different providers are highly but imperfectly correlated, consistent with the rating disagreement documented by Berg, Kölbel, and Rigobon (2022). Second, provider scores within a family are strongly correlated, consistent with vendor models that rely on common firm fundamentals (Aswani, Raghunandan, and Rajgopal, 2024). Both patterns motivate our design. If the return-relevant content of sustainability data is fundamentals in disguise, controlling for fundamentals nonparametrically should eliminate it.

Our merged sustainability sample spans January 2002 through November 2024. The merge of provider data with the returns panel is deliberately conservative. We require identifier agreement beyond ticker matching to rule out assigning one firm's ESG scores to another firm's returns. This costs us roughly 200 tickers relative to a naive ticker merge and

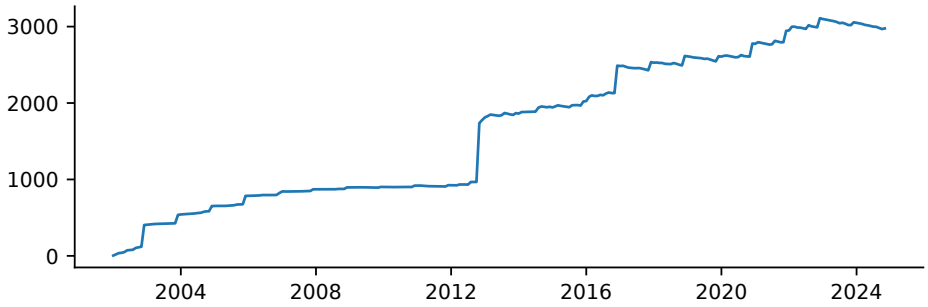


Figure 2: Number of securities over time. The figure shows the number of stocks in the sample over time. The expansions in 2013 and 2016 correspond to broader MSCI coverage and the Trucost acquisition by S&P.

leaves approximately 1,800 firms in January 2013, comparable to the roughly 2,000 firms in Pástor, Stambaugh, and Taylor (2022). MSCI coverage begins in 2007, which defines the feasible start of any analysis using MSCI-based measures, and S&P Global scores begin in September 2013. The final sample contains 4,500 stocks and 452,841 stock-month observations. Figure 2 shows the number of securities in the sample over time, and Appendix D documents the sources, the merge rules, and the attrition at each step.

2.3 Missing data and imputation

Sustainability data are extraordinarily sparse, with roughly 91% of stock-month-variable cells missing in the merged panel. Figure 3 shows the missingness structure. Coverage expands discretely in 2013 (MSCI universe expansion) and 2016 (Trucost expansion after the S&P acquisition), and most variables are updated annually at fiscal year-ends, which creates a seasonal pattern within each year. Sustainability characteristics are highly persistent, with twelve-month autocorrelations close to one for most variables, so our baseline imputation proceeds in four steps. First, we carry forward the most recent observation for the same stock-variable pair, which fills the months between annual updates. Second, we drop stock-months in which none of the 22 variables is observed, that is, firm-months before a firm enters any provider's coverage, because such a stock-month would consist entirely of imputed values and carry no firm-specific sustainability information. Third, we fill the remaining gaps with the industry (two-digit NAICS) median in that month and, where no industry peer is observed, with the cross-sectional median. Fourth, variables that no firm reports in a given month, for example S&P Global scores before September 2013 or MSCI scores before 2007, are set to zero in that month. Because such a value is constant across firms within the month, it carries no cross-sectional information, and at most it allows the model to distinguish the months before and after a provider's coverage begins. Appendix Table A.8

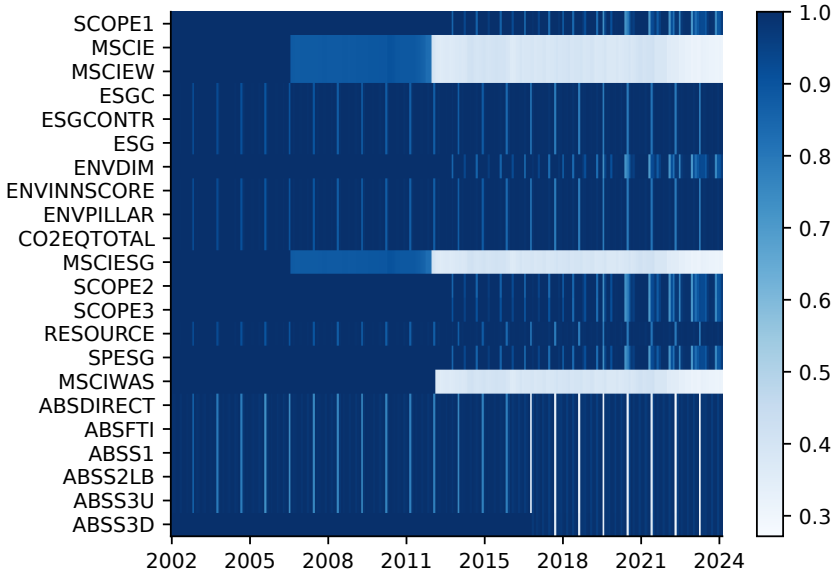


Figure 3: Missingness in the sustainability panel. The heatmap shows the share of missing observations by variable and month in the raw merged panel. Lighter shades correspond to lower shares of missing observations. The expansions in 2013 and 2016 correspond to broader MSCI coverage and the Trucost acquisition by S&P.

reports the number of cells and the share filled at each step, for the full sample and for the samples that start in November 2012. Because imputation is a substantive modeling choice at this level of missingness, Section 6.4 also estimates the models without any imputation, letting the learner handle missing values directly. A flexible learner can use the dependence among the covariates within the prediction, which amounts to imputing internally and is sufficient when the goal is prediction rather than parameter estimation. We also note that the green-minus-brown analysis of Section 5 is conducted both in a complete-case universe, in which the MSCI inputs of the greenness score are observed and nothing is imputed, and in the full imputed universe.

3 Methodology

3.1 A two-step orthogonalization design

The central identification problem is that sustainability characteristics are correlated with conventional firm characteristics. A finding that ESG scores predict returns is therefore uninformative about whether sustainability is priced, because the scores may simply proxy for size, profitability, investment, or momentum. Linear controls do not solve the problem if

the true relation between characteristics and expected returns is nonlinear with interactions, as the machine learning asset-pricing literature documents (Gu, Kelly, and Xiu, 2020; Chen, Pelger, and Zhu, 2024). Our design controls for conventional information nonparametrically.

Step 1 (conventional benchmark). We predict abnormal returns with the 45 conventional characteristics $z_{i,t}$ using a flexible function g_1 estimated by gradient boosting:

$$r^{\text{abn}}_{i,t+1} = g_1(z_{i,t}) + \varepsilon_{i,t+1}. \tag{2}$$

In equation (2), $g_1(z_{i,t}) = \mathbb{E}[r^{\text{abn}}_{i,t+1} \mid z_{i,t}]$ is the conditional mean of abnormal returns given the characteristics, so that the residual satisfies $\mathbb{E}[\varepsilon_{i,t+1} f(z_{i,t})] = 0$ for every function f of the characteristics with finite second moment. This is the general form of orthogonality. The residual of a linear regression is the special case in which g_1 is restricted to be linear. It is orthogonal only to linear functions of the characteristics, and in sample it is orthogonal by construction whether or not the linear model describes expected returns well.

The fitted model serves two purposes. It is the positive control that establishes the power of the protocol, and it delivers the out-of-sample residuals $\hat{\varepsilon}_{i,t+1} = r^{\text{abn}}_{i,t+1} - \hat{g}_1(z_{i,t})$, the return component that, based on our estimator, is orthogonal to the conventional characteristics. Because g_1 is estimated, this orthogonality holds up to the estimation error $g_1 - \hat{g}_1$. The purpose of a flexible and carefully tuned learner is to keep this error small, and the out-of-sample performance of the step-1 model (Section 4.1) indicates that $\hat{g}_1$ is a good approximation of the conditional mean, and Section 4.3 checks the orthogonality directly. Estimation error, if anything, works against our conclusion. Characteristic-related variation left in the residuals would be partly predictable from sustainability characteristics, which are correlated with conventional characteristics, and would make it easier for step 2 to detect predictability.

Step 2 (sustainability increment). We predict the step-1 residuals with the sustainability characteristics $h_{i,t}$:

$$\hat{\varepsilon}_{i,t+1} = g_2(h_{i,t}) + e_{i,t+1}. \tag{3}$$

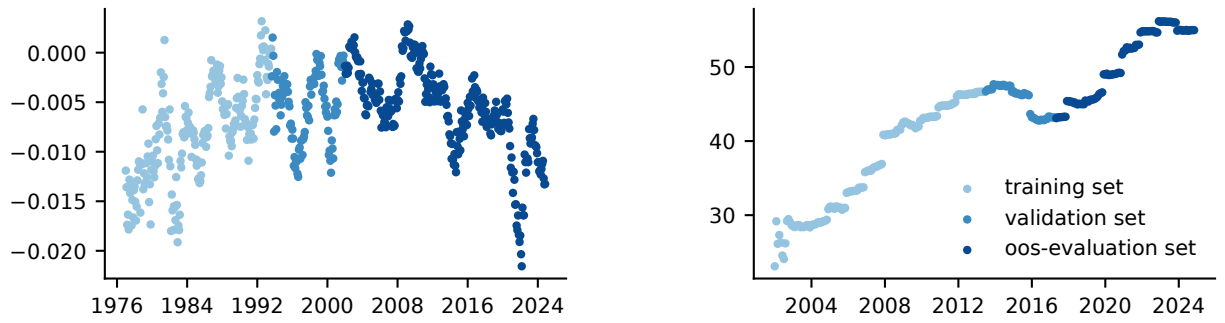
If sustainability information commands a premium beyond conventional sources of expected returns, g_2 should generate out-of-sample spread in the residuals. We complement step 2 with a "step 0" analysis that predicts abnormal returns directly with sustainability characteristics alone, which measures unconditional predictability and corresponds to the implicit experiment in much of the ESG literature.

3.2 Estimation: gradient boosting, ensembling, and tuning

Our baseline learner is extreme gradient boosting (XGBoost), an ensemble of regression trees built sequentially on the gradient of the loss. Tree ensembles are well suited to this problem. They capture nonlinearities and interactions, handle variables on heterogeneous scales, and extract weak signals from high-dimensional, noisy panels. To remove seed dependence we ensemble 20 models that differ only in the random seed and average their predictions. Hyperparameters are selected by grid search on a validation sample, and Appendix Table A.3 reports the grid. The model-selection criterion is economic rather than statistical. We choose the configuration that maximizes the Sharpe ratio of the value-weighted long–short decile portfolio on the validation sample, tying model selection to the object of economic interest. The number of boosting rounds is chosen on the validation sample by early stopping (the validation loss must improve within 20 rounds) and then held fixed when the selected configuration is re-estimated on the combined training and validation data, so that no information from the evaluation fold enters estimation or tuning. Internet Appendix Section IA.7 describes the tree ensemble and the two alternative learners of Section 6.2 in more detail.

3.3 Sampling schemes and out-of-sample evaluation

The two prediction problems have very different time-series lengths, and we tailor the evaluation scheme to each. For the conventional-characteristics model (step 1), we use a standard chronological split. The early sample (1977–2001) is divided into training and validation folds for estimation and tuning, and the model is evaluated out of sample on the subsequent period (2002–2024), which overlaps the sustainability panel. All reported step-1 performance is therefore out of sample. We deliberately do not re-estimate the characteristics model over the evaluation period. A model fixed in 2001 uses no information from the years in which sustainability information is tested, which keeps the step-1 residuals clean, and re-estimation over a rolling window could only strengthen the benchmark. For the sustainability models (steps 0 and 2), the panel is short (2002–2024, with MSCI-based measures from 2007 and S&P-based measures from September 2013), and a single chronological split would waste most of the data and confound the evaluation with a particular subperiod. We therefore use the cross-out-of-sample scheme of Kaniel, Lin, Pelger, and Van Nieuwerburgh (2023). The sample is divided into three chronological folds, each fold serves once as the evaluation set while the other two are used for training and validation, and rotating over folds yields an out-of-sample prediction for *every* stock-month in the sample. Because each fold's model is estimated on data from both before and after the evaluation period, the scheme measures predictability rather than the performance of a real-time strategy, and it gives the sustain-



(a) Conventional characteristics (step 1)

(b) Sustainability characteristics (steps 0 and 2)

Figure 4: Sampling schemes. Both panels plot the time series of a representative variable. Light, medium, and dark blue denote training, validation, and out-of-sample evaluation folds. The left panel shows the chronological split for the conventional-characteristics model. The right panel shows one of the three fold assignments of the cross-out-of-sample scheme, and the roles of the three folds rotate so that every observation is evaluated out of sample.

ability models more data than a real-time investor would have. Figure 4 illustrates both schemes.

3.4 Portfolios and inference

Each month we sort stocks into deciles by their out-of-sample prediction and form value-weighted (baseline), equal-weighted, and prediction-weighted portfolios, and the long–short portfolio buys the top and sells the bottom decile. We report annualized mean returns, annualized Sharpe ratios with 95% confidence intervals, and t-statistics for the mean long–short return based on Newey–West standard errors with four lags. The 95% confidence intervals for the annualized Sharpe ratio use the delta method (Lo, 2002), combining a Newey–West variance of the mean return with the sampling error of the volatility estimate. The information coefficient (IC, the monthly cross-sectional rank correlation between predictions and realizations) and its time-series stability (the IC ratio, the ratio of the mean to the standard deviation of the monthly ICs, annualized) complement the portfolio metrics with a measure that does not depend on portfolio construction.

Because the paper's thesis is that the literature's significance partly reflects a search over designs, our own inference must be conservative. We use four tools. First, when we examine families of related specifications, we report familywise corrections (Holm, 1979) and false-discovery-rate corrections (Benjamini and Hochberg, 1995), pooled over the entire family. Second, we compare forecast accuracy across models with Diebold and Mariano (1995) tests. Because many stocks respond to the same shock in a given month, stock-month loss differentials are not independent. We therefore average the squared-error differential

across stocks within each month and test the resulting monthly series with Newey–West standard errors. Third, we compare economic performance with the Sharpe-ratio test of Ledoit and Wolf (2008), which is robust to non-normality and autocorrelation. Fourth, we calibrate subsample claims with a bootstrap design described in Section 5.2. Unless stated otherwise, tests of long–short returns are one-sided tests of a positive spread, the natural alternative for a strategy that buys the predicted winners, and we report two-sided results where they differ. Internet Appendix Section IA.8 gives the formulas for the performance measures, the multiple-testing adjustments, and the tests.

4 Sustainability Does Not Predict Returns

Table 1 summarizes the three headline models, the conventional-characteristics model (step 1), the sustainability model for abnormal returns (step 0), and the sustainability model for characteristic-orthogonal residuals (step 2). Its last row is a check on the orthogonalization discussed in Section 4.3. All models are evaluated on the same universe and the same months, so that differences in performance reflect the information content of the inputs and not differences in data availability.

4.1 Positive control: characteristics predict returns

Panels (a) and (b) of Figure 5 show the out-of-sample performance of decile portfolios sorted on the conventional-characteristics prediction. The results confirm the consensus in the machine learning asset-pricing literature (Gu, Kelly, and Xiu, 2020). The long–short decile portfolio earns an annualized mean excess return of 13.9% with a Sharpe ratio of 0.72 ($t = 2.80$),[6] and the deciles line up nearly monotonically in realized performance. The model also loads on economically sensible predictors. The variable-importance analysis in Panel (a) of Figure 6 shows that earnings yield, unexplained trading volume, short-term reversal, book-to-market,

[6]The Sharpe ratio is lower than the headline numbers of Gu, Kelly, and Xiu (2020), who report 1.35 for the value-weighted and 2.45 for the equal-weighted long–short decile portfolio of their best neural network, for reasons of design. Our universe excludes micro-cap and penny stocks, whereas theirs is the full CRSP universe. Moreover, our portfolios are value weighted and most of the gains of machine learning portfolios come from the smallest stocks. Chen, Pelger, and Zhu (2024) show that value weighting alone roughly halves the Sharpe ratio of such long–short decile portfolios relative to equal weighting, and that the Sharpe ratios are lower in the later part of the sample than in the earlier training and validation years. Our evaluation period, 2002–2024, lies entirely in the period of lower anomaly returns documented by McLean and Pontiff (2016), while theirs starts in 1987. Additionally, our model is estimated once on 1977–2001 and never re-estimated. Consistent with the role of small stocks, equal weighting does not raise the Sharpe ratio in our universe (Section 6.3). What matters for this paper is the comparison rather than the level, since the sustainability models are evaluated with the same protocol, universe, weighting, and period.

Table 1: Long–short decile portfolio performance of the headline models. Value-weighted decile portfolios are formed monthly on out-of-sample predictions. The conventional-characteristics model predicts abnormal returns with 45 firm characteristics, estimated and tuned on the 1977–2001 sample and evaluated out of sample thereafter. The sustainability models predict abnormal returns (step 0) and step-1 residuals (step 2) with 22 sustainability variables under the cross-out-of-sample scheme on the 2002–2024 panel. The last row is the orthogonality check of Section 4.3, the 45 characteristics predicting the step-1 residuals under the same scheme. μ is the annualized mean excess return of the long–short portfolio in percent, SR its annualized Sharpe ratio with a 95% confidence interval, and IC is the average monthly rank correlation between predictions and realizations. t-statistics use Newey–West standard errors with four lags. One-sided Ledoit and Wolf (2008) tests reject equality of the Sharpe ratio of the characteristics model with those of the two sustainability models ($p = 0.04$ and 0.02) and do not reject equality of the two sustainability models. The Newey–West t-statistics of the mean ICs are 6.58 for the characteristics model, -1.56 and -0.45 for the two sustainability models, and -2.29 for the orthogonality check.

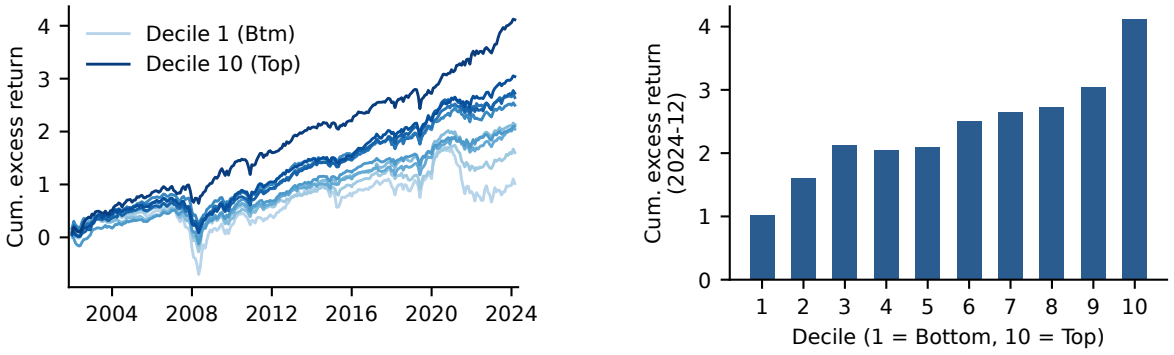
Model	μ (%)	SR	SR 95% CI	IC	t-stat
Characteristics → abnormal returns	13.9	0.72	[0.21, 1.22]	0.037	2.80
Sustainability → abnormal returns	−0.6	−0.06	[−0.49, 0.38]	−0.009	−0.26
Sustainability → residuals	−0.8	−0.08	[−0.52, 0.37]	−0.002	−0.33
Characteristics → residuals (check)	−2.2	−0.12	[−0.52, 0.27]	−0.014	−0.61

and size account for the largest shares of the model's predictions, consistent with the variables that the cross-sectional literature identifies as strong predictors. Two implications follow. First, the protocol (abnormal-return target, Sharpe-ratio-based tuning, ensembling, and out-of-sample evaluation) has power to detect predictability of the magnitude documented in the literature. Second, the step-1 residuals are purged of an economically large component of expected returns, so step 2 is a meaningful test.

4.2 Sustainability does not predict abnormal returns

Panels (c) and (d) of Figure 5 repeat the exercise with the 22 sustainability characteristics as the only inputs (step 0). The contrast with Panels (a) and (b) is stark. The deciles do not separate, the cumulative-return paths of the top and bottom deciles cross repeatedly, and the long–short portfolio earns an annualized mean of -0.6% with a Sharpe ratio of -0.06 ($t = -0.26$) and an IC of -0.009 (Table 1). The point estimate is zero for practical purposes, and its sign varies across the specifications of Section 6, which is the signature of noise. What is stable is the absence of the monotonic decile separation that genuine predictors produce. The variable-importance analysis in Panel (b) of Figure 6 shows that the model concentrates on Scope 3 emissions and on the S&P Global ESG score, and even the variables the model relies on most do not translate into predictive power. Early stopping on the validation sample selects a single boosting round, that is, a single tree, for both sustainability models, against 949 rounds for the characteristics model (Appendix Table A.4), because additional trees do not reduce the validation loss of the sustainability models. A model of this form

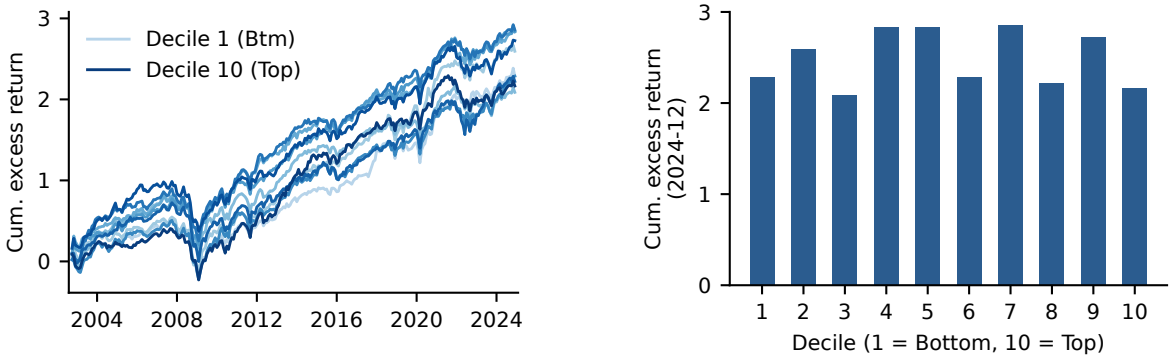
Characteristics → abnormal returns (step 1)



(a) Cumulative returns by prediction decile

(b) Terminal cumulative return by decile

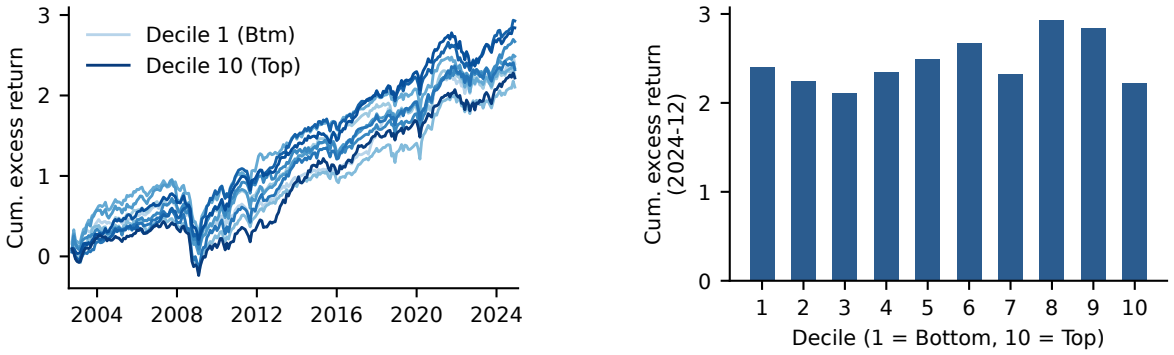
Sustainability → abnormal returns (step 0)



(c) Cumulative returns by prediction decile

(d) Terminal cumulative return by decile

Sustainability → residuals (step 2)



(e) Cumulative returns by prediction decile

(f) Terminal cumulative return by decile

Figure 5: Conventional characteristics predict abnormal returns, sustainability characteristics do not. Cumulative excess returns of value-weighted decile portfolios formed on out-of-sample XGBoost predictions. Panels (a) and (b): abnormal returns predicted from 45 conventional characteristics (step 1). Panels (c) and (d): abnormal returns predicted from 22 sustainability characteristics (step 0). Panels (e) and (f): step-1 residuals predicted from the same 22 sustainability characteristics (step 2). The sustainability models are evaluated with the cross-out-of-sample scheme on the 2002–2024 panel, and their portfolios begin once sufficient cross-sectional coverage is available.

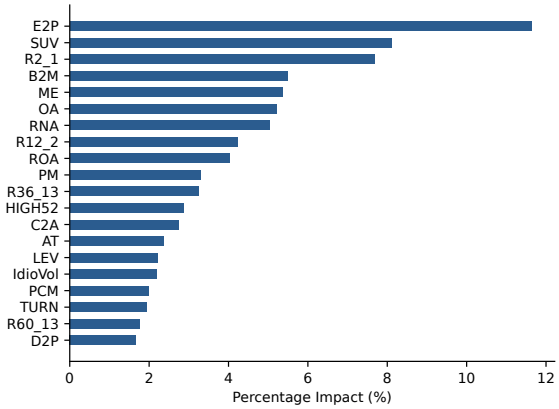
still ranks stocks, because each of its 20 members is a tree that splits on the inputs, but its predictions are coarse. The placebo calibration of Section 4.4 applies the same tuning and early-stopping rules, so its estimates of power account for this regularization. When a signal is present, the validation loss keeps improving and the pipeline keeps adding trees.

4.3 Sustainability does not predict the step-1 residuals

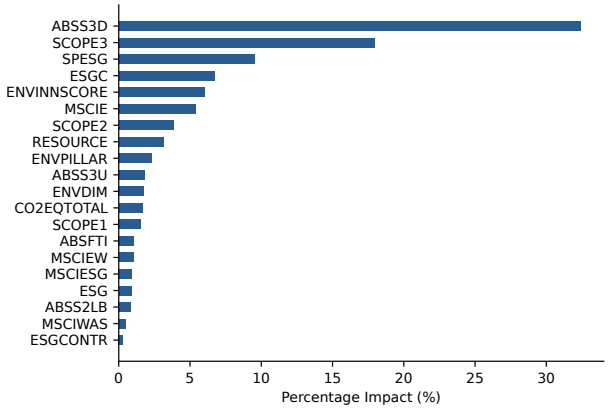
A natural objection to the unconditional analysis is that sustainability premia could be masked by correlated exposures to conventional characteristics. The residual analysis (step 2) addresses this concern directly. The step-1 residuals are, based on our estimate of the conditional mean, orthogonal to all 45 characteristics (Section 3.1), so predictability in them would reflect information beyond what conventional characteristics contain. Panels (e) and (f) of Figure 5 show no separation of deciles, a long–short Sharpe ratio of -0.08 ($t = -0.33$), and an IC of -0.002 (Table 1). Sustainability information neither predicts returns unconditionally nor predicts the return variation left unexplained by conventional characteristics. The residual model relies on the same Scope 3 emission variables as the returns model (Panel (c) of Figure 6).

The residual test is only as good as the orthogonalization, and we check it directly. An XGBoost model that predicts the step-1 residuals with the 45 characteristics themselves, tuned and evaluated with the same cross-out-of-sample scheme on 2002–2024, earns a long–short Sharpe ratio of -0.12 ($t = -0.61$, 95% confidence interval $[-0.52, 0.27]$), and its deciles show no monotonic pattern (Internet Appendix Section IA.5). Its IC of -0.014 (last row of Table 1) is small but significant ($t = -2.29$), and its sign is informative. A negative relation means that the step-1 model, which is fixed in 2001, slightly over-predicts along the characteristics in the later sample, in line with the decay of published predictors documented by McLean and Pontiff (2016). The residuals therefore lean slightly against the characteristics rather than retaining characteristic-related expected returns. This leftover is too small to exploit, and its sign works against a spurious sustainability premium in step 2, because a sustainability variable that is correlated with the characteristics would inherit a negative rather than a positive relation with the residuals. For the purpose of step 2 the residuals can be treated as orthogonal to the characteristics.

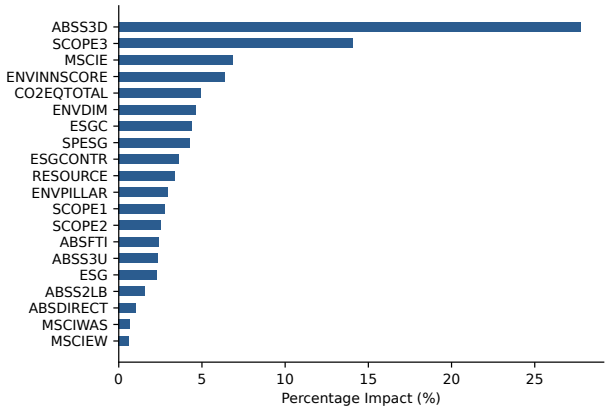
The IC columns of Table 1 provide a complementary view that does not depend on portfolio construction. The characteristics model achieves an average monthly rank IC of 0.037, an economically meaningful value by the standards of the cross-sectional prediction literature, while the ICs of both sustainability models are negative (-0.009 and -0.002) and not significant ($t = -1.56$ and $t = -0.45$). The information content of the inputs, not the



(a) Characteristics → abnormal returns (step 1)



(b) Sustainability → abnormal returns (step 0)



(c) Sustainability → residuals (step 2)

Figure 6: Variable importance of the three headline models. Mean absolute SHAP value of each input in the XGBoost ensemble, expressed as the percentage share of total feature impact. Panel (a) shows the 20 most important of the 45 conventional characteristics in the step-1 model, and Panels (b) and (c) show the 22 sustainability characteristics in the step-0 and step-2 models.

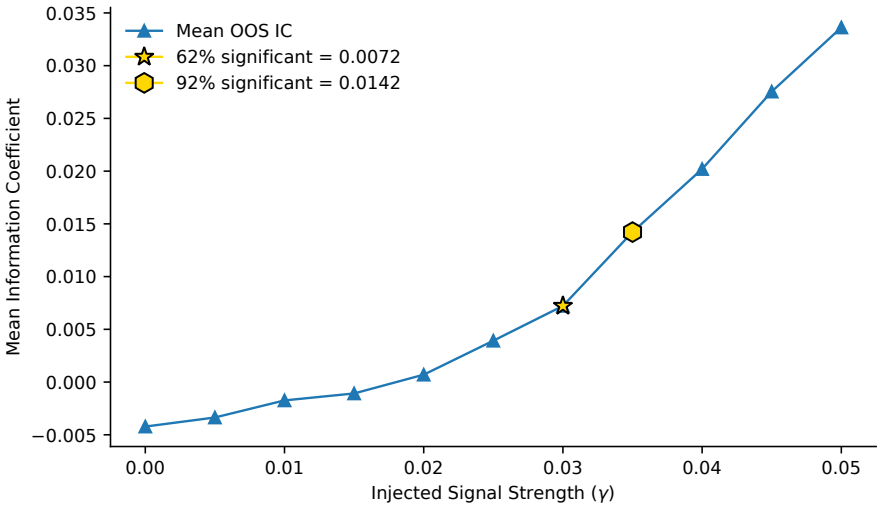


Figure 7: Placebo detection curve. Mean out-of-sample information coefficient (IC) of the full pipeline when a synthetic feature with population correlation γ to next month's abnormal return is added to the 22 sustainability variables, averaged over 50 draws per value of γ. The star marks $\gamma = 0.03$ (mean IC 0.0072), where the mean IC of the planted signal is significant at the 5% level (Newey–West t-statistic) in 62% of the draws, and the hexagon marks $\gamma = 0.035$ (mean IC 0.014), where it is significant in 92% of the draws. At $\gamma = 0.05$ (mean IC 0.034) it is significant in every draw.

portfolio construction, drives the performance gap. The next subsection asks how small a signal the protocol could have found.

4.4 Power: what the protocol can detect

Three pieces of evidence measure the power of the protocol. First, the characteristics model shows that it detects predictability of the magnitude documented for conventional predictors, and the Ledoit–Wolf tests of Table 1 confirm that its Sharpe ratio is significantly higher than those of the sustainability models. Second, the confidence intervals in Table 1 bound what the sustainability models can earn, as they exclude Sharpe ratios above 0.38, about half of the 0.72 of the characteristics model. Third, a placebo calibration measures how weak a signal the full pipeline can still find. For each stock i and month t we generate a synthetic feature

$$X_{i,t}^{\text{synthetic}} = \gamma \cdot y_{i,t+1} + \sqrt{1-\gamma^2} \cdot \sigma_{y,t+1}\, e_{i,t}, \tag{4}$$

where $y_{i,t+1}$ is next month's abnormal return, $e_{i,t} \sim \mathcal{N}(0,1)$ is white noise, and $\sigma_{y,t+1}$ is the cross-sectional standard deviation of abnormal returns in month $t+1$, so that γ is the population correlation between the feature and the target. The feature is added to the 22 sustainability variables, so the model has 23 inputs and has to find the signal among them. For each $\gamma \in \{0, 0.005, \ldots, 0.03, 0.05\}$ we draw 50 independent noise realizations and

re-estimate the full pipeline, including hyperparameter tuning, early stopping, and cross-out-of-sample evaluation, for each draw. In each draw we test the mean IC with the same Newey–West t-statistic that we use for the ICs in Table 1. Figure 7 plots the mean IC against γ. At $\gamma = 0.030$ the planted signal has a mean IC of 0.007, about a fifth of the 0.037 of the characteristics model, and it is detected at the 5% level in 62% of the draws. The detection rate reaches 92% at $\gamma = 0.035$, where the mean IC is 0.014, about a third of the characteristics model. At $\gamma = 0.05$ its mean IC is 0.034, comparable to the characteristics model, and it is detected in every draw. Without a planted signal ($\gamma = 0$) the mean IC of the pipeline is -0.004, in the range of the sustainability models' own ICs of -0.009 and -0.002. The planted signal is a favorable case. Its correlation with returns is the same in every month and its noise is idiosyncratic, so it is easier to detect than a real signal of the same average strength whose IC varies with market conditions. Even so, the evidence is clear. A signal as strong as conventional characteristics would not have been missed, and the null result of this section is not a statement about the limits of the method.

5 The Anatomy of Apparent Predictability

If sustainability information carries no detectable signal, why does a prominent literature find large green outperformance? This section takes the leading result as a laboratory. Pástor, Stambaugh, and Taylor (2022) construct a greenness score from two MSCI variables, the environmental pillar score and the environmental pillar weight, and form a tercile-based green-minus-brown (GMB) long–short portfolio over November 2012 to December 2020, reporting an annualized Sharpe ratio of approximately 1.14. PST interpret this outperformance as the realized consequence of unanticipated increases in climate concerns, not as evidence of high expected returns on green stocks, and their equilibrium model (Pástor, Stambaugh, and Taylor, 2021) predicts the opposite sign for expected returns. Subsequent empirical work and investment practice have nonetheless often read green outperformance as a return premium or as an investable signal. We ask how robust that reading is under the standards of the anomalies literature.

PST's design embeds four choices, each defensible in isolation. The universe is restricted to stocks with complete MSCI data, the sample begins in late 2012, the sorts are terciles, and the object of interest is realized returns rather than characteristic-adjusted returns. We reproduce their finding under their choices and then perturb each choice toward an equally standard alternative. A fifth choice, the search over specifications that precedes publication, is invisible in any single published table, and we make it explicit in Section 5.6. Tables 2 and 3 contain the full grid of two universes $\times$ two sample starts $\times$ two sample ends $\times$

two granularities, for portfolios sorted on the raw greenness score (Panels A and B) and on XGBoost predictions that use the score as the only input (Panels C and D), for returns and for characteristic-orthogonal residuals. The prediction-sorted panels apply the cross-out-of-sample scheme of Section 3.3 within each sample, so they show what a flexible learner extracts from the score when it is allowed to fit the mapping from the score to returns inside the window under study.

Table 2: Green-minus-brown portfolio performance in the restricted (complete-case) universe. Long–short tercile and decile portfolios. Panels A and B sort on the Pástor, Stambaugh, and Taylor (2022) greenness score and report realized excess returns (A) and characteristic-orthogonal residuals (B). Panels C and D sort on out-of-sample XGBoost predictions that use the score as the only input, of returns (C) and of step-1 residuals (D), and report realized excess returns. Portfolios are formed with samples starting in November 2012 or January 2007 and ending in December 2024 ("Full sample") or December 2020. The universe keeps stocks with non-missing MSCI inputs (1,627 tickers in January 2013, against roughly 2,000 in PST). Means and volatilities (in parentheses) are annualized and in percent, and Sharpe ratios are annualized. *, **, *** denote 10%, 5%, 1% significance of the mean long–short return in one-sided tests of a positive spread (Section 3.4), with Newey–West standard errors with four lags (three lags for the 2012–2020 samples). Two-sided Ledoit–Wolf tests do not reject equality of the Sharpe ratios of the score-sorted (Panel A) and prediction-sorted (Panel C) return portfolios in any cell.

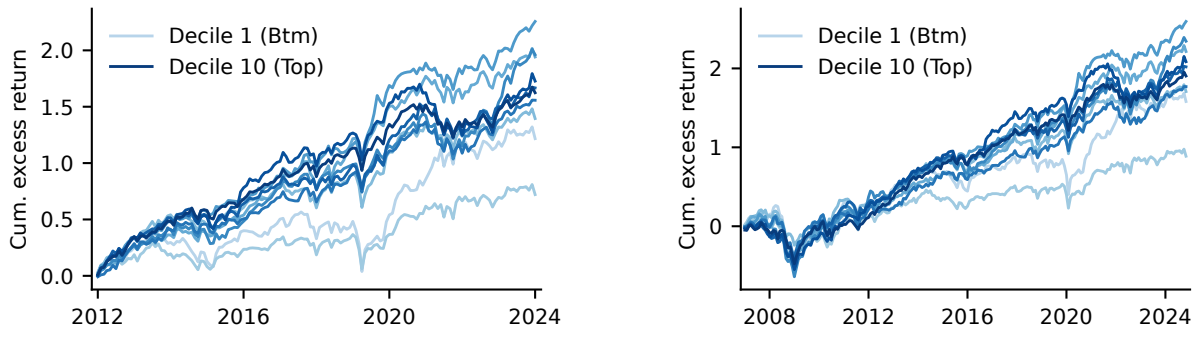
	Full sample (ending 2024)				**Ending 2020**			
	Start 11/2012		Start 2007		Start 11/2012		Start 2007	
Granularity	Mean	SR	Mean	SR	Mean	SR	Mean	SR
Panel A: Sorting on greenness score (returns)								
Terciles	5.1**	0.57	3.6*	0.39	8.0***	1.02	4.9**	0.54
	(9.0)		(9.5)		(7.8)		(9.0)	
Deciles	3.4	0.25	1.8	0.12	9.3**	0.77	4.9	0.32
	(13.8)		(15.4)		(12.2)		(15.1)	
Panel B: Sorting on greenness score (residuals)								
Terciles	2.8	0.32	1.9	0.20	4.2*	0.53	2.4	0.27
	(8.9)		(9.4)		(7.9)		(9.0)	
Deciles	0.9	0.06	−1.2	−0.08	3.8	0.31	−0.1	−0.00
	(13.6)		(15.3)		(12.3)		(15.2)	
Panel C: Sorting on XGB return prediction								
Terciles	2.6**	0.51	−0.1	−0.02	8.1***	1.09	4.7**	0.58
	(5.2)		(7.7)		(7.4)		(8.1)	
Deciles	−0.5	−0.04	−0.5	−0.05	7.3**	0.72	5.7*	0.42
	(11.1)		(12.0)		(10.1)		(13.5)	
Panel D: Sorting on XGB residual prediction								
Terciles	−1.3	−0.20	−2.3	−0.33	8.2***	1.05	2.5*	0.43
	(6.2)		(6.9)		(7.8)		(5.9)	
Deciles	−1.0	−0.14	0.9	0.09	8.0**	0.75	1.7	0.21
	(7.6)		(9.2)		(10.6)		(7.9)	

Table 3: Green-minus-brown portfolio performance in the full imputed universe. As Table 2, for the full imputed universe (1,813 tickers in January 2013). Two-sided Ledoit–Wolf tests reject equality of the Sharpe ratios of the score-sorted (Panel A) and prediction-sorted (Panel C) return portfolios only for the tercile portfolios in the 2012–2024 sample ($p = 0.016$), where the score-sorted portfolio has the higher Sharpe ratio, and the corresponding decile comparison is marginal ($p = 0.062$). No other return-based comparison rejects.

	Full sample (ending 2024)				**Ending 2020**			
	Start 11/2012		Start 2007		Start 11/2012		Start 2007	
Granularity	Mean	SR	Mean	SR	Mean	SR	Mean	SR
Panel A: Sorting on greenness score (returns)								
Terciles	4.8**	0.62	2.2	0.27	8.0***	1.13	3.4*	0.41
	(7.9)		(8.4)		(7.1)		(8.2)	
Deciles	3.8	0.31	2.0	0.15	10.2***	0.88	5.2*	0.38
	(12.4)		(13.5)		(11.6)		(13.5)	
Panel B: Sorting on greenness score (residuals)								
Terciles	2.6	0.34	0.2	0.02	4.2**	0.59	0.5	0.06
	(7.8)		(8.3)		(7.1)		(8.1)	
Deciles	1.7	0.14	−0.8	−0.06	4.8	0.41	0.3	0.02
	(12.2)		(13.4)		(11.7)		(13.5)	
Panel C: Sorting on XGB return prediction								
Terciles	0.5	0.07	1.3	0.18	7.4***	1.15	5.1***	0.74
	(7.0)		(7.1)		(6.4)		(6.9)	
Deciles	−2.3	−0.19	−0.5	−0.06	9.4***	1.13	6.7***	0.67
	(12.2)		(9.7)		(8.3)		(9.9)	
Panel D: Sorting on XGB residual prediction								
Terciles	−0.1	−0.01	1.3	0.19	8.0***	1.19	3.3*	0.40
	(7.1)		(6.9)		(6.8)		(8.2)	
Deciles	−1.3	−0.12	−0.4	−0.05	10.3***	1.11	4.7*	0.44
	(10.8)		(8.2)		(9.3)		(10.8)	

5.1 Replication

Under PST's choices (restricted universe, post-2012 sample ending in 2020, terciles, realized returns), we reproduce their finding. The GMB portfolio earns a significant annualized mean of 8.0% with a Sharpe ratio of 1.02 (Table 2, Panel A), close to their reported 1.14 given our slightly smaller universe. In the full imputed universe the corresponding Sharpe ratio is 1.13 (Table 3). The replication matters, because what follows is not a failure to reproduce but a demonstration of how much the interpretation of the number depends on choices that are invisible in the headline.



(a) Sample starting 11/2012

(b) Sample starting 2007

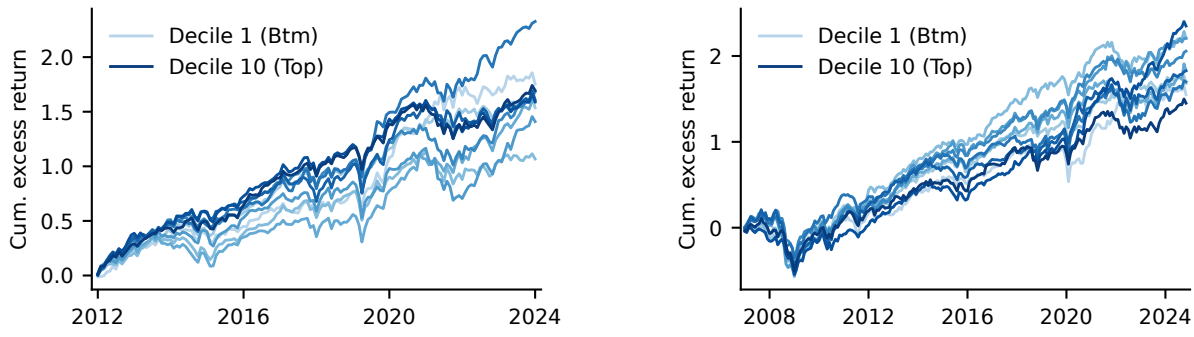
Figure 8: Greenness-score deciles and the sample period. Cumulative excess returns of value-weighted decile portfolios sorted on PST's greenness score, in the restricted universe, for samples starting November 2012 (left) and January 2007 (right). Decile 10 contains the greenest firms.

5.2 Design choice 1: the sample period

The sample period is the single most important choice. Extending the sample back to January 2007, the first month for which the underlying MSCI data exist, roughly halves the tercile Sharpe ratio (1.02 to 0.54) and renders the decile sort insignificant in the restricted universe (Table 2, Panel A). Extending the sample forward to December 2024 has the same effect (1.02 to 0.57), and the sample that uses all available data, 2007 to 2024, delivers a Sharpe ratio of 0.39 that is significant only at the 10% level. The outperformance is a feature of the window PST study, not of the periods before or after it. Figure 8 shows why. Sorted on the raw score, the deciles are tightly intertwined, before and after 2012. The top (greenest) decile sits in the middle of the pack, and the strongest single pattern is that the second decile lags, not that the greenest firms lead. Prediction-sorted portfolios based on out-of-sample XGBoost forecasts from the score behave in the same way (Figure 9).

The signal-strength analysis in Figures 10 and 11 makes the same point without portfolios. In the samples ending in 2020, the monthly rank IC of the greenness score and its IC ratio are positive for every design (highest for the 2012 start, with ICs of 0.015 and 0.017 and IC ratios of 0.53 and 0.63 in the two universes). In the samples ending in 2024, the IC is negative for every design except the 2012 start in the restricted universe, where it is positive but below 0.001, and for the 2007 start the IC ratio turns negative, reaching -0.05 in the restricted universe and -0.19 in the full universe. Whatever ranking ability the score had in the 2010s disappears once the sample includes 2021 to 2024.

Is discarding 2007 to 2011 justified by the thin early cross-section, as is often argued? The early period has a median of only 273 firms per month in the restricted universe, versus thousands later. We test this justification directly with a bootstrap. We repeatedly draw



(a) Sample starting 11/2012

(b) Sample starting 2007

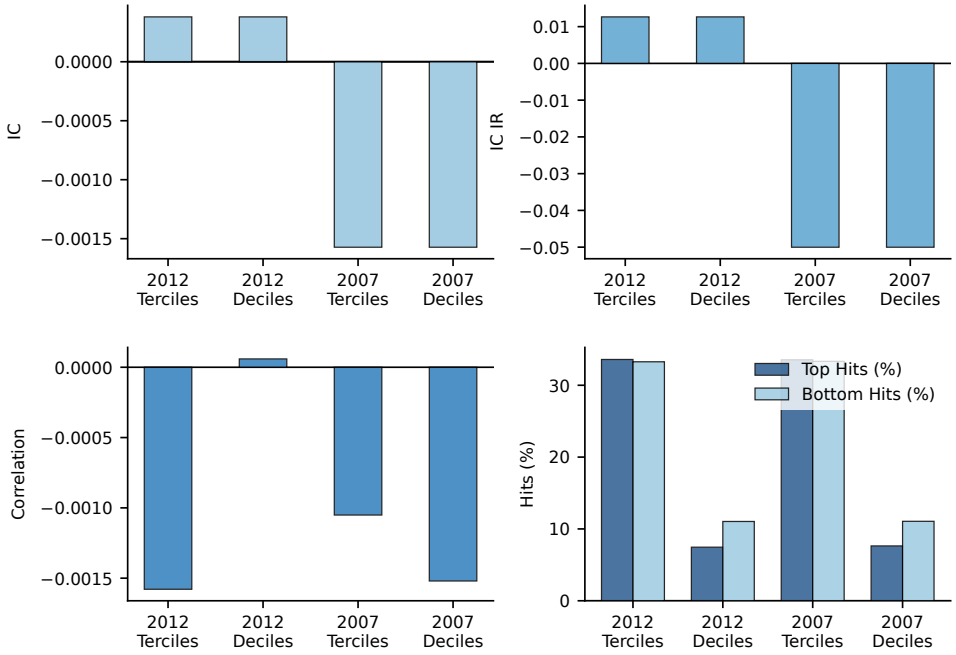
Figure 9: Prediction-sorted deciles and the sample period. Cumulative excess returns of value-weighted decile portfolios formed on out-of-sample XGBoost predictions of abnormal returns from the greenness score, in the restricted universe, for samples starting November 2012 (left) and January 2007 (right).

273 firms per month *within the post-2012 period* (1,000 draws), recompute the GMB tercile portfolio, and compare the resulting Sharpe-ratio distribution to the statistic observed with the full post-2012 cross-section. Figure 12 shows the result. The observed Sharpe ratio lies above the bootstrap mean but well within the 95% band, with a one-tailed p-value of 0.16 in the restricted universe and 0.20 in the full imputed universe (where the median early cross-section is 899 firms). Cross-sectional thinness of the early-period magnitude does not, by itself, destroy a true signal of the claimed size. The pre-2012 sample is not too small to detect green outperformance. Green outperformance is simply absent before 2012. Excluding the early years therefore selects a favorable window rather than removing an uninformative one.

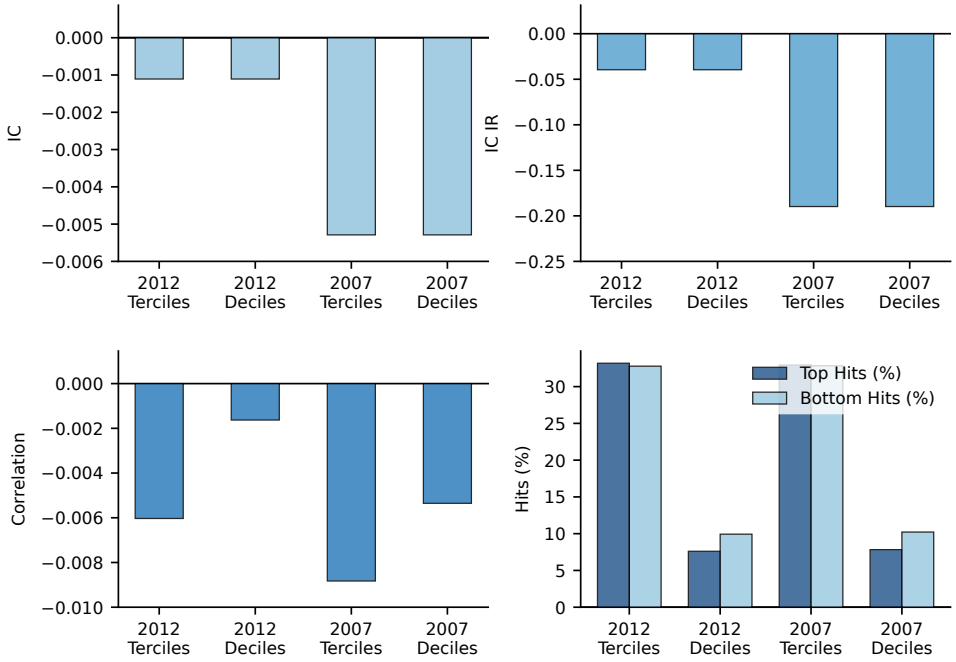
5.3 Design choice 2: the investment universe

Comparing Tables 2 and 3 isolates the universe choice. Within PST's window the choice matters little, as the full imputed universe delivers a tercile Sharpe ratio of 1.13 against 1.02 in the complete-case universe. In the samples that start in 2007, however, moving to the full universe weakens the score-sorted results. The 2007–2024 tercile Sharpe ratio falls from 0.39 to 0.27 for returns and loses significance, and the corresponding residual portfolios, at 0.20 and 0.02, are insignificant in both universes. Complete-case selection conditions on MSCI coverage, which expanded over time and is not random with respect to firm size and industry. The sensitivity of the longer-sample results to this conditioning is itself evidence of fragility, since a robust premium should not depend on whether the econometrician keeps or imputes partially covered firms.

The prediction-sorted panels add two observations. First, the XGBoost-sorted GMB

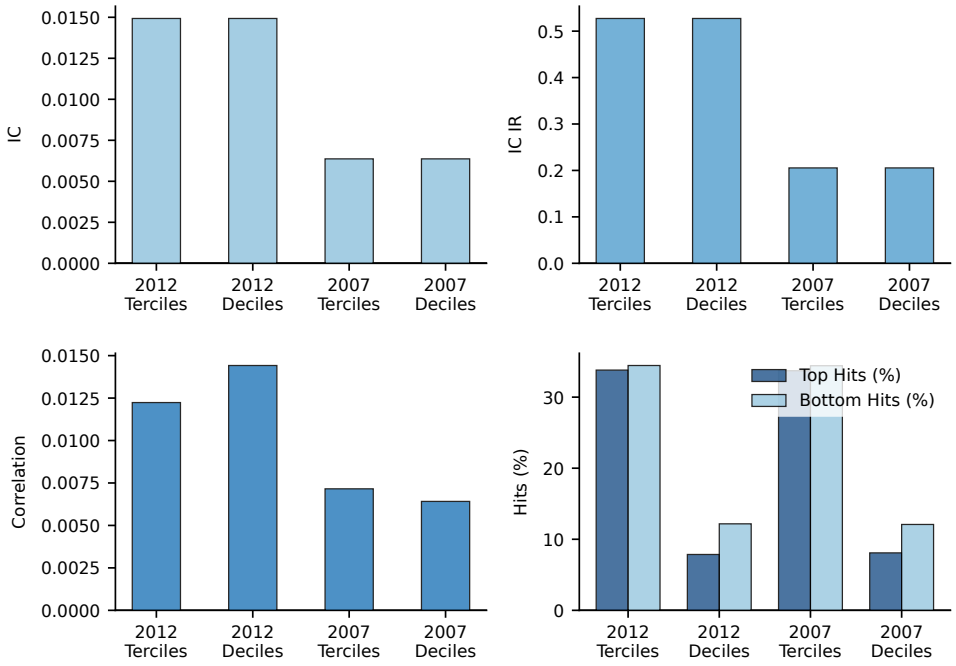


(a) Restricted universe

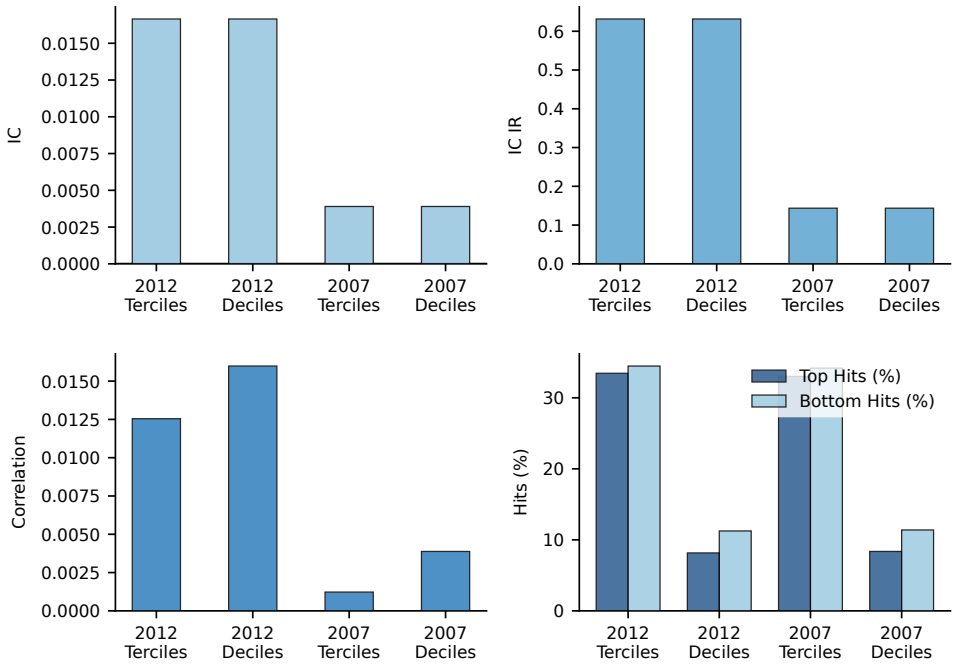


(b) Full imputed universe

Figure 10: Signal strength and bucketing quality across designs, samples ending in 2024. For tercile and decile sorts on the greenness score with samples starting 2007 and 2012, the top panels show the information coefficient (left) and the IC ratio (right), and the bottom panels show the rank correlation between predicted and realized buckets (left) and the top- and bottom-bucket hit rates (the share of stocks correctly assigned to the extreme buckets) (right). Under random assignment the expected hit rate equals the bucket share, one third for terciles and one tenth for deciles.

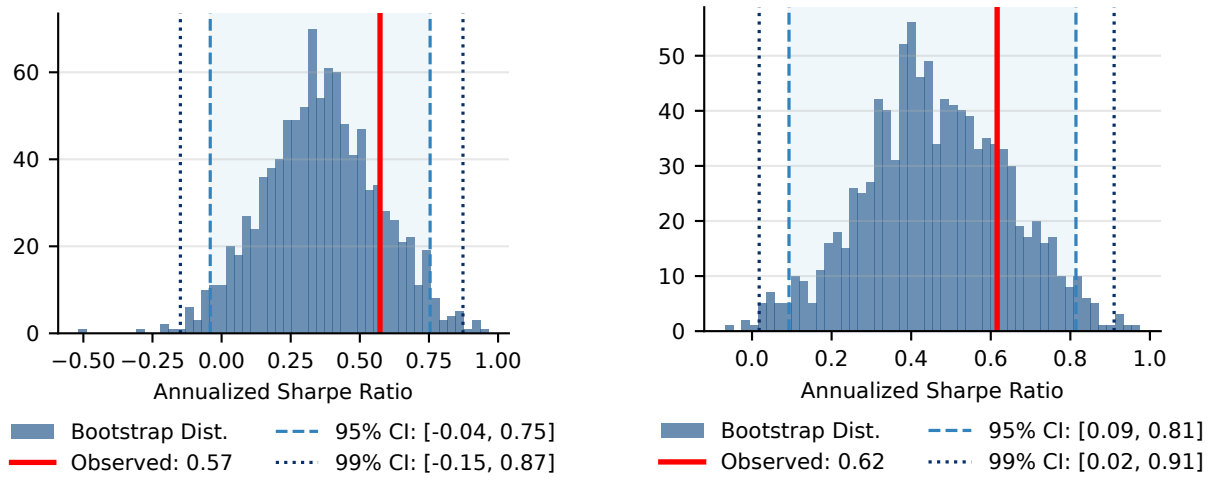


(a) Restricted universe



(b) Full imputed universe

Figure 11: Signal strength and bucketing quality across designs, samples ending in 2020. As Figure 10, for samples ending in December 2020.



(a) Restricted universe ($p = 0.160$)

(b) Full imputed universe ($p = 0.203$)

Figure 12: Bootstrap test of the thin early cross-section. Distribution of annualized GMB tercile Sharpe ratios when 273 (restricted universe) and 899 (full imputed universe) firms, the median 2007–2011 cross-section, are repeatedly sampled within the post-2012 period (1,000 draws), compared with the Sharpe ratio observed with the full post-2012 cross-section (vertical red line). Dashed and dotted lines mark the 95% and 99% bands of the bootstrap distribution, and p-values are one-tailed.

portfolios are significantly positive in every cell of PST's window and in seven of the eight cells of the 2007–2020 sample (at least at the 10% level), but in only one of the sixteen cells of the samples that end in 2024 (return terciles starting in 2012 in the restricted universe, at the 5% level), and that cell does not survive the move to the full universe (Sharpe ratio 0.51 against 0.07). A signal whose significance disappears once the last four years of data are included is behaving like fitted noise rather than like a premium. Second, the Ledoit–Wolf tests reject equality of the score-sorted and prediction-sorted return Sharpe ratios only in the 2012–2024 sample of the full universe, where it is the score-sorted portfolio that does better (0.62 against 0.07 for terciles with a two-sided $p = 0.016$, and 0.31 against -0.19 for deciles with a marginal $p = 0.062$). The learner fits the mapping from the score to returns over a sample in which that mapping turns from positive to negative once the years after 2020 are included (Section 5.2), and ends up with nothing, whereas a fixed long-green rule inherits the 2012–2020 run. Neither is the behavior of a stable predictor. In the samples ending in 2020, the two sorts are statistically indistinguishable in every cell of both universes.

5.4 Design choice 3: portfolio granularity

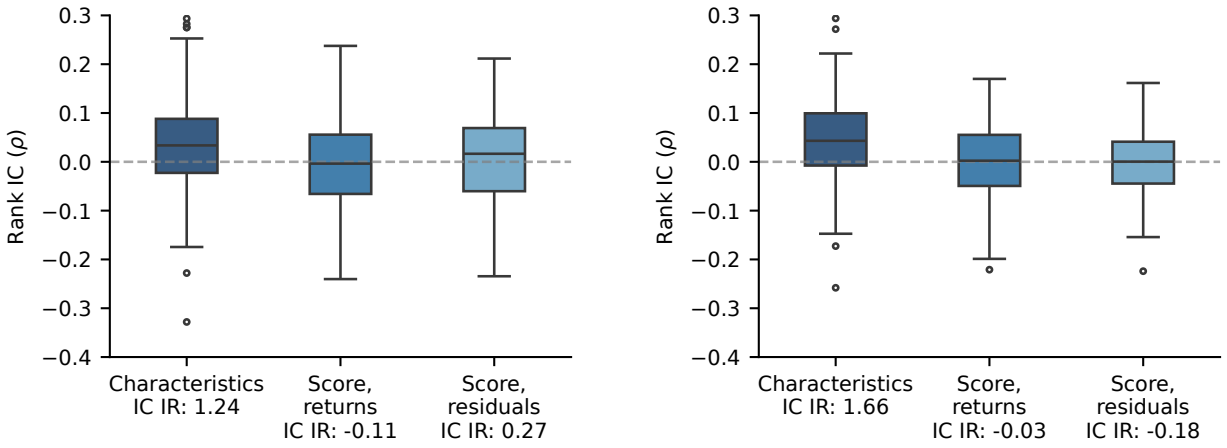
PST sort into terciles, whereas the standard in the anomalies literature is deciles. Table 2 shows that outside PST's window the decile sort on the greenness score is never significant, whereas the tercile sort is. In the 2012–2024 sample, for example, the tercile sort is significant

at the 5% level (mean 5.1%) while the decile sort is not (mean 3.4%), despite the decile sort taking more extreme positions. The mechanics are visible in the table itself. Moving from terciles to deciles reduces the mean spread by a third (5.1% to 3.4%) while raising the annualized volatility from 9.0% to 13.8%, so the Sharpe ratio falls from 0.57 to 0.25. A genuine, cross-sectionally graded signal behaves differently. Concentrating on the extremes raises the mean spread by enough to compensate for the lost diversification, as it does for the characteristics model, whose Sharpe ratio rises from 0.60 with terciles to 0.72 with deciles (Section 6). The bucketing-quality analysis in the bottom panels of Figure 10 explains why the greenness score cannot do this. The share of stocks that the score assigns correctly to the top and bottom buckets equals the random-assignment benchmark at both granularities (about one third for terciles and one tenth for deciles), and the rank correlation between predicted and realized buckets is negative or indistinguishable from zero in the samples ending in 2024. Ending the sample in 2020 (Figure 11) turns these rank correlations positive but tiny, and the tail hit rates stay at the random benchmark, so even in the favorable window the score does not identify the extreme performers. Coarse sorts flatter weak signals because a highly diversified tercile portfolio converts a broad tilt into a respectable Sharpe ratio, and granular sorts reveal that the score cannot identify the extremes.

5.5 Design choice 4: the prediction target

Sorting on the greenness score but evaluating characteristic-orthogonal *residuals* (Panel B of both tables) removes about half of the effect inside PST's window and all of its significance outside it. In PST's window and universe, the residual GMB portfolio earns 4.2% with a Sharpe ratio of 0.53, against 8.0% and 1.02 for realized returns, and is significant only at the 10% level. In the full universe the Sharpe ratio falls from 1.13 to 0.59. In the twelve score-sorted cells outside PST's window, not one residual portfolio is significant in either universe. Green stocks differ systematically from brown stocks in conventional characteristics (industry composition, size, investment, and profitability), and conventional characteristics predict returns, so sorting on greenness inherits part of that predictability. The part that survives residualization is the part that lives in PST's window, which Section 5.2 has already shown to be a feature of that window.

The prediction-sorted residual portfolios (Panel D) make the same point from the other side. Inside PST's window, an XGBoost model that is fitted to the window, uses nothing but the score, and predicts residuals rather than returns forms portfolios whose realized excess returns have Sharpe ratios of 1.05 and 1.19 in the two universes, so it recovers the window's pattern whatever the target. In every sample that extends beyond the window, the same



(a) Restricted (complete-case) universe

(b) Full imputed universe

Figure 13: Information coefficients of the characteristics model and the greenness-score models. Boxplots of the monthly cross-sectional information coefficient (rank correlation between out-of-sample predictions and realized outcomes) for the conventional-characteristics model, the XGBoost return model with the greenness score as input, and the XGBoost residual model with the greenness score as input. All three models are evaluated on a common universe so that differences cannot be attributed to differences in data availability. The number below each label (IC IR) is the annualized IC ratio.

model earns Sharpe ratios between -0.33 and 0.44, and only three of the twelve cells are significant, all at the 10% level. A pattern that a flexible learner can fit inside a window but cannot find outside it is a realization, not a predictor.

Figure 13 summarizes the information content of the greenness score across prediction targets in a way that is independent of portfolio construction. It plots the distribution of monthly cross-sectional rank ICs for three models evaluated on a common universe, the conventional-characteristics model, the XGBoost model predicting returns from the greenness score, and the XGBoost model predicting step-1 residuals from the greenness score. In both the restricted and the full imputed universe, the conventional-characteristics model yields a consistently positive distribution with the highest median IC and annualized IC ratios of 1.24 and 1.66, respectively. The greenness-score return and residual models are centered near zero, with IC ratios ranging from -0.18 to 0.27. The pattern is the same for the 22 sustainability variables (Figure 14), where the characteristics model has an IC ratio of 1.24, while the sustainability return and residual models have IC ratios of -0.46 and 0.01. Monthly IC distributions are wide for every model, which illustrates that single-month correlations are dominated by noise, and the IC ratio ranks the models exactly as the portfolio results do.

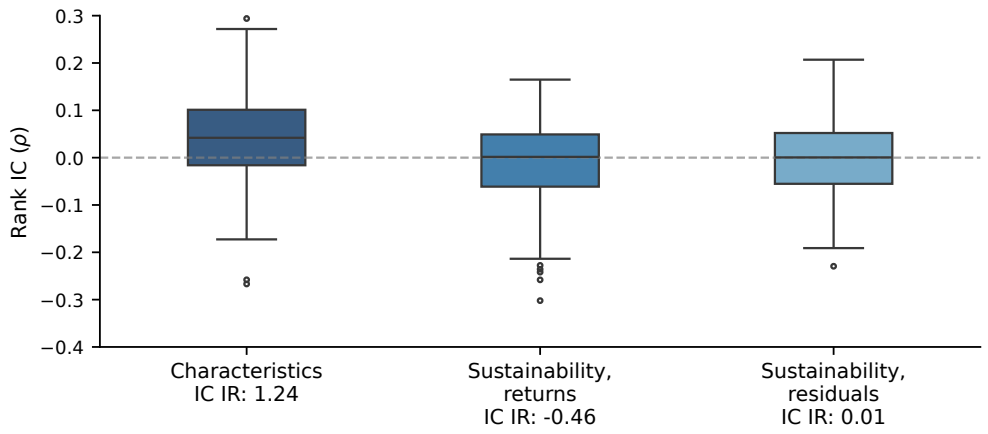


Figure 14: Information coefficients of the characteristics model and the sustainability models. Boxplots of the monthly cross-sectional information coefficient for the conventional-characteristics model, the XGBoost return model with the 22 sustainability variables as input, and the XGBoost residual model with the same inputs, all evaluated on a common universe. The number below each label (IC IR) is the annualized IC ratio. The vertical axis omits one month in which the IC of the characteristics model was −0.8.

5.6 Design choice 5: specification search and multiple testing

The choices above are visible in published papers. The final ingredient is invisible, the search over specifications that precedes publication. We make the search explicit. We estimate sustainability models on nested sets of $1, 2, \ldots, 22$ variables, added in the order of their post-2012 data availability and held fixed across samples, for three sample starts (2002, 2007, November 2012) and for both abnormal returns and residuals. This yields a family of 132 long–short portfolios that mimics a researcher's exploration space. Table 4 reports the 2012-start family, and the 2002 and 2007 families are in Appendix Tables A.6 and A.5.

Unadjusted, the family contains scattered "significant" results. t-statistics above two appear for particular variable counts in particular periods, in either direction, the kind of cells that, picked individually, would support a publishable claim. Adjusted, nothing survives. Pooling all 132 specifications, no Holm–Bonferroni-corrected p-value falls below conventional levels (the minimum is 0.78, at $k = 9$ for the sample starting in 2002), and the Benjamini–Hochberg procedure leaves no rejections at the 5% false-discovery rate. The $k = 9$ model, which combines the four MSCI variables with five of the six Trucost emission variables, illustrates how little a single cell means. Predicting characteristic-orthogonal residuals with the sample starting in 2002, it earns a Sharpe ratio of 0.58 ($t = 2.52$, unadjusted one-sided $p = 0.006$). With the same nine variables and the same target, it earns -0.29 ($t = -1.11$) when the sample starts in 2007 and -0.44 ($t = -1.49$) when it starts in November 2012. Nothing changes across the three cells except the start date. Two-sided tests tell the same

Table 4: Incremental information analysis for the sample starting November 2012. Each row is an XGBoost model using the k sustainability variables with the fewest missing observations after 2012, predicting abnormal returns (left) or step-1 residuals (right). Long–short decile portfolios, value weighted. μ is the annualized mean excess return in percent, SR the annualized Sharpe ratio, and t-stat the Newey–West t-statistic of the mean. p-values are one-sided, and HB and BH denote Holm–Bonferroni and Benjamini–Hochberg adjusted p-values, pooled over the family of all 132 specifications (22 models × three sample starts × two targets).

	Returns						Residuals					
k	μ (%)	SR	t-stat	p	HB	BH	μ (%)	SR	t-stat	p	HB	BH
1	−0.5	−0.06	−0.23	0.59	1.00	1.00	−4.8	−0.68	−2.65	1.00	1.00	1.00
2	−0.5	−0.06	−0.19	0.57	1.00	1.00	−3.8	−0.46	−1.62	0.95	1.00	1.00
3	−0.3	−0.02	−0.06	0.52	1.00	1.00	−1.5	−0.14	−0.50	0.69	1.00	1.00
4	−0.6	−0.04	−0.12	0.55	1.00	1.00	−6.4	−0.64	−2.37	0.99	1.00	1.00
5	−11.8	−0.54	−1.80	0.96	1.00	1.00	−10.1	−0.59	−1.91	0.97	1.00	1.00
6	−8.0	−0.48	−1.51	0.93	1.00	1.00	−7.3	−0.41	−1.32	0.91	1.00	1.00
7	−13.2	−0.68	−2.15	0.98	1.00	1.00	−6.4	−0.32	−1.03	0.85	1.00	1.00
8	−9.9	−0.47	−1.58	0.94	1.00	1.00	−6.9	−0.35	−1.19	0.88	1.00	1.00
9	−11.6	−0.55	−1.82	0.97	1.00	1.00	−8.5	−0.44	−1.49	0.93	1.00	1.00
10	−11.2	−0.52	−1.66	0.95	1.00	1.00	−9.9	−0.43	−1.32	0.91	1.00	1.00
11	−10.4	−0.57	−1.75	0.96	1.00	1.00	−7.5	−0.46	−1.53	0.94	1.00	1.00
12	−10.9	−0.53	−1.66	0.95	1.00	1.00	−5.8	−0.34	−1.19	0.88	1.00	1.00
13	−4.0	−0.24	−0.76	0.78	1.00	1.00	−5.6	−0.36	−1.19	0.88	1.00	1.00
14	−6.1	−0.37	−1.16	0.88	1.00	1.00	−8.2	−0.54	−1.61	0.95	1.00	1.00
15	−9.0	−0.55	−1.88	0.97	1.00	1.00	−5.4	−0.36	−1.25	0.89	1.00	1.00
16	−3.8	−0.22	−0.70	0.76	1.00	1.00	−4.2	−0.27	−0.93	0.83	1.00	1.00
17	−4.0	−0.29	−1.05	0.85	1.00	1.00	−3.7	−0.24	−0.79	0.79	1.00	1.00
18	−5.7	−0.39	−1.47	0.93	1.00	1.00	−5.5	−0.34	−1.13	0.87	1.00	1.00
19	−5.4	−0.36	−1.24	0.89	1.00	1.00	−2.9	−0.19	−0.63	0.73	1.00	1.00
20	−4.7	−0.26	−0.88	0.81	1.00	1.00	−3.0	−0.19	−0.61	0.73	1.00	1.00
21	−3.2	−0.22	−0.77	0.78	1.00	1.00	−1.6	−0.11	−0.38	0.65	1.00	1.00
22	−6.2	−0.42	−1.43	0.92	1.00	1.00	−2.3	−0.16	−0.51	0.69	1.00	1.00

story. Before correction, six cells are significantly negative and two significantly positive at the 5% level, and after correction none survives (the smallest two-sided Holm–Bonferroni and Benjamini–Hochberg adjusted p-values are 1.00 and 0.48). That significantly negative cells outnumber positive ones is what one expects when the relations a model learns in some periods reverse in others.

Formal model-comparison tests reinforce this conclusion. Diebold–Mariano tests on monthly average loss differentials cannot distinguish the forecast accuracy of any of the 22 nested sustainability models from a univariate benchmark that forecasts abnormal returns with a linear function of the greenness score estimated by OLS (the lowest p-value is 0.85, Internet Appendix Figure IA.39), and the Ledoit–Wolf test cannot distinguish their Sharpe ratios (Figure 15), where all raw p-values exceed 0.1 before any correction. The models differ only in how they fit noise.

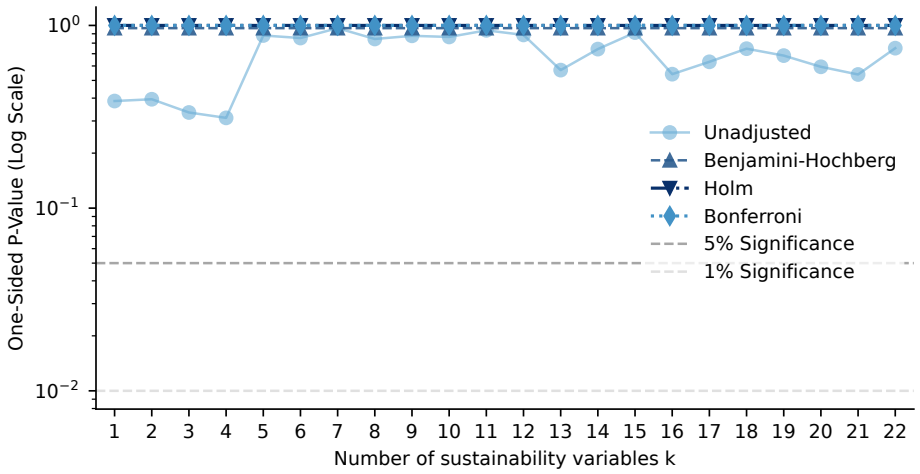


Figure 15: Sharpe-ratio comparisons across sustainability models. One-sided Ledoit and Wolf (2008) p-values for the difference between the Sharpe ratio of the k-variable sustainability model ($k = 1, \ldots, 22$, horizontal axis) and that of the univariate greenness-score benchmark, sample starting November 2012, together with the Holm–Bonferroni, Benjamini–Hochberg, and Bonferroni adjusted values. Dashed lines mark the 5% and 1% levels. The corresponding Diebold–Mariano tests on monthly average loss differentials are reported in the Internet Appendix.

5.7 Putting the pieces together

Table 5: Design choices and the green-minus-brown Sharpe ratio. Annualized Sharpe ratios of the score-sorted GMB portfolio from Tables 2 and 3, starting from PST's design (restricted universe, November 2012 to December 2020, terciles, realized returns) and changing one choice at a time, then all choices together. *, **, *** denote 10%, 5%, 1% significance of the mean long–short return in one-sided tests, as in Tables 2 and 3.

Design	Change relative to PST	Sharpe ratio
PST's design	none	1.02***
Sample period	start in January 2007	0.54**
	end in December 2024	0.57**
	all available data, 2007–2024	0.39*
Granularity	deciles instead of terciles	0.77**
Prediction target	characteristic-orthogonal residuals	0.53*
Universe	full imputed universe	1.13***
	full imputed universe, 2007–2024	0.27
All four choices	full universe, 2007–2024, deciles, residuals	−0.06

Table 5 collects the score-sorted Sharpe ratios from Tables 2 and 3 along the path from PST's design to the standard alternatives. The published Sharpe ratio above one requires the conjunction of all favorable choices.

Start the clock five years earlier, or let it run four years longer, and the Sharpe ratio halves. Use deciles, and significance disappears in every sample that extends beyond PST's window. Use characteristic-orthogonal returns, and the Sharpe ratio halves inside PST's

window and loses significance everywhere else. Use the full universe, and the results for the samples starting in 2007 weaken further. Account for the implicit search over specifications, and nothing remains anywhere in the design space. None of these perturbations is exotic, and each is the default choice in the broader cross-sectional literature. We conclude that the apparent predictability of stock returns from sustainability information is an artifact of research design in the operational sense that it does not survive standard robustness and inferential discipline. This conclusion is fully consistent with PST's own interpretation of green outperformance as the realized consequence of unanticipated shifts in climate concerns. What it rules out is the stronger reading that the greenness score, or the sustainability information behind it, predicts returns.

6 Robustness

Section 5 showed that the published positive result depends on design choices. This section asks the same question of our null result. The null of Section 4 rests on a particular set of variables, a particular learner, a particular portfolio construction, and a particular treatment of missing data. If sustainability information carried a signal that this design happens to miss, a different family of variables, a learner with a different balance of bias and variance, a portfolio that weights different stocks, or a different treatment of the gaps should reveal it. This section varies each of these choices in turn and explains what each alternative would detect. Table 6 collects the long–short statistics of all robustness specifications, 16 models for each of the two targets, together with Holm–Bonferroni and Benjamini–Hochberg adjusted p-values pooled over the 32 tests. Three of the 32 one-sided tests are nominally significant at the 5% level, and none survives the correction (the smallest adjusted p-values are 1.00 under Holm–Bonferroni and 0.34 under Benjamini–Hochberg). Two further specifications, the feedforward network for returns ($t = -2.61$) and the model without imputation for residuals ($t = -2.11$), have significantly *negative* spreads. With two-sided tests, these two are the only nominal rejections, and neither survives the correction (the smallest adjusted p-value is 0.29 under both procedures). A significantly negative out-of-sample spread is what a model produces when the relation it learns from two folds reverses in the third, and it is not exploitable, because the sign is not known in advance. The Internet Appendix reports the decile plots and full statistics for every specification.

Table 6: Robustness specifications and multiple-testing corrections. Each row is a sustainability model predicting abnormal returns (left) or step-1 residuals (right), and the headline model is the XGBoost model of Table 1. Long–short decile portfolios, value weighted unless stated otherwise. μ is the annualized mean excess return in percent, SR the annualized Sharpe ratio, t-stat the Newey–West t-statistic, p the one-sided p-value, and HB and BH the Holm–Bonferroni and Benjamini–Hochberg adjusted p-values pooled over the family of 16 models and two targets (32 tests). Rows are grouped by the subsections of Section 6, which also reports the two-sided results.

	Returns						Residuals					
Model	μ (%)	SR	t-stat	p	HB	BH	μ (%)	SR	t-stat	p	HB	BH
Headline model	−0.6	−0.06	−0.26	0.60	1.00	0.71	−0.8	−0.08	−0.33	0.63	1.00	0.72
Variable sets (Section 6.1)												
Emissions	0.7	0.06	0.30	0.38	1.00	0.53	2.2	0.19	0.88	0.19	1.00	0.53
ESG scores	1.3	0.14	0.65	0.26	1.00	0.53	3.0	0.33	1.84	0.03	1.00	0.34
Trucost	4.8	0.41	1.76	0.04	1.00	0.34	1.9	0.15	0.70	0.24	1.00	0.53
MSCI	2.9	0.23	0.82	0.21	1.00	0.53	−4.5	−0.50	−1.90	0.97	1.00	1.00
Refinitiv	0.6	0.06	0.31	0.38	1.00	0.53	0.2	0.02	0.12	0.45	1.00	0.58
S&P Global	0.8	0.09	0.48	0.32	1.00	0.53	0.9	0.10	0.46	0.32	1.00	0.53
All 513 variables	3.6	0.32	1.47	0.07	1.00	0.34	2.7	0.22	0.89	0.19	1.00	0.53
Learners and input transformations (Section 6.2)												
Elastic Net	3.1	0.32	1.45	0.07	1.00	0.34	−1.2	−0.10	−0.47	0.68	1.00	0.75
Feedforward network	−5.7	−0.54	−2.61	1.00	1.00	1.00	4.0	0.36	1.74	0.04	1.00	0.34
Ranked inputs	0.7	0.07	0.34	0.37	1.00	0.53	4.0	0.32	1.54	0.06	1.00	0.34
Portfolio construction (Section 6.3)												
Equal weighted	2.7	0.30	1.44	0.08	1.00	0.34	2.1	0.26	1.16	0.12	1.00	0.50
Prediction weighted	1.1	0.12	0.63	0.27	1.00	0.53	1.5	0.14	0.57	0.29	1.00	0.53
Terciles	1.4	0.22	1.03	0.15	1.00	0.53	−0.1	−0.02	−0.09	0.54	1.00	0.66
Quintiles	0.2	0.03	0.14	0.45	1.00	0.58	0.5	0.07	0.36	0.36	1.00	0.53
Imputation (Section 6.4)												
No imputation	0.8	0.07	0.32	0.38	1.00	0.53	−4.5	−0.41	−2.11	0.98	1.00	1.00

6.1 Variable sets

The baseline uses 22 variables that pool four providers. Three concerns motivate alternatives. First, the providers disagree with each other (Berg, Kölbel, and Rigobon, 2022), so pooling their variables may average away a signal that one provider's methodology captures and the others do not. Running steps 0 and 2 on the variables of S&P Global, Trucost, Refinitiv, and MSCI separately isolates each methodology. Second, different strands of the literature use different subsets of the information. The carbon-premium literature uses emissions alone (Bolton and Kacperczyk, 2021), and ratings-based studies and investment practice use the aggregated ESG scores, so we estimate the models on emissions only and on scores only. Third, our selection of 22 variables could have discarded the informative ones. Estimating the models on all 513 available variables lets the learner choose among them and tests whether the null is an artifact of our selection. The Internet Appendix (Sections IA.2 and IA.4) reports every specification.

No variable set predicts returns or residuals after the correction. Two produce nominally significant results on one-sided tests, the Trucost return model (Sharpe ratio 0.41, $t = 1.76$) and the ESG-score residual model (0.33, $t = 1.84$), and neither comes close to surviving the correction. Emissions alone, the variables behind the carbon premium of Bolton and Kacperczyk (2021), predict neither returns (0.06, $t = 0.30$) nor residuals (0.19, $t = 0.88$). This is the out-of-sample counterpart of the finding of Duan and Pelger (2025) that the carbon premium disappears once vendor-estimated emissions are treated as missing. Neither in sample with valid inference nor out of sample is there a return relation to exploit. The MSCI residual model has a negative spread (-0.50, $t = -1.90$). The model with all 513 inputs, the most flexible specification in the paper and the one most prone to finding spurious structure, comes no closer than the others (0.32 with $t = 1.47$ for returns and 0.22 with $t = 0.89$ for residuals).

The null therefore does not depend on pooling across providers, on the family of variables, or on our selection. No provider's methodology carries a signal that the pooled variables hide, and the nominal results that single providers produce come with both signs, which is the pattern of noise rather than of a signal that one methodology captures. Giving the learner every variable the providers sell does not change the answer either.

6.2 Alternative learners and input transformations

The baseline learner is a tree ensemble. A learner can miss a signal for two opposite reasons. A flexible method fits noise in a short panel, which raises variance, and a restrictive method cannot represent the true relation, which raises bias. This is the classical bias-variance tradeoff, and the right point on it depends on the sample and on the signal. With a short and sparse panel, a simpler model might learn a weak but stable structure that the tree ensemble regularizes away, and a more flexible model might capture a structure that trees represent poorly. We therefore repeat both steps with a penalized linear model, Elastic Net, which sits at the low-variance, high-bias end, and with a feedforward neural network, which is at least as flexible as the tree ensemble and fits smooth rather than piecewise constant functions (results in Internet Appendix Section IA.1, definitions and tuning in Section IA.7). Both learners are sensitive to the scale of their inputs, so the sustainability variables enter as rank quantiles. To separate the effect of the learner from that of the transformation, we also run XGBoost on the same transformed inputs (Internet Appendix Section IA.4).

Replacing XGBoost with Elastic Net leaves the conclusions unchanged. Characteristics predict abnormal returns (Sharpe ratio 0.64, $t = 2.91$), and sustainability variables predict neither returns (0.32, $t = 1.45$) nor residuals (-0.10). A feedforward neural network produces

the strongest positive control (0.76, $t = 3.23$) but its sustainability return model produces a significantly negative spread (-0.54, $t = -2.61$). Its residual model is nominally significant on a one-sided test (0.36, $t = 1.74$), but not after the correction (Holm–Bonferroni $p = 1.00$). XGBoost on the same transformed inputs delivers a return model with $t = 0.34$ and a residual model with $t = 1.54$ (nominal one-sided $p = 0.06$, Holm–Bonferroni $p = 1.00$), so the network result is specific to the learner rather than to the transformation.

Two implications follow. The positive control is robust to the learner, with Sharpe ratios of 0.64, 0.72, and 0.76 for the linear model, the tree ensemble, and the network, so the benchmark against which the sustainability models are judged does not hinge on the choice of XGBoost. The sustainability null is robust at both ends of the bias-variance tradeoff. The simpler model finds no stable linear structure, and the more flexible model finds relations that reverse out of sample, which is what fitted noise looks like. There is no functional form, simple or flexible, under which sustainability information predicts returns.

6.3 Portfolio construction

The baseline sorts stocks into value-weighted deciles. Each element of this construction decides which stocks drive the result and how much of the prediction is used. Value weighting lets large firms dominate the portfolio. Equal weighting gives a small firm the same weight as a large one, so it tilts the portfolio toward small and typically less liquid stocks, where return predictability is usually strongest and trading is most expensive. If sustainability information were priced only in such stocks, equal weighting would reveal it. Prediction weighting goes the other way. It weights the stocks within the deciles by the size of their forecast, so it uses the magnitude of the prediction and not only its rank, and it rewards a model whose extreme forecasts are informative. Coarser sorts, terciles and quintiles, diversify more and so lower volatility, but they dilute the signal by mixing extreme and moderate forecasts. Section 5.4 showed that PST's tercile sorts are what make the greenness score look informative, so the same test is due for the sustainability models. Each alternative is compared with the characteristics model under the same construction, and the equal- and prediction-weighted models are tuned on the corresponding objective (Internet Appendix Section IA.3).

Equal weighting weakens the characteristics model somewhat (Sharpe ratio 0.57, $t = 2.44$) and leaves the sustainability models insignificant (0.30 and 0.26, $t = 1.44$ and 1.16). Prediction weighting strengthens the characteristics model (0.84, $t = 3.63$) and leaves the sustainability models at 0.12 and 0.14. With tercile sorts the characteristics model earns 0.60 ($t = 2.81$) and the sustainability models about 0.2 and 0.0, and with quintile sorts 0.67 ($t = 2.90$) against 0.03 and 0.07.

The pattern is informative in both directions. The characteristics model behaves like a graded signal. It gains from finer sorts and from weights that use the magnitude of its forecasts (0.60 with terciles, 0.67 with quintiles, 0.72 with deciles, and 0.84 with prediction weights), and it loses a little when small stocks receive more weight, which is expected in a universe that already excludes micro-caps. The sustainability models gain from none of these changes. Their spreads do not grow when the sorts become finer or when the extreme forecasts receive more weight, and the small-stock tilt of equal weighting produces the highest point estimates among the constructions, 0.30 and 0.26, but both are insignificant and well inside the confidence intervals of Table 1. The null is therefore not an artifact of weighting large firms, of discarding the magnitude of the forecasts, or of the sort granularity, and it holds in the coarse sorts that flatter weak signals in Section 5.4.

6.4 Imputation

With 91% of the raw cells missing, the results should not depend on how the gaps are filled. How much the treatment of missing data matters depends on the purpose of the analysis. When the question is whether a set of inputs carries predictive information, as in this paper, a flexible learner that is told which values are missing does not need the gaps filled in advance. XGBoost treats a missing value as information in its own right. At every split it learns the branch to which observations with a missing value are sent, so the prediction for a firm with a missing variable is formed from the variables that are observed for that firm and from the fact that the variable is missing. The learner thereby imputes internally. Where the observed variables carry information about the missing one, the fitted trees exploit the dependence among the covariates as part of the prediction, which is what an explicit imputation step does before estimation. This requires a method flexible enough to represent such interactions, which a tree ensemble is and a linear model with missing-value indicators is not. When instead the goal is to estimate parameters or to draw inference, for example a risk premium, a regression coefficient, or the contribution of an individual variable, the treatment of missing data matters a great deal. Restricting the sample to firms with complete data selects on data availability, ad hoc fills such as the median bias the estimates (Bryzgalova, Lerner, Lettau, and Pelger, 2025), and imputed values treated as observed invalidate standard errors unless the imputation error is accounted for (Duan and Pelger, 2025). For such questions an imputation model that exploits the time-series and cross-sectional dependence of the panel, combined with inference that corrects for imputation error, is the right approach. Our headline question belongs to the first category, which is what makes the simple treatment valid as a robustness check. The variable-importance

measures of Figure 6 lean toward the second, which is why we read them as descriptive.

We therefore re-estimate the headline models without any of the imputation steps of Section 2.3 and let XGBoost handle the missing values (Internet Appendix Section IA.4). The return model earns a Sharpe ratio of 0.07 ($t = 0.32$). The residual model earns -0.41 ($t = -2.11$), a significantly negative spread that does not survive the correction. A model that also learns from the pattern of missing values, which reflects which firms the providers cover and when, can fit relations that reverse out of sample.

In a prediction test, then, missing data are a question of power rather than of bias. An imputation that uses neither returns nor firm characteristics cannot manufacture out-of-sample predictability, but it makes the inputs noisier and so attenuates a real signal. Missingness therefore works against the sustainability variables and in favor of the null, and the placebo of Section 4.4, whose planted signal is fully observed, does not account for this loss of power. Two further facts limit the role of imputation. The baseline imputation does not use firm characteristics, so it cannot import their predictive content into the sustainability variables, and the complete-case analysis of Section 5, in which the inputs of the greenness score are observed for every firm, involves no imputation at all, at the price of a universe selected on data availability, which is why Section 5 reports both universes. Two extensions are natural. The first is a placebo signal that is masked with the missingness pattern of a real variable and passed through the imputation, which measures the power lost to the gaps. The second is explicit model-based imputation of the sustainability panel before estimation (Bryzgalova, Lerner, Lettau, and Pelger, 2025; Duan and Pelger, 2025), which would also allow a regression analysis of the sustainability variables with valid inference.

7 Conclusion

We set out to measure how much sustainability is in stock prices and find that the answer is too little to detect, and far less than headline results in the literature suggest. A machine learning protocol powerful enough to recover the well-documented predictability in conventional firm characteristics, to detect planted signals about a third as strong in 92% of simulated samples, and to exclude sustainability Sharpe ratios above half of what the characteristics deliver, finds no predictability, unconditional or incremental, in a comprehensive panel of sustainability characteristics from the four major data providers. Replicating the most prominent evidence of green outperformance, we show that its reading as a return predictor requires a specific constellation of research-design choices and does not survive standard alternatives or standard multiple-testing discipline.

Three interpretations are worth distinguishing. First, our results are consistent with

equilibrium models in which green preferences lower the cost of capital of green firms (Pástor, Stambaugh, and Taylor, 2021), and with the interpretation that PST themselves give of their evidence, that realized green outperformance reflects unanticipated shifts in tastes or climate concerns, which are by nature unpredictable, and that expected green returns are if anything slightly negative, an effect too small to detect in two decades of data. What our results reject is the stronger claim, implicit in much commercial and some academic work, that sustainability data contain exploitable forecasting information. Second, our results sharpen the measurement critique of Berg, Kölbel, and Rigobon (2022) and Aswani, Raghunandan, and Rajgopal (2024), since whatever firm-level information sustainability scores contain, its return-relevant component is already spanned by conventional characteristics. Third, for the replication agenda in finance (Harvey, Liu, and Zhu, 2016; Hou, Xue, and Zhang, 2020; Jensen, Kelly, and Pedersen, 2023), climate finance offers a live example of how a young literature with strong priors, short samples, and many discretionary choices can converge on fragile stylized facts. The discipline we apply here, a prediction protocol with verified power, design-space perturbation, bootstrap calibration of subsample claims, and multiple-testing corrections over the design space, travels beyond ESG.

A Variable Definitions

Table A.1: Conventional firm characteristics. The 45 stock characteristics, following Bryzgalova, Lerner, Lettau, and Pelger (2025).

#	Acronym	Name	#	Acronym	Name
1	A2ME	Assets to market cap	24	OA	Operating accruals
2	AC	Accrual	25	OL	Operating leverage
3	AT	Total assets	26	OP	Operating profitability
4	ATO	Net sales over lagged net operating assets	27	PCM	Price to cost margin
5	B2M	Book to market ratio	28	PM	Profit margin
6	Beta_d	CAPM beta (daily)	29	PROF	Profitability
7	Beta_m	Market beta (monthly)	30	Q	Tobin's Q
8	C2A	Cash to total assets	31	R_60_13	Long-term reversal (60–13m lookback)
9	CF2B	Free cash flow to book value	32	R12_2	Momentum
10	CF2P	Cash flow to price	33	R12_7	Intermediate momentum
11	CTO	Capital turnover	34	R2_1	Short-term reversal
12	D2A	Capital intensity	35	R36_13	Long-term reversal (36–13m lookback)
13	D2P	Dividend yield	36	RNA	Return on net operating assets
14	DPI2A	Change in PP&E	37	ROA	Return on assets
15	E2P	Earnings to price	38	ROE	Return on equity
16	FC2Y	Fixed costs to sales	39	RVAR	Residual variance
17	HIGH52	Closeness to 52-week high	40	S2P	Sales to price
18	IdioVol	Idiosyncratic volatility	41	SGA2S	SG&A to sales
19	INV	Investment	42	SPREAD	Bid–ask spread
20	LEV	Leverage	43	SUV	Standard unexplained volume
21	ME	Size	44	TURN	Turnover
22	NI	Net share issues	45	VAR	Variance
23	NOA	Net operating assets			

Table A.2: Sustainability characteristics. The 22 sustainability variables, by source.

#	Acronym	Name	Source
1	ABSDIRECT	Absolute GHG: direct	Trucost
2	ABSFTI	Absolute GHG: first-tier indirect	Trucost
3	ABSS1	Absolute GHG: Scope 1	Trucost
4	ABSS2LB	Absolute GHG: Scope 2 (location-based)	Trucost
5	ABSS3U	Absolute GHG: Scope 3 upstream	Trucost
6	ABSS3D	Absolute GHG: Scope 3 downstream	Trucost
7	ESGC	ESG combined score	Refinitiv
8	ESGCONTR	ESG controversies score	Refinitiv
9	ESG	ESG score	Refinitiv
10	ENVINNSCORE	Environmental innovation score	Refinitiv
11	ENVPILLAR	Environmental pillar score	Refinitiv
12	RESOURCE	Resource use score	Refinitiv
13	CO2EQTOTAL	Estimated CO_2-equivalent emissions, total	Refinitiv
14	ENVDIM	Environmental dimension	S&P Global
15	SCOPE1	Direct GHG emissions (Scope 1)	S&P Global
16	SCOPE2	Indirect GHG emissions (Scope 2)	S&P Global
17	SCOPE3	Indirect GHG emissions (Scope 3)	S&P Global
18	SPESG	S&P Global ESG score	S&P Global
19	MSCIE	Environmental pillar score	MSCI
20	MSCIEW	Environmental pillar weight	MSCI
21	MSCIWAS	Weighted average score	MSCI
22	MSCIESG	Industry-adjusted score	MSCI

B Implementation Details

Inference definitions. For a family of m null hypotheses with ordered p-values $p_{(1)} \leq \cdots \leq p_{(m)}$, the Holm–Bonferroni adjusted p-value is $\tilde{p}_{(i)} = \max_{j \leq i} \min\{(m-j+1)p_{(j)}, 1\}$ and controls the familywise error rate, and the Benjamini–Hochberg procedure controls the false discovery rate at level q by rejecting all hypotheses up to the largest i with $p_{(i)} \leq iq/m$. The Diebold–Mariano statistic tests equal squared-error loss of two forecasts. We compute the loss differential for each stock-month, average it across stocks within each month, and test the mean of the resulting monthly series with Newey–West standard errors. The Ledoit–Wolf test compares two Sharpe ratios using a heteroskedasticity- and autocorrelation-consistent estimator of the variance of the studentized difference.

Table A.3: Hyperparameter grid. Grid values for the XGBoost cross-validation. The selection criterion is the validation-sample Sharpe ratio of the value-weighted long–short decile portfolio.

Parameter	Grid values	Description
colsample_bytree	{0.6, 0.8}	Fraction of features sampled per tree
max_bin	{10, 50}	Histogram bins for continuous features
learning_rate	0.01	Step-size shrinkage
subsample	{0.3, 0.6}	Fraction of observations sampled per tree
alpha (ω)	{1, 10}	L1 penalty on the leaf values
lambda (κ)	{1, 10}	L2 penalty on the leaf values
max_depth	{2, 3}	Maximum tree depth
min_child_weight	{30, 50}	Minimum child weight
n_estimators	$\leq$200 for sustainability models $\leq$1000 for characteristic model	Maximum number of boosting rounds. The number used is chosen by early stopping on the validation sample (patience 20) and held fixed in the final fit

Table A.4: Selected hyperparameters. Cross-validated configurations of the three headline XGBoost models. Each model is an ensemble of 20 members differing only in the random seed. The number of boosting rounds is chosen by early stopping on the validation sample with a patience of 20 rounds and a cap of 200 (for sustainability models) or 1000 (for the characteristics model), and held fixed when the model is re-estimated on the combined training and validation data.

Parameter	Characteristics	Sustainability (returns)	Sustainability (residuals)
max_bin	10	10	10
max_depth	3	2	3
learning_rate	0.01	0.01	0.01
n_estimators (cap)	1000	200	200
subsample	0.30	0.30	0.60
colsample_bytree	0.60	0.80	0.60
min_child_weight	30	30	50
alpha (ω)	10	1	1
lambda (κ)	10	1	1
early stopping patience (validation)	20	20	20
boosting rounds selected	949	1	1
ensemble members	20	20	20

C Additional Incremental-Analysis Results

Tables A.5 and A.6 report the incremental information analysis of Section 5.6 for the samples starting in 2007 and 2002.

Table A.5: Incremental information analysis for the sample starting 2007. See Table 4 for the design. HB and BH are pooled over all 132 specifications.

	Returns						Residuals					
k	μ (%)	SR	t-stat	p	HB	BH	μ (%)	SR	t-stat	p	HB	BH
1	−0.6	−0.08	−0.30	0.62	1.00	1.00	−1.2	−0.16	−0.61	0.73	1.00	1.00
2	1.8	0.24	1.07	0.14	1.00	1.00	−2.3	−0.30	−1.30	0.90	1.00	1.00
3	−0.9	−0.10	−0.43	0.67	1.00	1.00	−6.4	−0.63	−2.38	0.99	1.00	1.00
4	1.6	0.13	0.49	0.31	1.00	1.00	−5.8	−0.61	−2.45	0.99	1.00	1.00
5	0.5	0.04	0.12	0.45	1.00	1.00	−2.5	−0.19	−0.82	0.79	1.00	1.00
6	−5.0	−0.38	−1.35	0.91	1.00	1.00	−3.3	−0.29	−1.14	0.87	1.00	1.00
7	−2.9	−0.21	−0.80	0.79	1.00	1.00	−2.9	−0.24	−0.90	0.82	1.00	1.00
8	−4.5	−0.34	−1.12	0.87	1.00	1.00	−5.9	−0.51	−1.96	0.98	1.00	1.00
9	−5.2	−0.37	−1.30	0.90	1.00	1.00	−3.4	−0.29	−1.11	0.87	1.00	1.00
10	−4.2	−0.34	−1.22	0.89	1.00	1.00	−5.0	−0.43	−1.66	0.95	1.00	1.00
11	−3.6	−0.27	−0.98	0.84	1.00	1.00	−4.7	−0.40	−1.55	0.94	1.00	1.00
12	−4.7	−0.32	−1.24	0.89	1.00	1.00	−3.9	−0.31	−1.26	0.90	1.00	1.00
13	−2.5	−0.21	−0.84	0.80	1.00	1.00	−1.2	−0.11	−0.39	0.65	1.00	1.00
14	−0.5	−0.04	−0.16	0.56	1.00	1.00	−4.3	−0.36	−1.47	0.93	1.00	1.00
15	−2.8	−0.23	−0.75	0.77	1.00	1.00	0.0	0.00	0.01	0.50	1.00	1.00
16	−0.1	−0.01	−0.03	0.51	1.00	1.00	3.6	0.31	1.18	0.12	1.00	1.00
17	−0.9	−0.09	−0.35	0.64	1.00	1.00	0.9	0.08	0.32	0.38	1.00	1.00
18	−2.4	−0.24	−0.84	0.80	1.00	1.00	−0.9	−0.08	−0.29	0.61	1.00	1.00
19	0.9	0.09	0.32	0.37	1.00	1.00	−2.8	−0.17	−0.64	0.74	1.00	1.00
20	−2.0	−0.16	−0.57	0.72	1.00	1.00	−3.2	−0.18	−0.70	0.76	1.00	1.00
21	0.8	0.08	0.28	0.39	1.00	1.00	−2.5	−0.23	−0.79	0.79	1.00	1.00
22	0.8	0.09	0.34	0.37	1.00	1.00	−2.1	−0.20	−0.73	0.77	1.00	1.00

Table A.6: Incremental information analysis for the sample starting 2002. See Table 4 for the design. HB and BH are pooled over all 132 specifications. The $k = 22$ row is the headline model of Table 1.

	Returns						Residuals					
k	μ (%)	SR	t-stat	p	HB	BH	μ (%)	SR	t-stat	p	HB	BH
1	0.4	0.04	0.19	0.43	1.00	1.00	−1.3	−0.14	−0.71	0.76	1.00	1.00
2	−1.0	−0.11	−0.54	0.71	1.00	1.00	−0.1	−0.01	−0.04	0.52	1.00	1.00
3	1.5	0.17	0.80	0.21	1.00	1.00	0.1	0.01	0.09	0.47	1.00	1.00
4	0.7	0.07	0.29	0.39	1.00	1.00	0.5	0.06	0.23	0.41	1.00	1.00
5	2.4	0.21	0.96	0.17	1.00	1.00	−0.5	−0.05	−0.25	0.60	1.00	1.00
6	2.2	0.23	1.01	0.16	1.00	1.00	0.1	0.01	0.05	0.48	1.00	1.00
7	2.2	0.17	0.80	0.21	1.00	1.00	2.8	0.28	1.23	0.11	1.00	1.00
8	2.0	0.18	0.89	0.19	1.00	1.00	1.0	0.11	0.50	0.31	1.00	1.00
9	0.8	0.08	0.33	0.37	1.00	1.00	6.5	0.58	2.52	0.06	0.78	0.78
10	2.3	0.22	0.95	0.17	1.00	1.00	1.4	0.17	0.81	0.21	1.00	1.00
11	−0.6	−0.07	−0.32	0.62	1.00	1.00	−0.4	−0.05	−0.22	0.59	1.00	1.00
12	2.1	0.21	1.02	0.15	1.00	1.00	−2.8	−0.32	−1.46	0.93	1.00	1.00
13	2.6	0.27	1.23	0.11	1.00	1.00	−1.2	−0.13	−0.57	0.72	1.00	1.00
14	−0.1	−0.02	−0.07	0.53	1.00	1.00	1.2	0.13	0.60	0.28	1.00	1.00
15	0.2	0.02	0.11	0.46	1.00	1.00	4.4	0.47	2.16	0.02	1.00	1.00
16	2.6	0.33	1.50	0.07	1.00	1.00	3.7	0.40	1.76	0.04	1.00	1.00
17	1.3	0.17	0.72	0.24	1.00	1.00	1.4	0.16	0.60	0.28	1.00	1.00
18	0.2	0.03	0.13	0.45	1.00	1.00	−1.7	−0.16	−0.73	0.77	1.00	1.00
19	2.6	0.27	1.39	0.08	1.00	1.00	1.3	0.12	0.55	0.29	1.00	1.00
20	3.3	0.35	1.61	0.05	1.00	1.00	2.4	0.24	0.99	0.16	1.00	1.00
21	2.9	0.32	1.49	0.07	1.00	1.00	2.0	0.19	0.87	0.19	1.00	1.00
22	−0.6	−0.06	−0.26	0.60	1.00	1.00	−0.8	−0.08	−0.33	0.63	1.00	1.00

D Data Appendix

D.1 Data provenance

The empirical dataset combines several proprietary and public data feeds.

Stock characteristics. Stock characteristics are constructed from CRSP and Compustat data downloaded through WRDS and imputed following Bryzgalova, Lerner, Lettau, and Pelger (2025). Relative to the original dataset of that paper, the only difference is in the ranking of the characteristics, where ties now receive the same rank, whereas previously they were split randomly. The correlation between the updated and the original characteristics exceeds 0.97 for all but one characteristic, and the median correlation across characteristics is 0.99. The primary identifier is `permno`.

Sustainability variables. The sustainability dataset merges data from four providers.

- **S&P Global:** 5 variables, 2,956 companies, September 2013 to 2024. The primary identifier is the `institutionid`. Retrieved from WRDS.

- **Trucost:** 6 variables, 4,133 companies, 2002 to 2024. The primary identifier is the `institutionid`. Retrieved from WRDS.
- **Refinitiv:** 7 variables, 1,035 companies, 2002 to 2024. The primary identifier is the `ticker`. Retrieved from LSEG.
- **MSCI:** 4 variables, 3,969 companies, 2007 to 2024. The primary identifier is the `ticker`. Retrieved from the Stanford Libraries through the Redivis platform.

D.2 Merge rules and match rates

The components are integrated sequentially under strict hierarchical matching rules, which minimize data loss while ensuring that each sustainability record is attached to the correct security.

Stock characteristics.

- **Ticker mapping:** We use the CRSP mapping table to link `permno` to `ticker` symbols.
- **Cross-sectional deduplication:** Duplicate records for a given (ticker, date) pair are resolved by keeping the record with the highest market capitalization and, if duplicates remain, the most recent one.

Trucost data.

- **Download:** We download companies domiciled in the United States.
- **Identifier matching:** We map `ticker`s to Trucost identifiers (`institutionid`).
- **Hierarchical duplicate resolution:** Cases in which one ticker maps to several `institutionid`s are resolved in stages.

    1. We use the CRSP mapping table to match `gvkey` and `ticker` to `permno`, and keep the observations for which `gvkey` and `permno` match the stock characteristics. If no match is found, all observations are kept.
    2. Remaining duplicates are resolved by the similarity of the company name in the CRSP mapping table and in the Trucost database, and ties keep all observations.
    3. Remaining duplicates are resolved by keeping the `ticker`–`institutionid` pair with the most data (fewest "disclosure rejected" entries).
    4. Remaining duplicates are resolved by matching `gvkey`. This step comes last because `gvkey` is sparse in the mapping table and would otherwise restrict the sample.

5. Four tickers remain associated with multiple `institutionid`s (CRK, MDRX, MTCH, SRT), and we verify manually that they refer to the same company.

S&P Global data.

- **Download:** We download the companies for which Trucost data are available, based on the `institutionid`.
- **Duplicate resolution:** We add the `ticker` from the same mapping table and keep the `ticker`–`institutionid` pairs that are included in the Trucost dataset.
- **Raw versus modeled scores:** When both a raw and a modeled score are available for the same ticker, we average them.

MSCI data.

- **Initial filtering:** We keep tickers for which stock characteristics are available and companies domiciled in the United States.
- **CUSIP matching:** For tickers with a CUSIP, we keep the `ticker`–`cusip` pairs that are present in the mapping table and, among remaining duplicates, the pair with the most non-missing data.
- For tickers without a CUSIP, we keep only those for which other sustainability variables are available.
- We keep companies whose name has a similarity score of at least 0.8 with the name in the Trucost database.
- One ticker remains associated with multiple companies and is dropped.

Refinitiv data. Refinitiv data require no pre-processing, because the data are clean and the primary identifier is the `ticker`.

D.3 Industry classification

- **Download:** NAICS codes are obtained from WRDS (CRSP/Compustat Merged (CCM) Link History Table).
- **Resolution:** We keep the tickers in our sample and the first two digits of the NAICS code. Duplicate entries for a ticker are resolved by checking, in order, whether a `ticker`–`permno` combination exists in the stock characteristics, whether a

`ticker`–`gvkey` combination exists, and whether the `ticker`–`gvkey` pair matches the Trucost database. Three tickers with duplicate NAICS codes remain (IRDM, POWW, VHI) and are resolved manually.

D.4 Final datasets

- **Scaling:** S&P Global variables and Refinitiv scores lie between 0 and 100 and MSCI scores between 0 and 10. Estimated CO_2-equivalent emissions from Refinitiv and the Trucost emission variables are raw quantities and enter in logs, in line with the literature.
- **Imputation:** The baseline imputation proceeds as described in Section 2.3, that is, last observation carried forward, deletion of stock-months without any observed variable, industry (two-digit NAICS) median, cross-sectional median, and zero for variables that no firm reports in a given month. Table A.8 reports the share of missing cells after each step.
- **Robustness:** Section 6 compares the baseline model with a specification that does not impute missing values.

D.5 Match-rate summary

Table A.7 details the sequential attrition at each phase of the database integration.

D.6 Imputation steps

Table A.8 reports the number of stock-month-variable cells and the share of missing cells after each step of the baseline imputation, for the full sample (Panel A) and for the samples that start in November 2012 and end in December 2020 (Panel B, the sample period of Pástor, Stambaugh, and Taylor (2022)) or December 2024 (Panel C). Step 2 removes stock-months in which none of the 22 variables is observed after step 1. It changes the sample rather than filling cells, so the shares from step 2 onward refer to the cells of the remaining stock-months. The cells that remain missing after the cross-sectional median are those of variables that no firm reports in the given month (for example S&P Global scores before September 2013 or MSCI scores before 2007). They are set to zero and carry no cross-sectional information. Coverage is better after 2012 (86% of raw cells missing, against 91% in the full sample), but because more variables are reported by at least one firm, a larger share of the remaining gaps is filled with industry medians and a smaller share with zeros. The complete-case analysis

Table A.7: Sequential merge and match-rate summary.

Step	Dataset	Merge phase	Tickers	Rows
1	Stock characteristics	Initial stage	21,503	2,623,822
2	MSCI	Initial stage	3,969	330,571
3	S&P Global	Initial stage	2,956	23,303
4	Refinitiv	Initial stage	1,035	14,175
5	Trucost	Initial stage	4,133	38,274
6	Sustainability	Merge sustainability files	4,853	393,793
7	Sustainability	Interpolate missing months	4,853	1,339,428
8	Sustainability	Eliminate tickers not in stock characteristics	4,799	1,324,524
9	NAICS	Initial stage	23,040	29,729
10	NAICS	Keep only tickers in sustainability data	4,797	7,150
11	NAICS	Resolve duplicates	4,797	4,797
12	Sustainability	Merge with NAICS	4,799	1,324,524
13	Sustainability	Keep (date, ticker) in stock characteristics	4,716	714,984
14	Returns	Add excess return	21,502	2,623,882
15	Returns	Lag returns	21,478	2,600,062
16	Returns	Add market capitalization	21,475	2,599,933
17	Sustainability	Align sustainability and returns	4,711	710,074
18	Stock characteristics	Align characteristics and returns	21,475	2,599,933
19	Sustainability	Impute missing data	4,500	452,841
20	Stock characteristics	Final	21,475	2,599,933
21	Returns	Final	21,475	2,599,933
22	Sustainability	Final	4,500	452,841

of Section 5, in which the inputs of the greenness score are observed for every firm, involves no imputation.

Table A.8: Missing cells after each imputation step. Number of stock-month-variable cells (22 variables), share of missing cells after each step of the baseline imputation, and share of cells filled in each step (percentage points). Steps 0 and 1 are computed on all cells of the merged panel, and steps 2 to 5 on the cells of the stock-months that remain after step 2.

Step	Operation	Cells	Missing after step (%)	Filled in step (pp)
Panel A: full sample, 2002–2024				
0	Raw merged panel	15,621,628	90.65	
1	Last observation carried forward	15,621,628	66.70	23.95
2	Delete stock-months with no observed variable	9,962,502	47.79	(sample change)
3	Industry (NAICS) median	9,962,502	9.49	38.30
4	Cross-sectional median	9,962,502	9.41	0.08
5	Zero for variables not reported by any firm	9,962,502	0.00	9.41
Panel B: November 2012 to December 2020				
0	Raw merged panel	6,040,936	86.28	
1	Last observation carried forward	6,040,936	59.94	26.34
2	Delete stock-months with no observed variable	4,843,674	50.04	(sample change)
3	Industry (NAICS) median	4,843,674	4.04	46.00
4	Cross-sectional median	4,843,674	3.97	0.07
5	Zero for variables not reported by any firm	4,843,674	0.00	3.97
Panel C: November 2012 to December 2024				
0	Raw merged panel	9,349,142	85.63	
1	Last observation carried forward	9,349,142	52.67	32.96
2	Delete stock-months with no observed variable	7,910,474	44.06	(sample change)
3	Industry (NAICS) median	7,910,474	2.47	41.59
4	Cross-sectional median	7,910,474	2.43	0.04
5	Zero for variables not reported by any firm	7,910,474	0.00	2.43

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Internet Appendix

How Much Sustainability is Really in Stock Prices?

This Internet Appendix reports the long–short statistics and decile plots for the robustness specifications summarized in Table 6 of the paper. In every table, the first row repeats the conventional-characteristics benchmark evaluated under the same specification, and the second and third rows report the sustainability models for abnormal returns (step 0) and for step-1 residuals (step 2). μ is the annualized mean excess return of the long–short portfolio in percent, SR the annualized Sharpe ratio with a 95% confidence interval, IC the average monthly rank correlation between predictions and realizations, and t-stat the Newey–West t-statistic of the mean return (four lags). Unless stated otherwise, portfolios are value-weighted deciles. Section IA.5 reports the direct check that the 45 characteristics do not predict the step-1 residuals, Section IA.6 the Diebold–Mariano tests of Section 5.6, Section IA.7 the three learners and their tuning, and Section IA.8 the performance measures, the multiple-testing adjustments, and the tests.

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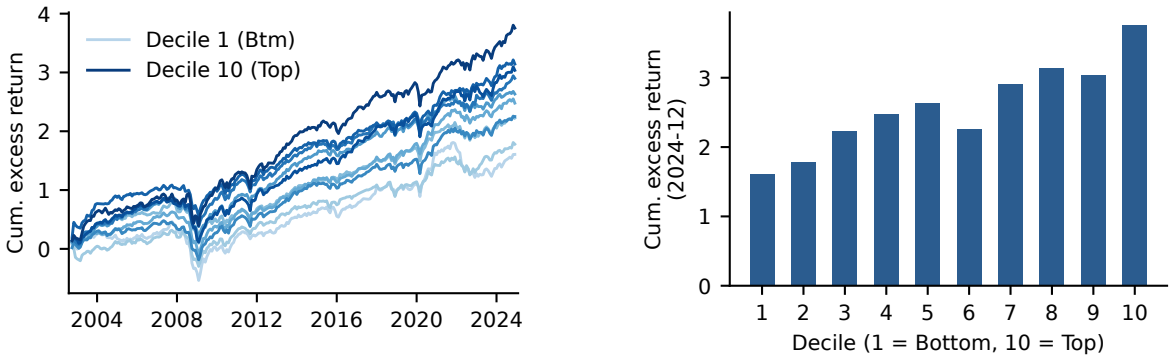
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IA.1 Alternative learners

Tables IA.1 and IA.2 and Figures IA.1 to IA.6 report the results when Elastic Net or a feedforward neural network replaces XGBoost in both steps, so that the residuals of step 2 are those of the alternative learner's own step-1 model.

Table IA.1: Elastic Net. Long–short decile portfolio performance when Elastic Net replaces XGBoost. Sustainability variables enter as rank quantiles.

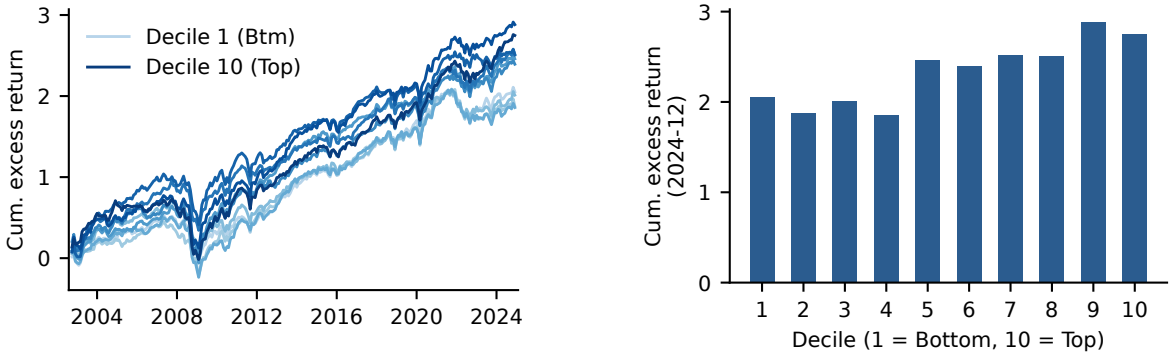
Model	μ (%)	SR	SR 95% CI	IC	t-stat
Characteristics → abnormal returns	9.7	0.64	[0.21, 1.07]	0.034	2.91
Sustainability → abnormal returns	3.1	0.32	[−0.11, 0.75]	0.017	1.45
Sustainability → residuals	−1.2	−0.10	[−0.54, 0.33]	0.001	−0.47



(a) Cumulative returns by prediction decile

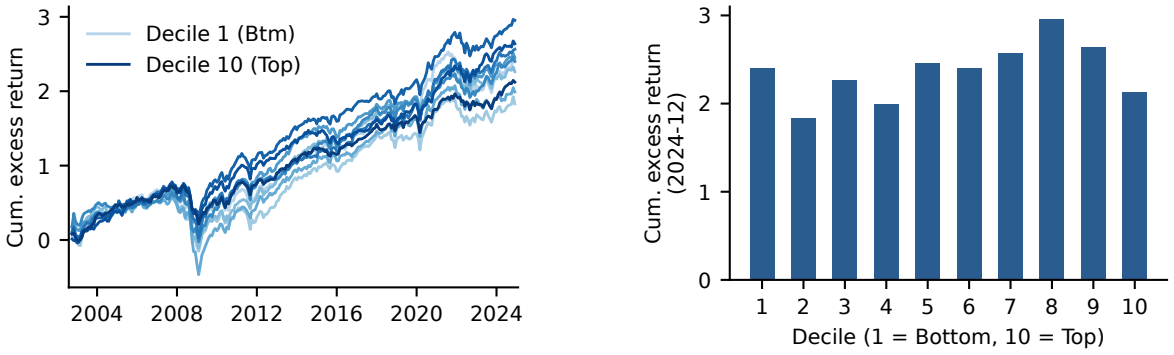
(b) Terminal cumulative return by decile

Figure IA.1: Conventional characteristics predict abnormal returns (step 1). Cumulative excess returns of value-weighted decile portfolios formed on out-of-sample Elastic Net predictions of abnormal returns from 45 conventional characteristics.



(a) Cumulative returns by prediction decile **(b)** Terminal cumulative return by decile

Figure IA.2: Rank quantiles of sustainability characteristics do not predict abnormal returns (step 0). Cumulative excess returns of value-weighted decile portfolios formed on out-of-sample Elastic Net predictions of abnormal returns from 22 sustainability characteristics transformed in rank quantiles, cross-out-of-sample evaluation on the 2002–2024 sustainability panel. Portfolios begin once sufficient cross-sectional coverage is available.

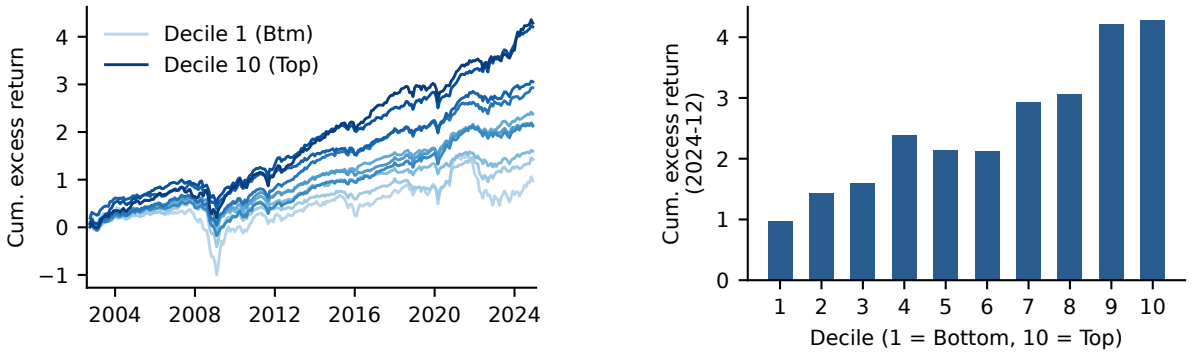


(a) Cumulative returns by prediction decile **(b)** Terminal cumulative return by decile

Figure IA.3: Rank quantiles of sustainability characteristics do not predict characteristic-orthogonal residuals (step 2). Cumulative excess returns of value-weighted decile portfolios formed on out-of-sample Elastic Net predictions of step-1 residuals from 22 sustainability characteristics transformed in rank quantiles, cross-out-of-sample evaluation on the period for which step-1 residuals are available.

Table IA.2: Feedforward neural network. Long–short decile portfolio performance when a feedforward neural network replaces XGBoost. Sustainability variables enter as rank quantiles.

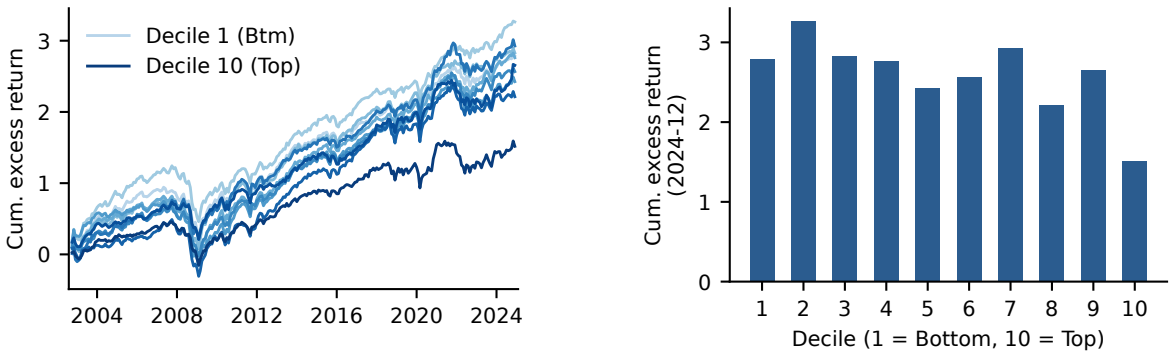
Model	μ (%)	SR	SR 95% CI	IC	t-stat
Characteristics → abnormal returns	14.9	0.76	[0.29, 1.22]	0.046	3.23
Sustainability → abnormal returns	−5.7	−0.54	[−0.95, −0.13]	−0.016	−2.61
Sustainability → residuals	4.0	0.36	[−0.05, 0.76]	0.008	1.74



(a) Cumulative returns by prediction decile

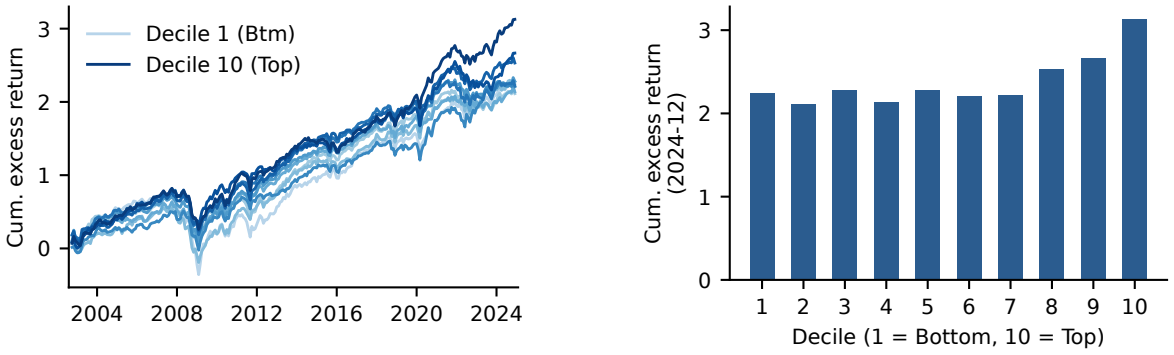
(b) Terminal cumulative return by decile

Figure IA.4: Conventional characteristics predict abnormal returns (step 1). Cumulative excess returns of value-weighted decile portfolios formed on out-of-sample feedforward neural network predictions of abnormal returns from 45 conventional characteristics.



(a) Cumulative returns by prediction decile **(b)** Terminal cumulative return by decile

Figure IA.5: Rank quantiles of sustainability characteristics and abnormal returns (step 0), feedforward network. Cumulative excess returns of value-weighted decile portfolios formed on out-of-sample feedforward neural network predictions of abnormal returns from 22 sustainability characteristics transformed in rank quantiles, cross-out-of-sample evaluation on the 2002–2024 sustainability panel. Portfolios begin once sufficient cross-sectional coverage is available.



(a) Cumulative returns by prediction decile **(b)** Terminal cumulative return by decile

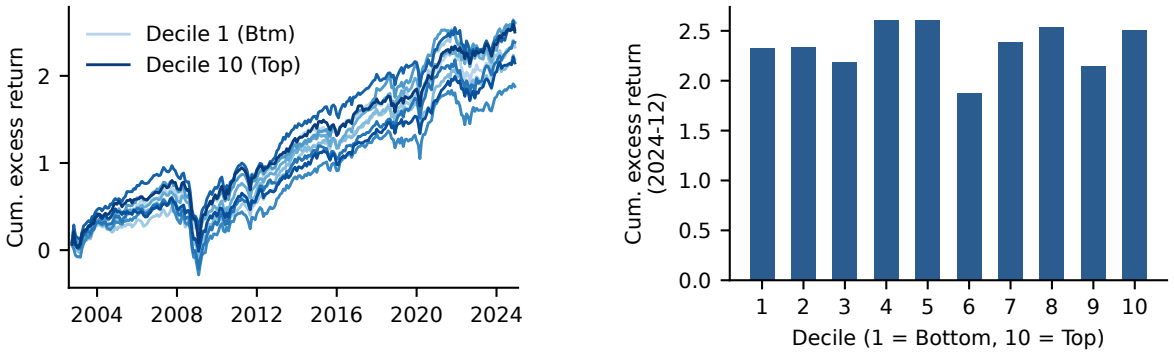
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IA.2 Provider subsets, emissions, and ESG scores

This section reports the results for the XGBoost models estimated on each provider's variables separately, on emission variables only, and on ESG scores only.

Table IA.3: S&P Global variables only. Long–short decile portfolio performance of XGBoost models that use the five S&P Global variables as the only sustainability inputs.

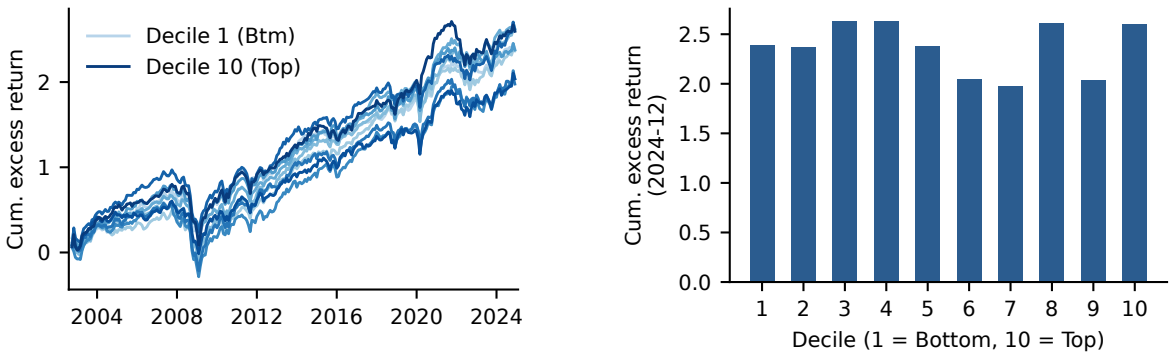
Model	μ (%)	SR	SR 95% CI	IC	t-stat
Characteristics → abnormal returns	13.9	0.72	[0.21, 1.22]	0.037	2.80
Sustainability → abnormal returns	0.8	0.09	[−0.28, 0.46]	0.005	0.48
Sustainability → residuals	0.9	0.10	[−0.33, 0.54]	−0.005	0.46



(a) Cumulative returns by prediction decile

(b) Terminal cumulative return by decile

Figure IA.7: S&P Global sustainability characteristics do not predict abnormal returns (step 0). Cumulative excess returns of value-weighted decile portfolios formed on out-of-sample XGBoost predictions of abnormal returns from 5 sustainability characteristics from S&P Global, cross-out-of-sample evaluation on the 2002–2024 sustainability panel. Portfolios begin once sufficient cross-sectional coverage is available.



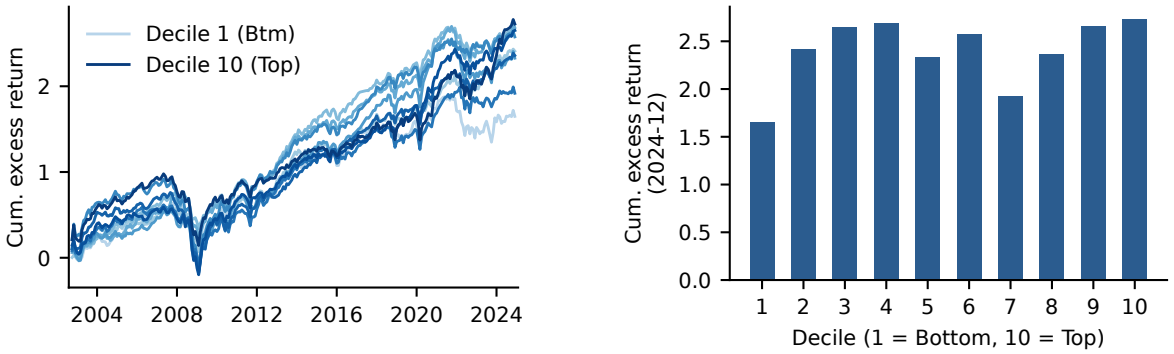
(a) Cumulative returns by prediction decile

(b) Terminal cumulative return by decile

Figure IA.8: S&P Global sustainability characteristics do not predict characteristic-orthogonal residuals (step 2). Cumulative excess returns of value-weighted decile portfolios formed on out-of-sample XGBoost predictions of step-1 residuals from 5 sustainability characteristics from S&P Global, cross-out-of-sample evaluation on the period for which step-1 residuals are available.

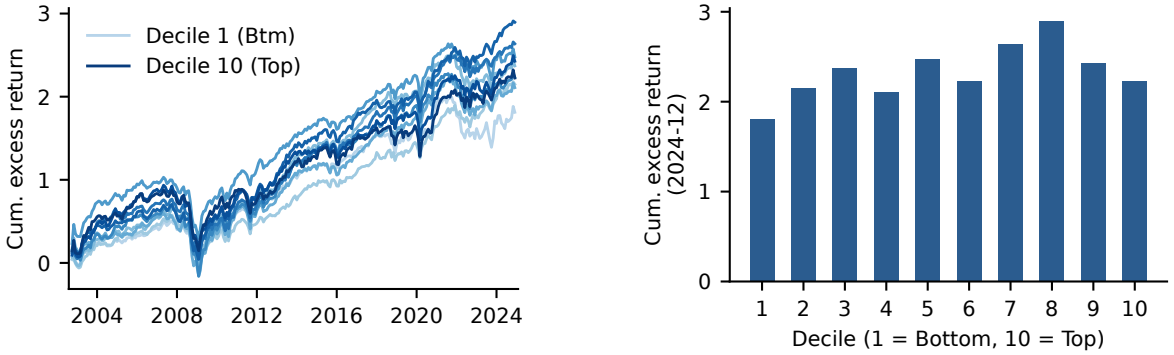
Table IA.4: Trucost variables only. Long–short decile portfolio performance of XGBoost models that use the six Trucost emission variables as the only sustainability inputs.

Model	μ (%)	SR	SR 95% CI	IC	t-stat
Characteristics $\rightarrow$ abnormal returns	13.9	0.72	[0.21, 1.22]	0.037	2.80
Sustainability $\rightarrow$ abnormal returns	4.8	0.41	[−0.05, 0.86]	0.008	1.76
Sustainability $\rightarrow$ residuals	1.9	0.15	[−0.27, 0.57]	0.011	0.70



(a) Cumulative returns by prediction decile **(b)** Terminal cumulative return by decile

Figure IA.9: Trucost sustainability characteristics do not predict abnormal returns (step 0). Cumulative excess returns of value-weighted decile portfolios formed on out-of-sample XGBoost predictions of abnormal returns from 6 sustainability characteristics from Trucost, cross-out-of-sample evaluation on the 2002–2024 sustainability panel. Portfolios begin once sufficient cross-sectional coverage is available.

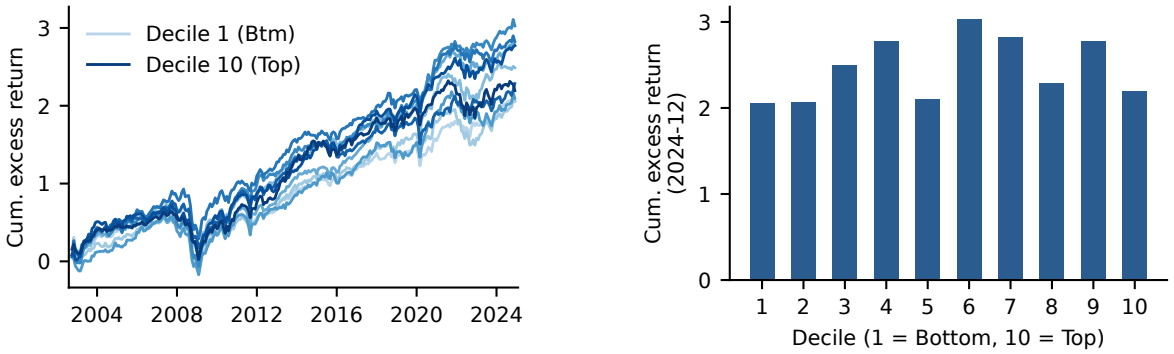


(a) Cumulative returns by prediction decile **(b)** Terminal cumulative return by decile

Figure IA.10: Trucost sustainability characteristics and characteristic-orthogonal residuals (step 2). Cumulative excess returns of value-weighted decile portfolios formed on out-of-sample XGBoost predictions of step-1 residuals from 6 sustainability characteristics from Trucost, cross-out-of-sample evaluation on the period for which step-1 residuals are available.

Table IA.5: Refinitiv variables only. Long–short decile portfolio performance of XGBoost models that use the seven Refinitiv variables as the only sustainability inputs.

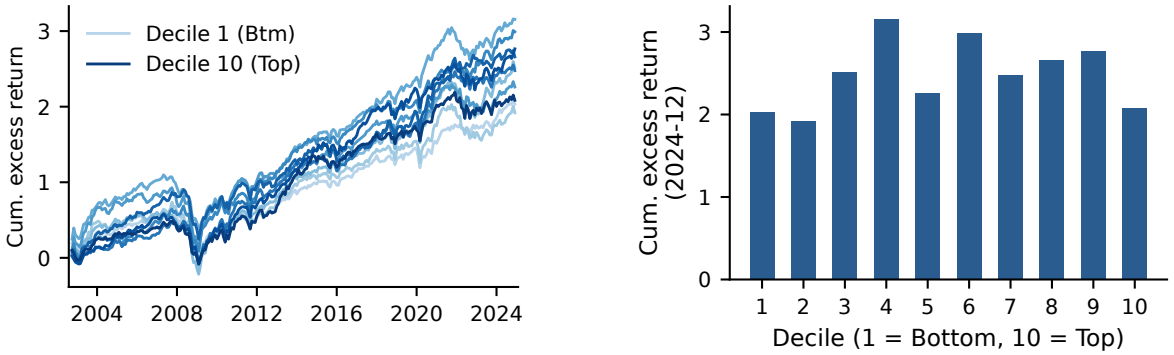
Model	μ (%)	SR	SR 95% CI	IC	t-stat
Characteristics → abnormal returns	13.9	0.72	[0.21, 1.22]	0.037	2.80
Sustainability → abnormal returns	0.6	0.06	[−0.32, 0.44]	−0.013	0.31
Sustainability → residuals	0.2	0.02	[−0.34, 0.39]	−0.009	0.12



(a) Cumulative returns by prediction decile

(b) Terminal cumulative return by decile

Figure IA.11: Refinitiv sustainability characteristics do not predict abnormal returns (step 0). Cumulative excess returns of value-weighted decile portfolios formed on out-of-sample XGBoost predictions of abnormal returns from 7 sustainability characteristics from Refinitiv, cross-out-of-sample evaluation on the 2002–2024 sustainability panel. Portfolios begin once sufficient cross-sectional coverage is available.



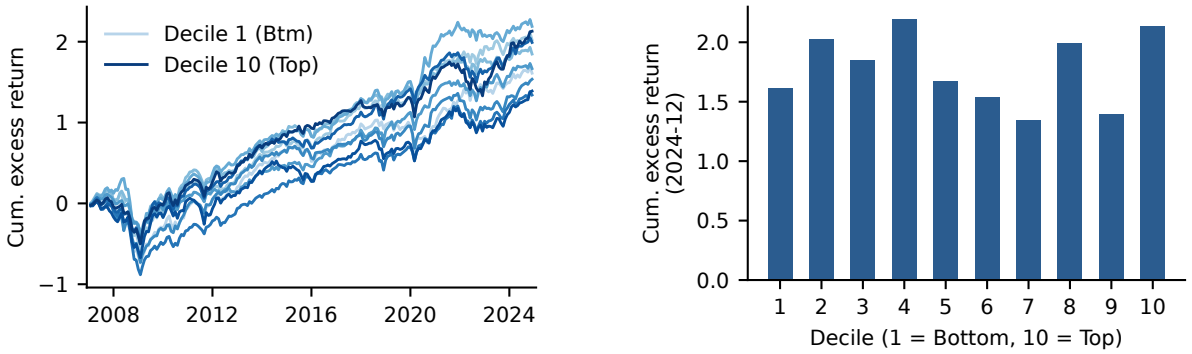
(a) Cumulative returns by prediction decile

(b) Terminal cumulative return by decile

Figure IA.12: Refinitiv sustainability characteristics do not predict characteristic-orthogonal residuals (step 2). Cumulative excess returns of value-weighted decile portfolios formed on out-of-sample XGBoost predictions of step-1 residuals from 7 sustainability characteristics from Refinitiv, cross-out-of-sample evaluation on the period for which step-1 residuals are available.

Table IA.6: MSCI variables only. Long–short decile portfolio performance of XGBoost models that use the four MSCI variables as the only sustainability inputs.

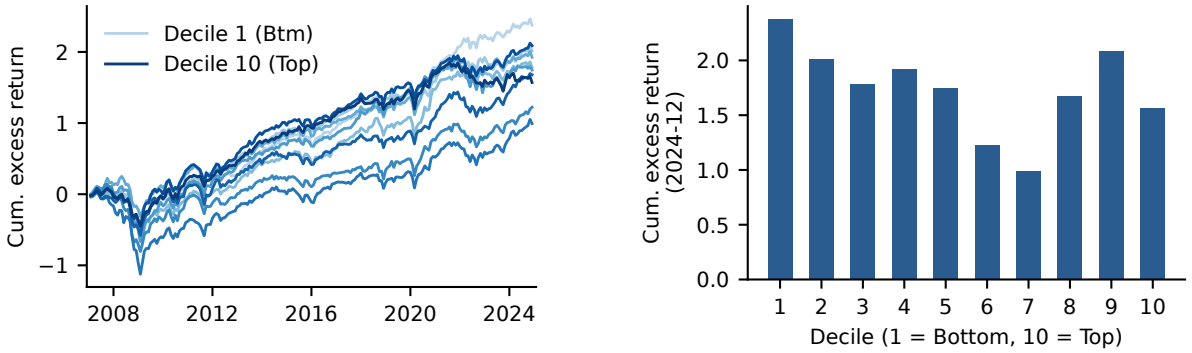
Model	μ (%)	SR	SR 95% CI	IC	t-stat
Characteristics → abnormal returns	13.9	0.72	[0.21, 1.22]	0.037	2.80
Sustainability → abnormal returns	2.9	0.23	[−0.32, 0.77]	−0.013	0.82
Sustainability → residuals	−4.5	−0.50	[−1.02, 0.02]	−0.014	−1.90



(a) Cumulative returns by prediction decile

(b) Terminal cumulative return by decile

Figure IA.13: MSCI sustainability characteristics do not predict abnormal returns (step 0). Cumulative excess returns of value-weighted decile portfolios formed on out-of-sample XGBoost predictions of abnormal returns from 4 sustainability characteristics from MSCI, cross-out-of-sample evaluation on the 2002–2024 sustainability panel. Portfolios begin once sufficient cross-sectional coverage is available.



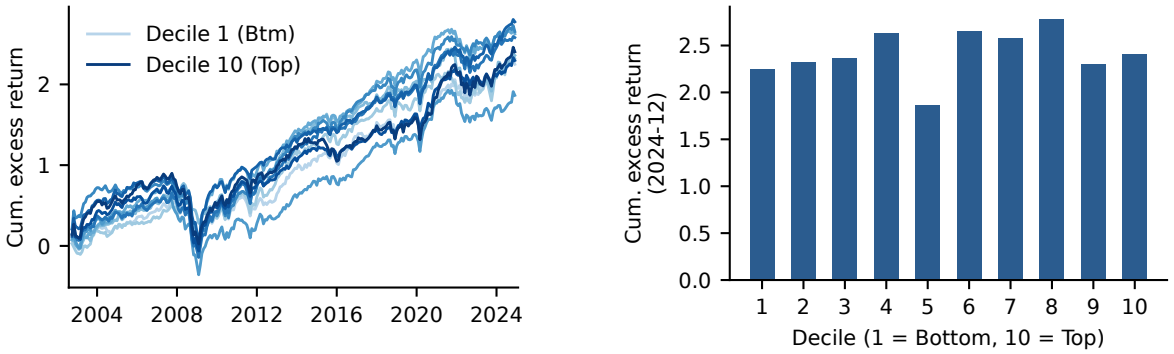
(a) Cumulative returns by prediction decile

(b) Terminal cumulative return by decile

Figure IA.14: MSCI sustainability characteristics do not predict characteristic-orthogonal residuals (step 2). Cumulative excess returns of value-weighted decile portfolios formed on out-of-sample XGBoost predictions of step-1 residuals from 4 sustainability characteristics from MSCI, cross-out-of-sample evaluation on the period for which step-1 residuals are available.

Table IA.7: Emission variables only. Long–short decile portfolio performance of XGBoost models that use four emission variables as the only sustainability inputs.

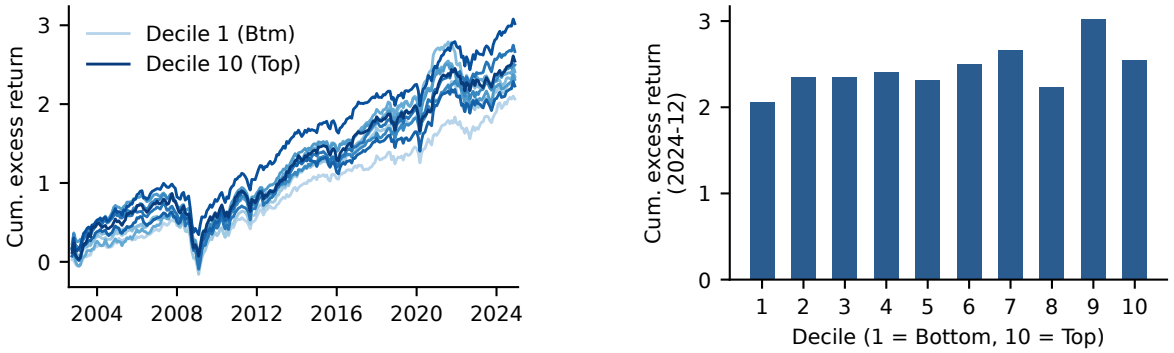
Model	μ (%)	SR	SR 95% CI	IC	t-stat
Characteristics → abnormal returns	13.9	0.72	[0.21, 1.22]	0.037	2.80
Sustainability → abnormal returns	0.7	0.06	[−0.36, 0.48]	0.007	0.30
Sustainability → residuals	2.2	0.19	[−0.23, 0.62]	0.010	0.88



(a) Cumulative returns by prediction decile

(b) Terminal cumulative return by decile

Figure IA.15: Emission variables and abnormal returns (step 0). Cumulative excess returns of value-weighted decile portfolios formed on out-of-sample XGBoost predictions of abnormal returns from 4 emission variables, cross-out-of-sample evaluation on the 2002–2024 sustainability panel. Portfolios begin once sufficient cross-sectional coverage is available.



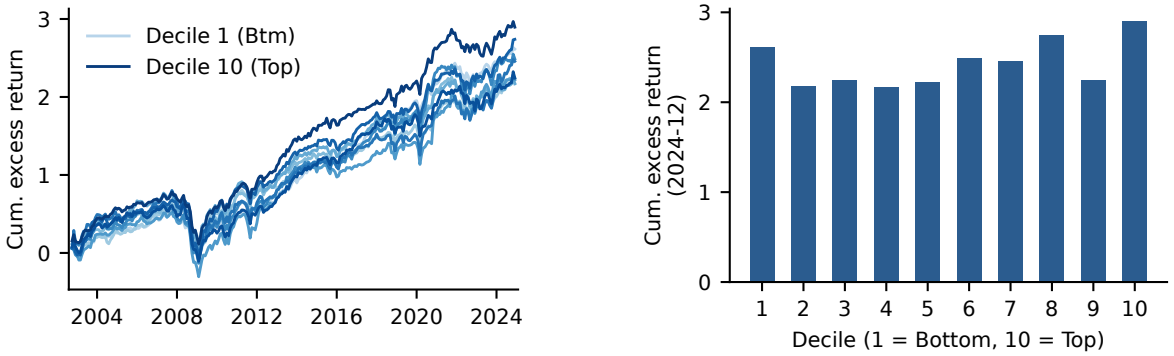
(a) Cumulative returns by prediction decile

(b) Terminal cumulative return by decile

Figure IA.16: Emissions do not predict characteristic-orthogonal residuals (step 2). Cumulative excess returns of value-weighted decile portfolios formed on out-of-sample XGBoost predictions of step-1 residuals from 4 emission variables, cross-out-of-sample evaluation on the period for which step-1 residuals are available.

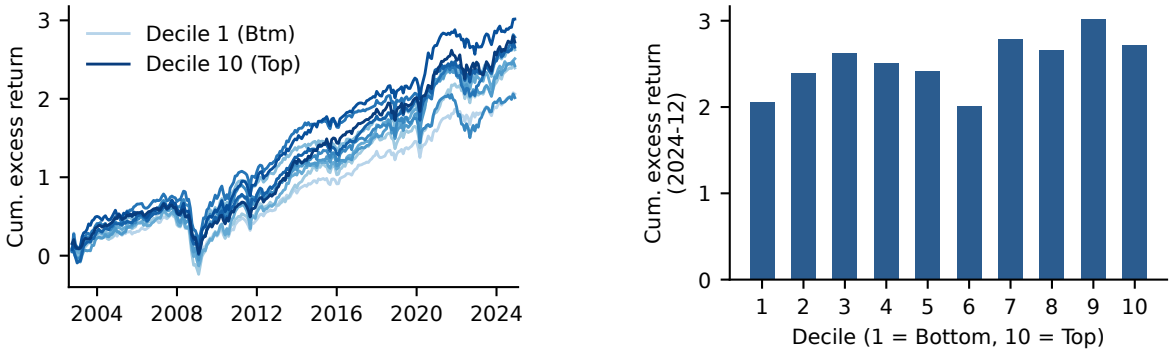
Table IA.8: ESG scores only. Long–short decile portfolio performance of XGBoost models that use the provider ESG scores as the only sustainability inputs.

Model	μ (%)	SR	SR 95% CI	IC	t-stat
Characteristics → abnormal returns	13.9	0.72	[0.21, 1.22]	0.037	2.80
Sustainability → abnormal returns	1.3	0.14	[−0.28, 0.56]	0.003	0.65
Sustainability → residuals	3.0	0.33	[−0.02, 0.68]	0.006	1.84



(a) Cumulative returns by prediction decile **(b)** Terminal cumulative return by decile

Figure IA.17: ESG scores do not predict abnormal returns (step 0). Cumulative excess returns of value-weighted decile portfolios formed on out-of-sample XGBoost predictions of abnormal returns from 3 ESG scores, cross-out-of-sample evaluation on the 2002–2024 sustainability panel. Portfolios begin once sufficient cross-sectional coverage is available.



(a) Cumulative returns by prediction decile **(b)** Terminal cumulative return by decile

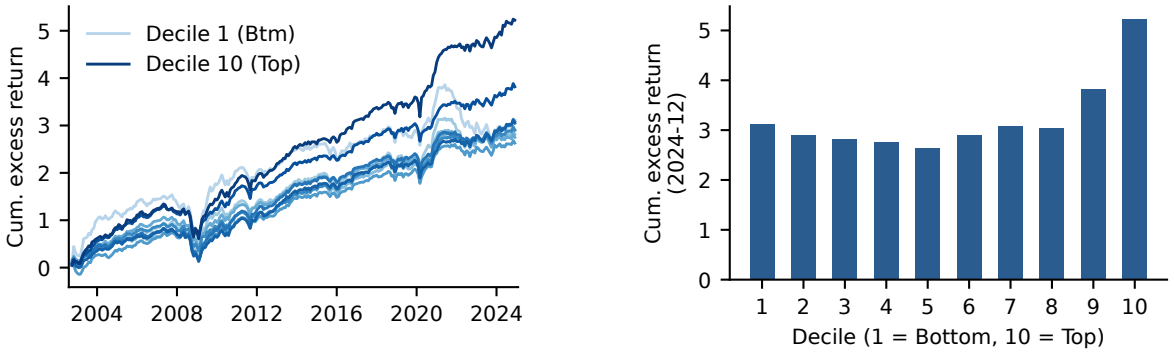
Figure IA.18: ESG scores do not predict characteristic-orthogonal residuals (step 2). Cumulative excess returns of value-weighted decile portfolios formed on out-of-sample XGBoost predictions of step-1 residuals from 3 ESG scores, cross-out-of-sample evaluation on the period for which step-1 residuals are available.

IA.3 Portfolio construction

This section reports the results for the headline models with equal-weighted, prediction-weighted, tercile, and quintile portfolios.

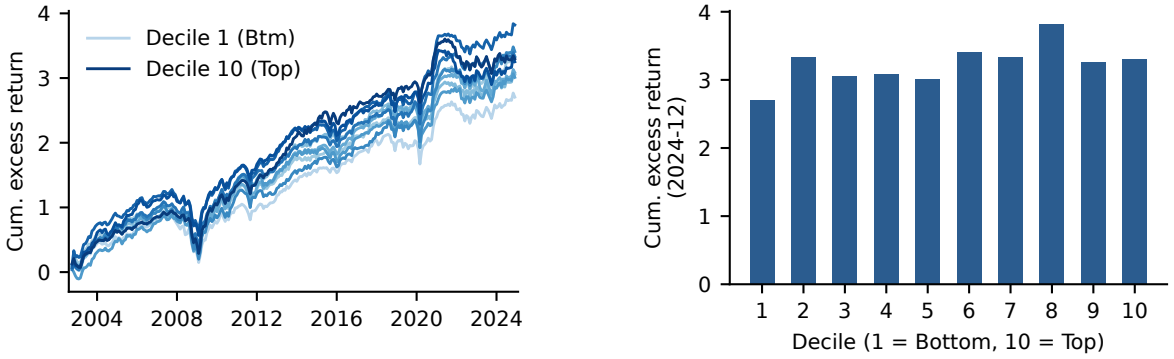
Table IA.9: Equal-weighted portfolios. Long–short decile portfolio performance of the headline models with equal-weighted deciles.

Model	μ (%)	SR	SR 95% CI	IC	t-stat
Characteristics → abnormal returns	9.5	0.57	[0.11, 1.02]	0.037	2.44
Sustainability → abnormal returns	2.7	0.30	[−0.11, 0.71]	−0.002	1.44
Sustainability → residuals	2.1	0.26	[−0.18, 0.69]	−0.007	1.16



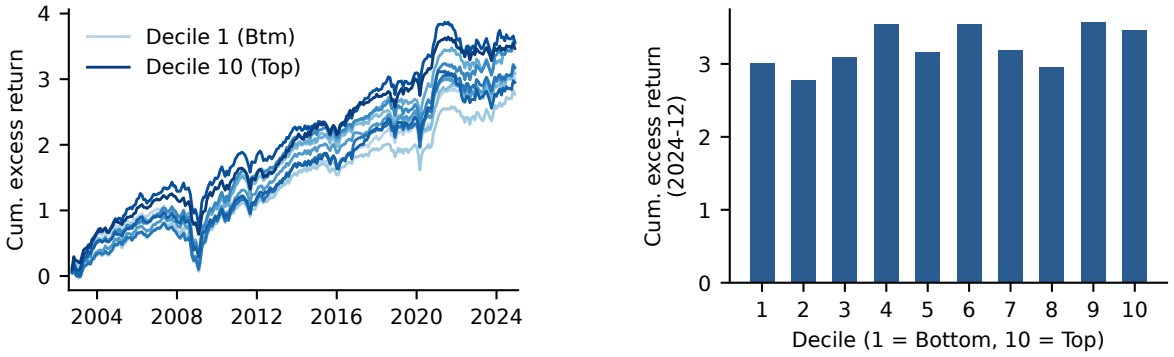
(a) Cumulative returns by prediction decile **(b)** Terminal cumulative return by decile

Figure IA.19: Conventional characteristics, equal-weighted portfolios (step 1). Cumulative excess returns of equal-weighted decile portfolios formed on out-of-sample XGBoost predictions of abnormal returns from 45 conventional characteristics.



(a) Cumulative returns by prediction decile **(b)** Terminal cumulative return by decile

Figure IA.20: Sustainability characteristics do not predict abnormal returns (step 0). Cumulative excess returns of equal-weighted decile portfolios formed on out-of-sample XGBoost predictions of abnormal returns from 22 sustainability characteristics, cross-out-of-sample evaluation on the 2002–2024 sustainability panel. Portfolios begin once sufficient cross-sectional coverage is available.

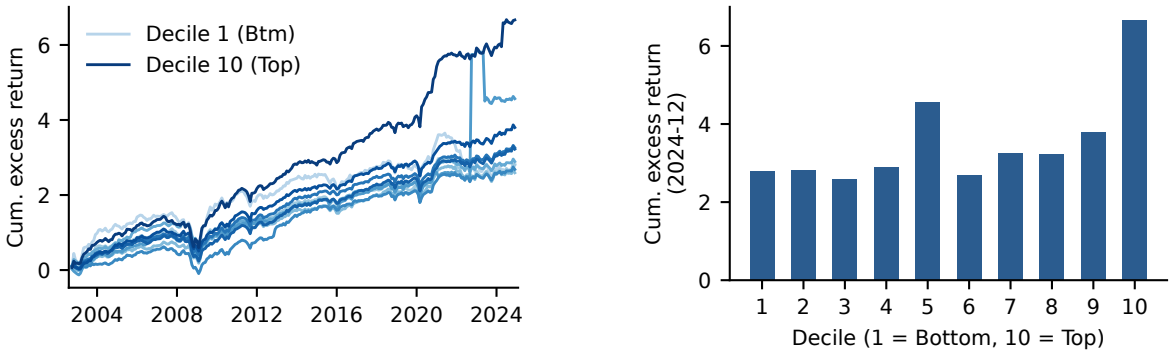


(a) Cumulative returns by prediction decile **(b)** Terminal cumulative return by decile

Figure IA.21: Sustainability characteristics do not predict characteristic-orthogonal residuals (step 2). Cumulative excess returns of equal-weighted decile portfolios formed on out-of-sample XGBoost predictions of step-1 residuals from 22 sustainability characteristics, cross-out-of-sample evaluation on the period for which step-1 residuals are available.

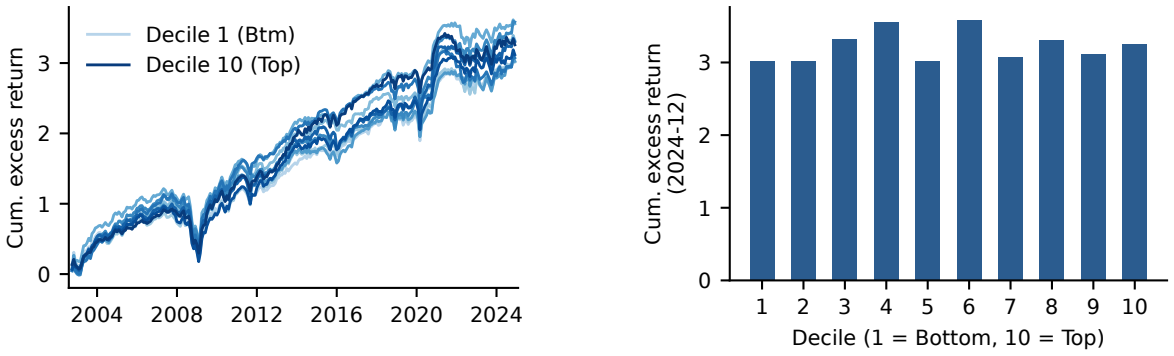
Table IA.10: Prediction-weighted portfolios. Long–short portfolio performance of the headline models with weights proportional to the out-of-sample prediction.

Model	μ (%)	SR	SR 95% CI	IC	t-stat
Characteristics → abnormal returns	17.4	0.84	[0.38, 1.30]	0.036	3.63
Sustainability → abnormal returns	1.1	0.12	[−0.26, 0.50]	−0.002	0.63
Sustainability → residuals	1.5	0.14	[−0.35, 0.63]	−0.008	0.57



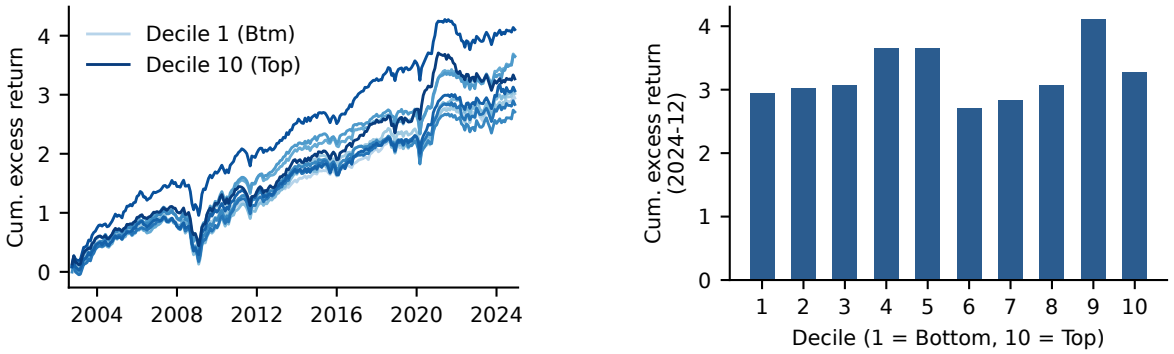
(a) Cumulative returns by prediction decile **(b)** Terminal cumulative return by decile

Figure IA.22: Conventional characteristics predict abnormal returns (step 1). Cumulative excess returns of prediction-weighted decile portfolios formed on out-of-sample XGBoost predictions of abnormal returns from 45 conventional characteristics.



(a) Cumulative returns by prediction decile **(b)** Terminal cumulative return by decile

Figure IA.23: Sustainability characteristics do not predict abnormal returns (step 0). Cumulative excess returns of prediction-weighted decile portfolios formed on out-of-sample XGBoost predictions of abnormal returns from 22 sustainability characteristics, cross-out-of-sample evaluation on the 2002–2024 sustainability panel. Portfolios begin once sufficient cross-sectional coverage is available.



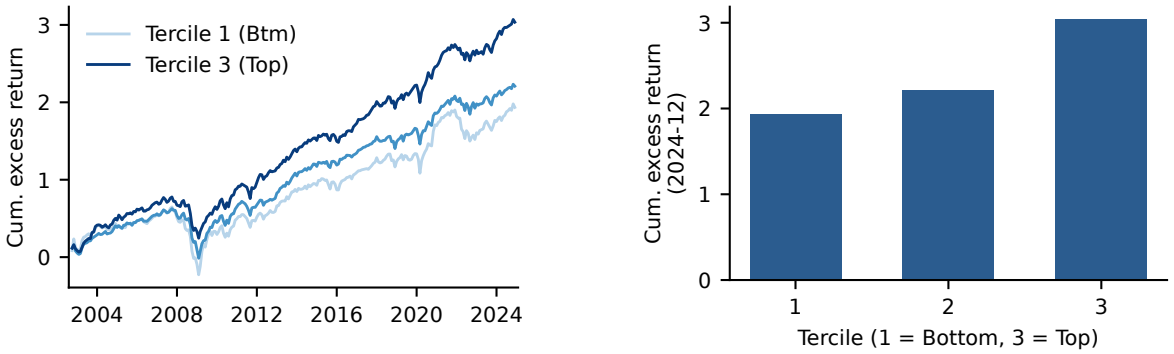
(a) Cumulative returns by prediction decile

(b) Terminal cumulative return by decile

Figure IA.24: Sustainability characteristics do not predict characteristic-orthogonal residuals (step 2). Cumulative excess returns of prediction-weighted decile portfolios formed on out-of-sample XGBoost predictions of step-1 residuals from 22 sustainability characteristics, cross-out-of-sample evaluation on the period for which step-1 residuals are available.

Table IA.11: Tercile sorts. Long–short performance of the headline models with value-weighted tercile portfolios.

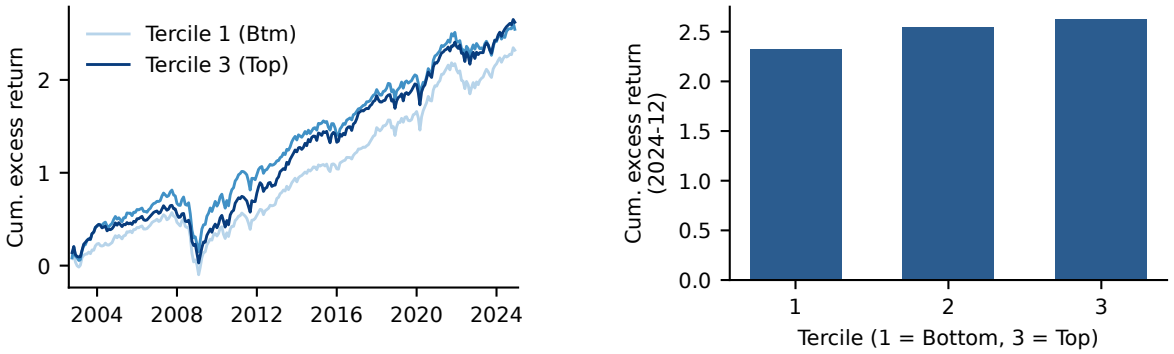
Model	μ (%)	SR	SR 95% CI	IC	t-stat
Characteristics → abnormal returns	4.9	0.60	[0.18, 1.02]	0.037	2.81
Sustainability → abnormal returns	1.4	0.22	[−0.20, 0.64]	0.004	1.03
Sustainability → residuals	−0.1	−0.02	[−0.45, 0.41]	−0.000	−0.09



(a) Cumulative returns by prediction decile

(b) Terminal cumulative return by decile

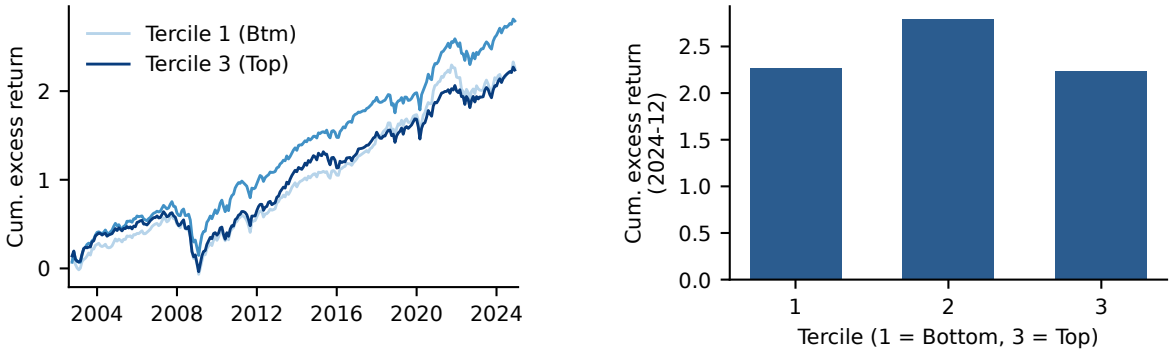
Figure IA.25: Conventional characteristics, tercile portfolios (step 1). Cumulative excess returns of value-weighted tercile portfolios formed on out-of-sample XGBoost predictions of abnormal returns from 45 conventional characteristics.



(a) Cumulative returns by prediction decile

(b) Terminal cumulative return by decile

Figure IA.26: Sustainability characteristics do not predict abnormal returns (step 0). Cumulative excess returns of value-weighted terciles portfolios formed on out-of-sample XGBoost predictions of abnormal returns from 22 sustainability characteristics, cross-out-of-sample evaluation on the 2002–2024 sustainability panel. Portfolios begin once sufficient cross-sectional coverage is available.



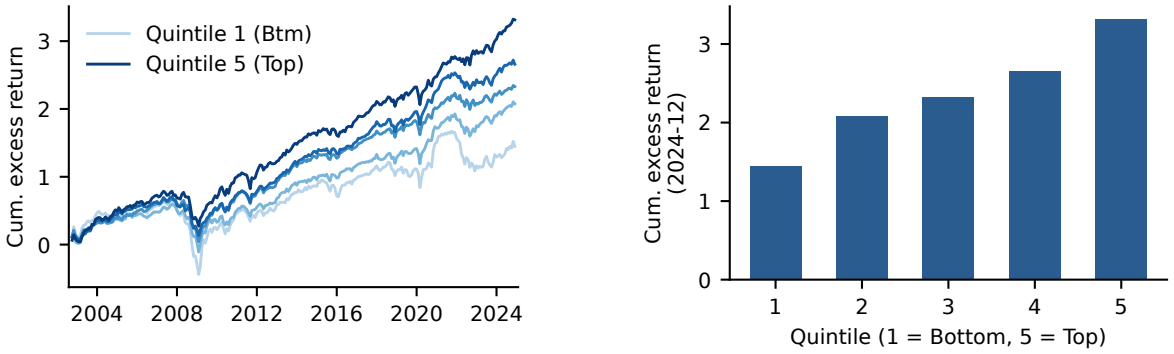
(a) Cumulative returns by prediction decile

(b) Terminal cumulative return by decile

Figure IA.27: Sustainability characteristics do not predict characteristic-orthogonal residuals (step 2). Cumulative excess returns of value-weighted terciles portfolios formed on out-of-sample XGBoost predictions of step-1 residuals from 22 sustainability characteristics, cross-out-of-sample evaluation on the period for which step-1 residuals are available.

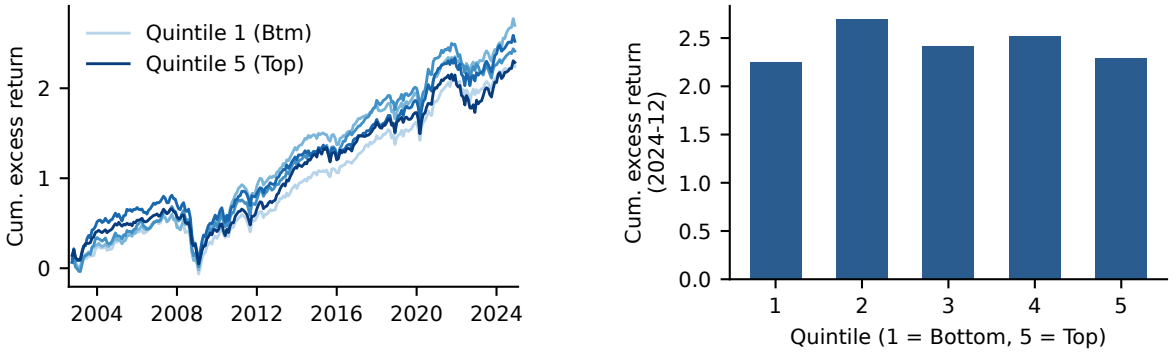
Table IA.12: Quintile sorts. Long–short performance of the headline models with value-weighted quintile portfolios.

Model	μ (%)	SR	SR 95% CI	IC	t-stat
Characteristics → abnormal returns	8.4	0.67	[0.21, 1.12]	0.037	2.90
Sustainability → abnormal returns	0.2	0.03	[−0.36, 0.42]	−0.008	0.14
Sustainability → residuals	0.5	0.07	[−0.31, 0.44]	0.002	0.36



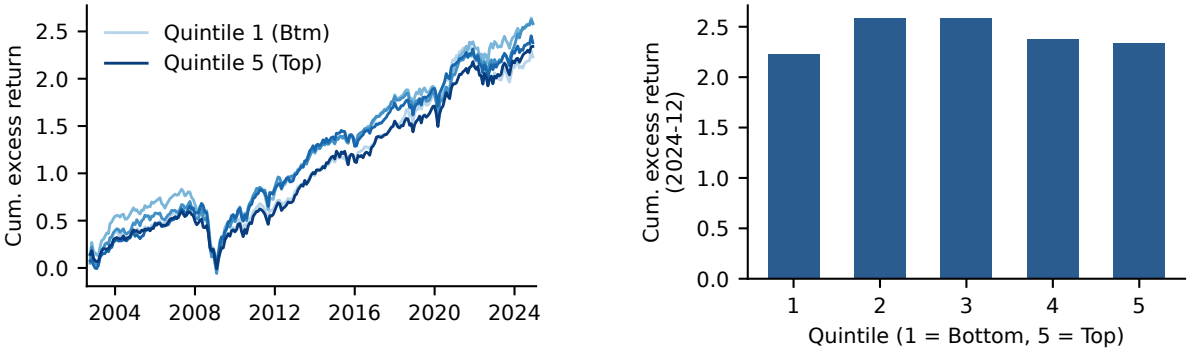
(a) Cumulative returns by prediction decile **(b)** Terminal cumulative return by decile

Figure IA.28: Conventional characteristics, quintile portfolios (step 1). Cumulative excess returns of value-weighted quintile portfolios formed on out-of-sample XGBoost predictions of abnormal returns from 45 conventional characteristics.



(a) Cumulative returns by prediction decile **(b)** Terminal cumulative return by decile

Figure IA.29: Sustainability characteristics do not predict abnormal returns (step 0). Cumulative excess returns of value-weighted quintile portfolios formed on out-of-sample XGBoost predictions of abnormal returns from 22 sustainability characteristics, cross-out-of-sample evaluation on the 2002–2024 sustainability panel. Portfolios begin once sufficient cross-sectional coverage is available.



(a) Cumulative returns by prediction decile **(b)** Terminal cumulative return by decile

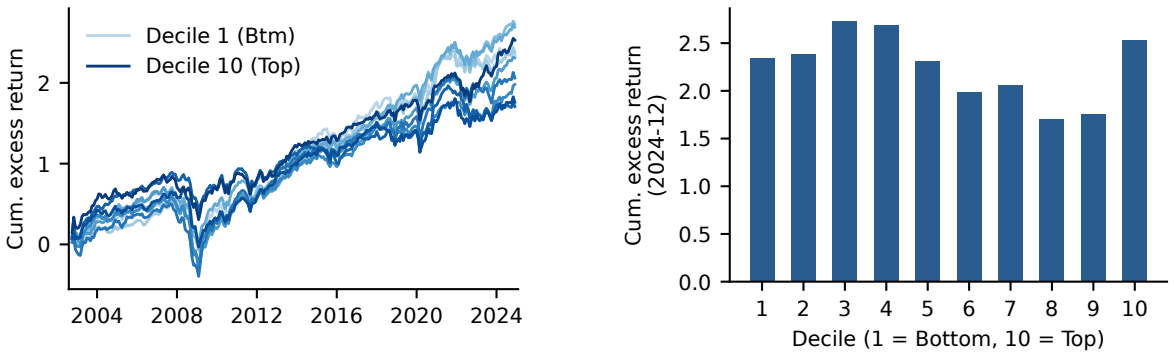
Figure IA.30: Sustainability characteristics do not predict characteristic-orthogonal residuals (step 2). Cumulative excess returns of value-weighted quintile portfolios formed on out-of-sample XGBoost predictions of step-1 residuals from 22 sustainability characteristics, cross-out-of-sample evaluation on the period for which step-1 residuals are available.

IA.4 No imputation, rank quantiles and the full variable set

This section reports results without imputation, for models that use rank quantiles of the 22 sustainability variables, and for models that use all 513 sustainability variables.

Table IA.13: No imputation. Long–short decile portfolio performance without imputation, letting XGBoost split directly on missing values. For comparability with the baseline, the table restricts the sample to the baseline sample.

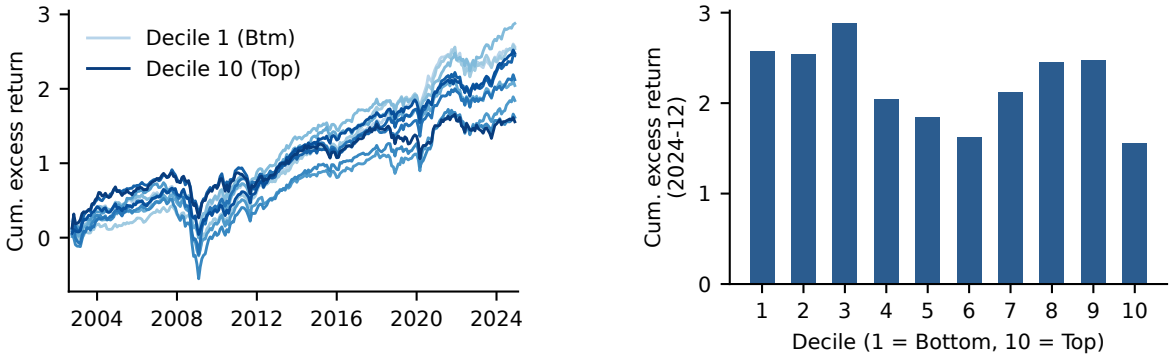
Model	μ (%)	SR	SR 95% CI	IC	t-stat
Characteristics → abnormal returns	13.9	0.72	[0.21, 1.22]	0.037	2.80
Sustainability → abnormal returns	0.8	0.07	[−0.37, 0.51]	−0.011	0.32
Sustainability → residuals	−4.5	−0.41	[−0.79, −0.03]	−0.006	−2.11



(a) Cumulative returns by prediction decile

(b) Terminal cumulative return by decile

Figure IA.31: Sustainability characteristics do not predict abnormal returns (step 0). Cumulative excess returns of value-weighted decile portfolios formed on out-of-sample XGBoost predictions of abnormal returns from 22 sustainability characteristics without imputation, cross-out-of-sample evaluation on the 2002–2024 sustainability panel. Portfolios begin once sufficient cross-sectional coverage is available.



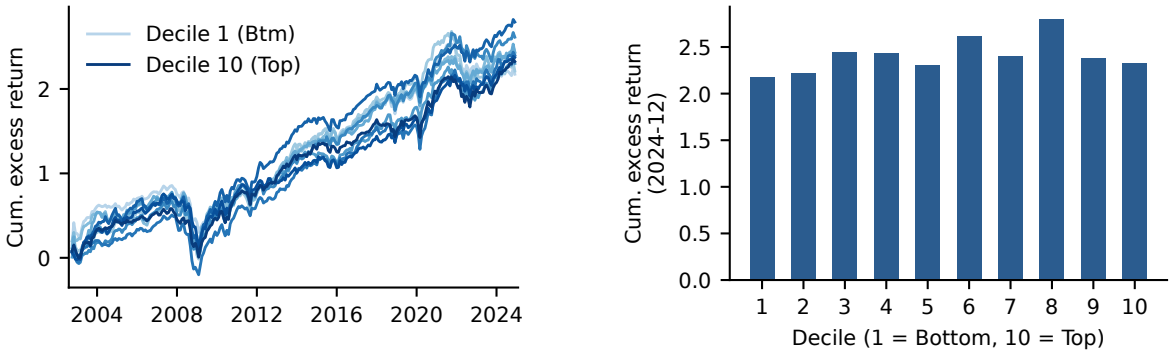
(a) Cumulative returns by prediction decile

(b) Terminal cumulative return by decile

Figure IA.32: Sustainability characteristics do not predict characteristic-orthogonal residuals (step 2). Cumulative excess returns of value-weighted decile portfolios formed on out-of-sample XGBoost predictions of step-1 residuals from 22 sustainability characteristics without imputation, cross-out-of-sample evaluation on the period for which step-1 residuals are available.

Table IA.14: Rank quantiles. Value-weighted decile portfolios are formed monthly on out-of-sample predictions. The conventional-characteristics model is as above. The sustainability models predict abnormal returns (step 0) and step-1 residuals (step 2) with 22 sustainability variables transformed in rank quantiles under the cross-out-of-sample scheme on the 2002–2024 panel.

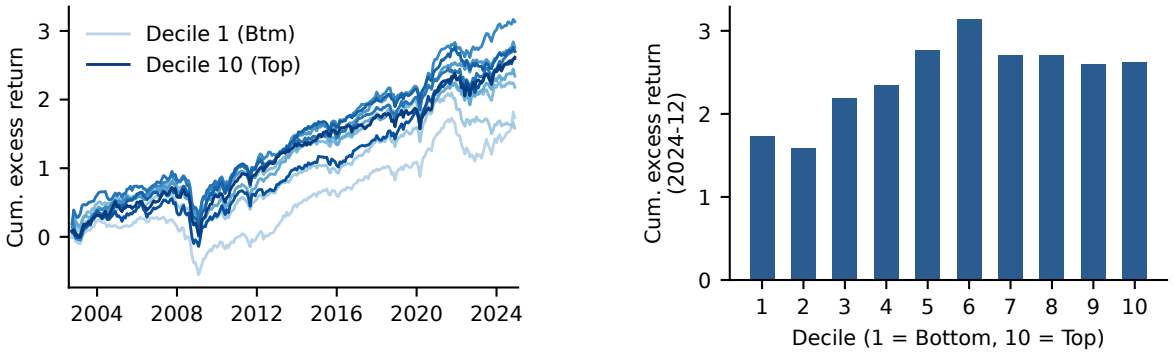
Model	μ (%)	SR	SR 95% CI	IC	t-stat
Characteristics $\rightarrow$ abnormal returns	13.9	0.72	[0.21, 1.22]	0.037	2.80
Sustainability $\rightarrow$ abnormal returns	0.7	0.07	[−0.34, 0.48]	0.010	0.34
Sustainability $\rightarrow$ residuals	4.0	0.32	[−0.09, 0.73]	0.008	1.54



(a) Cumulative returns by prediction decile

(b) Terminal cumulative return by decile

Figure IA.33: Rank quantiles of sustainability characteristics do not predict abnormal returns (step 0). Cumulative excess returns of value-weighted decile portfolios formed on out-of-sample XGBoost predictions of abnormal returns from 22 sustainability characteristics transformed in rank quantiles, cross-out-of-sample evaluation on the 2002–2024 sustainability panel. Portfolios begin once sufficient cross-sectional coverage is available.



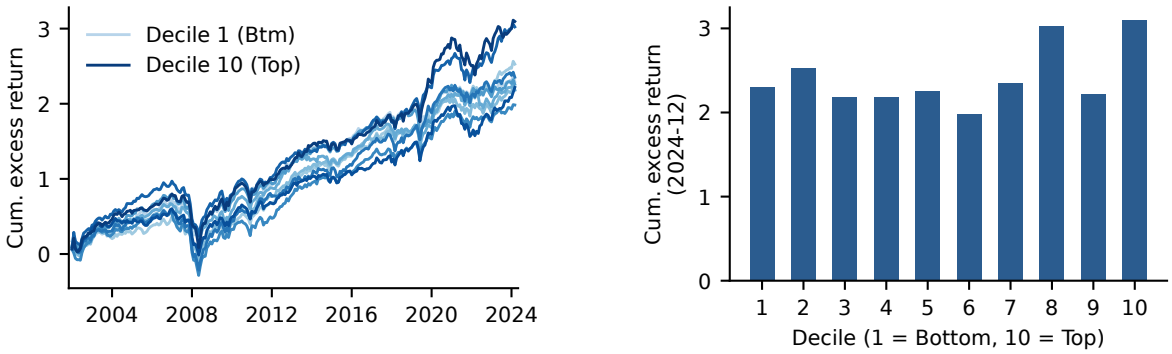
(a) Cumulative returns by prediction decile

(b) Terminal cumulative return by decile

Figure IA.34: Rank quantiles of sustainability characteristics do not predict characteristic-orthogonal residuals (step 2). Cumulative excess returns of value-weighted decile portfolios formed on out-of-sample XGBoost predictions of step-1 residuals from 22 sustainability characteristics transformed in rank quantiles, cross-out-of-sample evaluation on the period for which step-1 residuals are available.

Table IA.15: All 513 variables. Value-weighted decile portfolios are formed monthly on out-of-sample predictions. The conventional-characteristics model is as above. The sustainability models predict abnormal returns (step 0) and step-1 residuals (step 2) with 513 sustainability variables under the cross-out-of-sample scheme on the 2002–2024 panel.

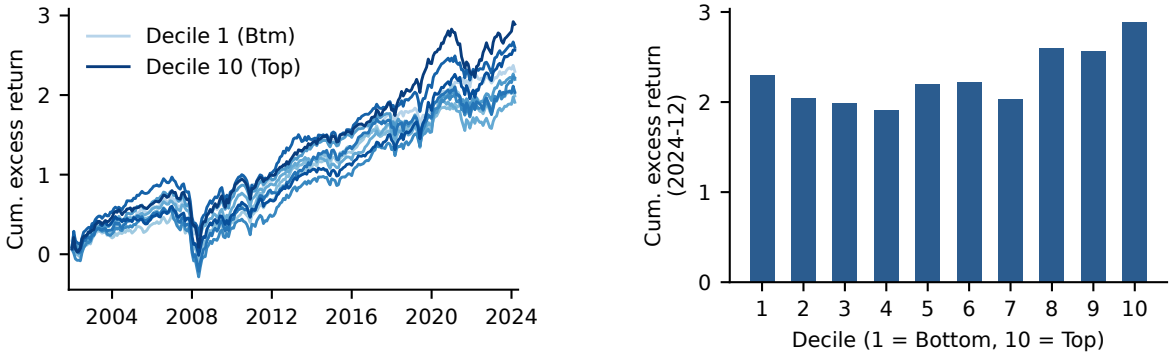
Model	μ (%)	SR	SR 95% CI	IC	t-stat
Characteristics → abnormal returns	13.9	0.72	[0.21, 1.22]	0.037	2.80
Sustainability → abnormal returns	3.6	0.32	[−0.11, 0.75]	−0.010	1.47
Sustainability → residuals	2.7	0.22	[−0.27, 0.72]	0.000	0.89



(a) Cumulative returns by prediction decile

(b) Terminal cumulative return by decile

Figure IA.35: Sustainability characteristics do not predict abnormal returns (step 0). Cumulative excess returns of value-weighted decile portfolios formed on out-of-sample XGBoost predictions of abnormal returns from 513 sustainability characteristics, cross-out-of-sample evaluation on the 2002–2024 sustainability panel. Portfolios begin once sufficient cross-sectional coverage is available.



(a) Cumulative returns by prediction decile

(b) Terminal cumulative return by decile

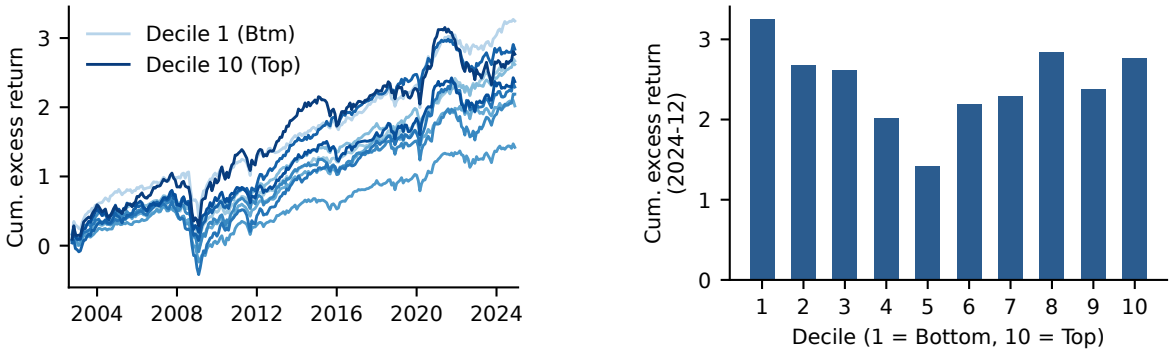
Figure IA.36: Sustainability characteristics do not predict characteristic-orthogonal residuals (step 2). Cumulative excess returns of value-weighted decile portfolios formed on out-of-sample XGBoost predictions of step-1 residuals from 513 sustainability characteristics, cross-out-of-sample evaluation on the period for which step-1 residuals are available.

IA.5 Orthogonality check: characteristics and the step-1 residuals

This section reports the direct check of the orthogonalization in Section 3.1. An XGBoost model predicts the step-1 residuals with the 45 conventional characteristics themselves, tuned and evaluated with the cross-out-of-sample scheme on the 2002–2024 panel, exactly as the sustainability models of step 2. If the step-1 model has removed the return-relevant information in the characteristics, this model should find nothing exploitable, and it does not. The long–short Sharpe ratio is −0.12 and insignificant. The IC of −0.014 is small and negative, and Section 4.3 explains why a negative sign cannot produce a spurious sustainability premium in step 2.

Table IA.16: Characteristics do not predict the step-1 residuals. Long–short performance of value-weighted decile portfolios formed on out-of-sample XGBoost predictions of the step-1 residuals from the 45 conventional characteristics, cross-out-of-sample evaluation on the 2002–2024 panel.

Model	μ (%)	SR	SR 95% CI	IC	t-stat
Characteristics → step-1 residuals	−2.2	−0.12	[−0.52, 0.27]	−0.014	−0.61



(a) Cumulative returns by prediction decile

(b) Terminal cumulative return by decile

Figure IA.37: Characteristics do not predict the step-1 residuals. Cumulative excess returns of value-weighted decile portfolios formed on out-of-sample XGBoost predictions of the step-1 residuals from the 45 conventional characteristics.

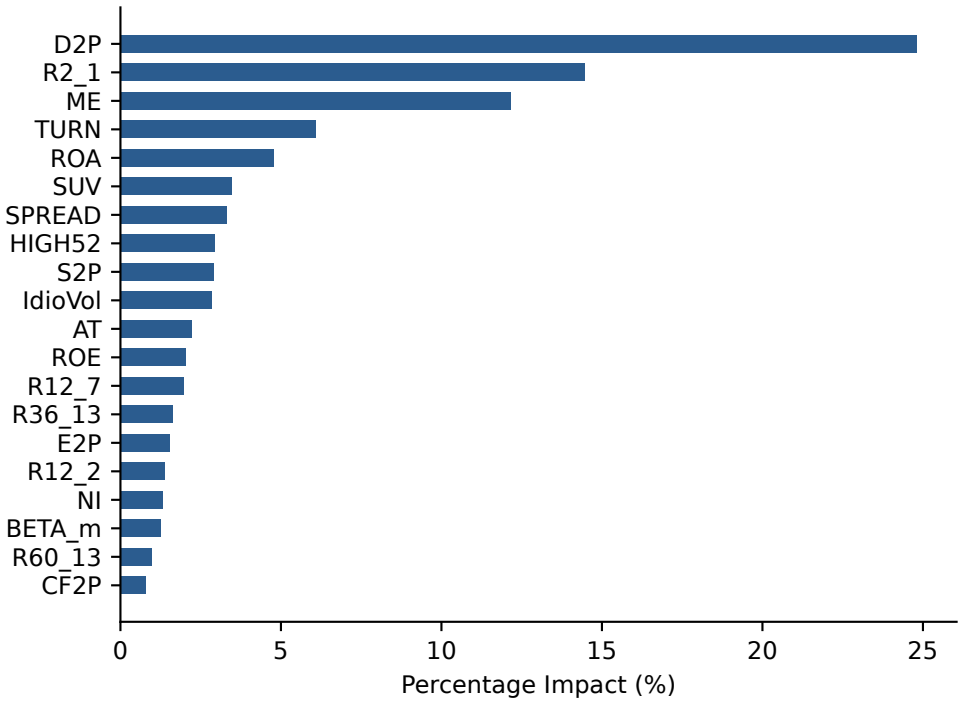


Figure IA.38: Variable importance of the orthogonality check. Mean absolute SHAP value of each characteristic in the XGBoost model that predicts the step-1 residuals, expressed as the percentage share of total feature impact.

IA.6 Diebold–Mariano tests

This section reports the Diebold and Mariano (1995) tests of Section 5.6, which compare the forecast accuracy of the k-variable sustainability models ($k = 1, \ldots, 22$) with that of a univariate benchmark that forecasts abnormal returns with a linear function of the greenness score estimated by OLS.

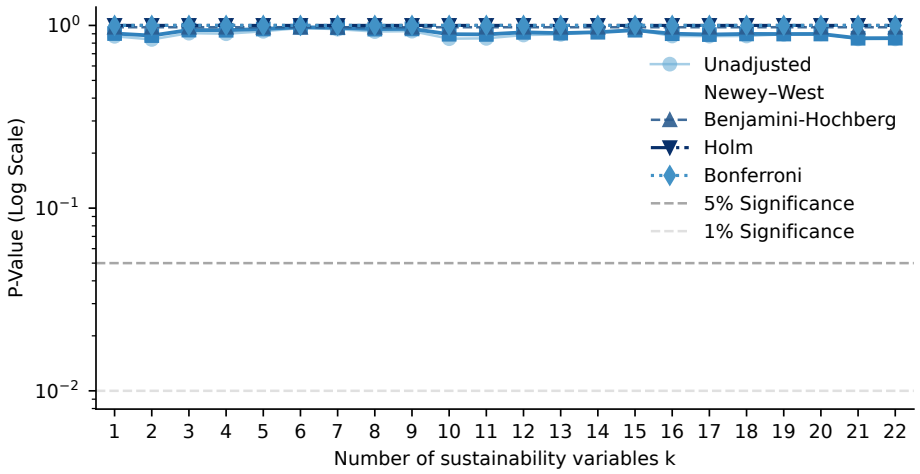


Figure IA.39: Diebold–Mariano comparisons across sustainability models. One-sided Diebold and Mariano (1995) p-values for the hypothesis that the k-variable sustainability model ($k = 1, \ldots, 22$, horizontal axis) has a lower mean squared forecast error for abnormal returns than the OLS forecast from the greenness score, sample starting November 2012. Loss differentials are averaged across stocks within each month and tested with Newey–West standard errors, and the figure also shows the Holm–Bonferroni, Benjamini–Hochberg, and Bonferroni adjusted values. Dashed lines mark the 5% and 1% levels. The corresponding Ledoit–Wolf tests are in Figure 15.

IA.7 The three learners

This section describes the three learners of the paper, the gradient boosted tree ensemble of Section 3.2 and the two alternatives of Section 6.2. All three estimate the conditional mean functions g_1 and g_2 of equations (2) and (3) by minimizing the mean squared prediction error on the training folds, and all three are tuned on the validation folds with the criterion of Section 3.2, the Sharpe ratio of the value-weighted long–short decile portfolio. They differ in the class of functions they can represent and in how they are regularized.

Gradient boosted regression trees

A regression tree partitions the input space into rectangles through a sequence of binary splits, each on one input and one threshold, and predicts the average training outcome within each rectangle. A tree of depth d has at most 2^d leaves and can represent interactions between up to d inputs. A single tree is a coarse and unstable estimator. Boosting builds an additive model from many shallow trees,

$$F_M(z) = \sum_{m=1}^{M} \nu \, f_m(z),$$

where each tree f_m is fitted to the negative gradient of the loss evaluated at the current model F_{m-1}, which for squared error is the vector of residuals, and the learning rate $\nu \in (0, 1]$ shrinks the contribution of each tree (Friedman, 2001). XGBoost (Chen and Guestrin, 2016) fits each tree with a second-order approximation of the loss and adds a penalty on the tree itself. For a tree with leaf values $w_1, \ldots, w_J$ the penalty is $\omega \sum_{j=1}^{J} |w_j| + \frac{\kappa}{2} \sum_{j=1}^{J} w_j^2$, an L1 penalty with weight ω and an L2 penalty with weight κ (the parameters `alpha` and `lambda` of the software, see Table A.3). Further regularization comes from a minimum weight per leaf, subsampling of rows and of columns for each tree, and histogram binning of the inputs. Each of these is a hyperparameter in Table A.3, and Table A.4 reports the selected values. The number of trees M is the main regularization parameter. It is chosen by early stopping on the validation fold, so a model whose trees do not improve the validation loss stops after a single tree, which is what happens to the sustainability models in Table A.4. Missing inputs require no imputation, because every split learns a default direction for observations whose splitting variable is missing, and the no-imputation check of Section 6.4 relies on this property. Tree ensembles are invariant to monotone transformations of the inputs, so the variables enter in their raw form. Variable importance in Figures 6 and IA.38 is measured by SHAP values (Lundberg and Lee, 2017). They are the Shapley values of a cooperative game in which the inputs are the players and the payoff of a coalition S of inputs is the expected prediction of the model when the inputs in S are held at their observed values and the remaining inputs are integrated out. For a stock-month with inputs z and a model with K inputs, the SHAP value of input j is

$$\phi_j(z) = \sum_{S \subseteq \{1,\ldots,K\} \setminus \{j\}} \frac{|S|!\,(K - |S| - 1)!}{K!} \Big[f_z\big(S \cup \{j\}\big) - f_z(S) \Big],$$

where $f_z(S)$ denotes the expected prediction given the inputs in S. The SHAP values are the unique attribution that assigns zero to an input the model does not use, treats inputs with identical contributions identically, and sums to the deviation of the prediction from the average prediction, $\sum_{j=1}^{K} \phi_j(z) = F_M(z) - \mathbb{E}[F_M]$, so each prediction is decomposed exactly into contributions of the inputs. For tree ensembles the SHAP values are computed exactly and in polynomial time with the TreeSHAP algorithm (Lundberg, Erion, Chen, DeGrave, Prutkin, Nair, Katz, Himmelfarb, Bansal, and Lee, 2020), which evaluates the coalitions by following the splits of the trees rather than by sampling. Because the ensemble prediction is the average of the predictions of its members, the SHAP values of the ensemble are the averages of the SHAP values of the members. The importance of input j is the mean of $|\phi_j(z)|$ over all stock-months in the out-of-sample evaluation folds, and the figures report it as a percentage of the sum over inputs, which is the percentage impact on the horizontal

axis. A large value means that the input moves the predictions of the model by a lot on average. It does not mean that the input predicts returns, which is why the sustainability models of Panels (b) and (c) of Figure 6 can concentrate on a few emission variables and still have no predictive power.

Feedforward neural network

A feedforward network with L hidden layers maps the inputs $z^{(0)} = z$ through

$$z^{(l)} = \sigma\big(W^{(l)} z^{(l-1)} + b^{(l)}\big), \quad l = 1, \ldots, L, \qquad g(z) = w^{(L+1)\top} z^{(L)} + b^{(L+1)},$$

where each hidden layer applies an affine transformation followed by the activation function σ, applied element by element, and the output layer is linear. We use the rectified linear unit $\sigma(x) = \max(x, 0)$. A network with one hidden layer and enough units can approximate any continuous function on a bounded set, and the network represents nonlinearities and interactions as smooth rather than piecewise constant functions, which is why it serves as the flexible alternative to trees in Section 6.2. The weights are estimated by minimizing the mean squared error with the Adam variant of stochastic gradient descent (Kingma and Ba, 2015) on mini-batches of 10,000 observations, with a learning rate of 0.001. The network is regularized by dropout with a rate of 0.05, by L1 and L2 penalties on the weights, and by early stopping once the validation loss has not improved for five epochs, with a cap of 50 epochs (Goodfellow, Bengio, and Courville, 2016). The inputs enter as rank quantiles, since the network is sensitive to their scale. The optimization problem is not convex and its solution depends on the random initialization, so we average the predictions of 20 networks trained from different random seeds, as we do for the tree ensemble. The architecture and the two penalty weights are chosen by grid search on the validation folds with the same criterion as for XGBoost, from three architectures with two, three, and five hidden layers (32–16, 32–16–8, and 32–16–8–4–2 units) and from penalty weights in $\{0, 0.01, 0.05\}$ for each penalty, while the learning rate, the batch size, the dropout rate, and the stopping rule are held fixed. Table IA.17 lists the grid and the selected values. The validation selects the two-layer network for the characteristics model and for the sustainability return model and the three-layer network for the residual model, with penalties of zero for the characteristics model and small positive penalties for the two sustainability models, and early stopping ends training after 40, 13, and 12 epochs.

Elastic Net

The Elastic Net is a linear model, $g(z) = \beta^\top z$, estimated by minimizing

$$\frac{1}{2N}\sum_{i=1}^{N}\left(y_i - \beta^\top z_i\right)^2 + \lambda\left[\rho\,\|\beta\|_1 + \frac{1-\rho}{2}\,\|\beta\|_2^2\right],$$

where $\lambda \geq 0$ sets the strength of the penalty and $\rho \in [0,1]$ the mix between the lasso penalty, which sets coefficients exactly to zero and thereby selects variables, and the ridge penalty, which shrinks the coefficients of correlated inputs toward each other (Zou and Hastie, 2005). The problem is convex and is solved by coordinate descent (Friedman, Hastie, and Tibshirani, 2010), with the coefficients updated in random order. The inputs enter as rank quantiles, which puts them on a common scale so that a single penalty applies to all of them. The Elastic Net can represent only additive linear effects in the transformed inputs, which places it at the low-variance, high-bias end of the range of learners. The two penalty parameters are chosen by grid search on the validation folds with the same criterion as for XGBoost, with λ and ρ each on the grid $\{0.0001, 0.001, 0.01, 0.1, 0.5\}$ (Table IA.17). The validation selects a lasso share of 0.001 or below for all three models, so the selected models are close to ridge regressions, with a penalty strength of 0.1 for the characteristics and the residual model and of 0.001 for the sustainability return model.

Table IA.17: Hyperparameters of the alternative learners. Grid values and selected values for the feedforward neural network and the Elastic Net of Section 6.2, for the three headline models. The selection criterion is the validation-sample Sharpe ratio of the value-weighted long–short decile portfolio, as for XGBoost (Tables A.3 and A.4). Parameters with a single grid value are held fixed and their selected columns are left empty. The number of epochs of the network is chosen by early stopping on the validation mean squared error with a patience of five epochs and a cap of 50.

Parameter	Grid values	Charact.	Sustain. (returns)	Sustain. (residuals)
Panel A: Feedforward neural network				
hidden units per layer	{32–16, 32–16–8, 32–16–8–4–2}	32–16	32–16	32–16–8
L1 penalty (weights)	{0, 0.01, 0.05}	0	0.05	0.01
L2 penalty (weights)	{0, 0.01, 0.05}	0	0.01	0.01
activation	ReLU			
optimizer, learning rate	Adam, 0.001			
dropout rate	0.05			
batch size	10,000			
maximum epochs, patience	50, 5			
epochs (early stopping)		40	13	12
ensemble members	20			
Panel B: Elastic Net				
penalty strength λ	{0.0001, 0.001, 0.01, 0.1, 0.5}	0.1	0.001	0.1
lasso share ρ	{0.0001, 0.001, 0.01, 0.1, 0.5}	0.0001	0.001	0.0001

IA.8 Performance measures, multiple-testing adjustments, and tests

This section collects the definitions behind the numbers in the tables. Throughout, r_t denotes the excess return of a long–short portfolio in month t, $t = 1, \ldots, T$, and $\hat{\mu}$ and $\hat{\sigma}$ its sample mean and sample standard deviation.

Sharpe ratio, confidence interval, and t-statistic

The annualized Sharpe ratio is $\widehat{\mathrm{SR}} = \sqrt{12}\,\hat{\mu}/\hat{\sigma}$, and the annualized mean return reported in the tables is $12\hat{\mu}$. The t-statistic of the mean long–short return is $\hat{\mu}/\widehat{\mathrm{se}}(\hat{\mu})$, where $\widehat{\mathrm{se}}(\hat{\mu})$ is the Newey–West standard error with four lags (three lags for the 2012–2020 samples). The 95% confidence interval for the annualized Sharpe ratio is $\widehat{\mathrm{SR}} \pm 1.96\,\widehat{\mathrm{se}}(\widehat{\mathrm{SR}})$ with

$$\widehat{\mathrm{se}}(\widehat{\mathrm{SR}})^2 = \frac{12\,\widehat{\mathrm{se}}(\hat{\mu})^2}{\hat{\sigma}^2} + \frac{\widehat{\mathrm{SR}}^2}{2T},$$

which is the delta-method variance of Lo (2002) with the Newey–West variance of the mean in place of the iid variance, so that the interval accounts for autocorrelation in the mean and for the sampling error of the volatility estimate.

Information coefficient and IC ratio

For each month t, the information coefficient IC_t is the Spearman rank correlation, across the stocks in the evaluation sample in that month, between the out-of-sample predictions and the realized targets, abnormal returns for the step-0 and step-1 models and step-1 residuals for the step-2 models. The IC reported in the tables is the time-series mean $\overline{\mathrm{IC}} = T^{-1}\sum_t \mathrm{IC}_t$, its t-statistic uses Newey–West standard errors with four lags, and the annualized IC ratio is

$$\text{IC ratio} = \sqrt{12}\,\frac{\overline{\mathrm{IC}}}{\mathrm{sd}(\mathrm{IC}_t)},$$

the Sharpe ratio of the monthly IC series. The IC measures whether the predictions rank stocks correctly, independently of portfolio weights, and the IC ratio measures how stable that ranking ability is over time.

Multiple-testing adjustments

A family of m tests with p-values $p_1, \ldots, p_m$ produces false rejections at the 5% level in about $0.05\,m$ tests even when every null hypothesis is true. The adjusted p-values in Tables 4, 6, A.5 and A.6 and in Figures 15 and IA.39 correct for this. Let $p_{(1)} \leq \cdots \leq p_{(m)}$ denote the ordered p-values. The Bonferroni adjustment is $\tilde{p}_{(k)} = \min(1, m\,p_{(k)})$, and rejecting when $\tilde{p} \leq \alpha$ keeps the probability of at least one false rejection in the family at or below α. The Holm (1979) adjustment, labeled HB in the tables, is

$$\tilde{p}_{(k)} = \min\Big(1,\ \max_{j \leq k} (m - j + 1)\, p_{(j)}\Big),$$

which controls the same familywise error rate but is uniformly more powerful than Bonferroni, because the k-th smallest p-value is multiplied by $m - k + 1$ rather than by m. The Benjamini and Hochberg (1995) adjustment, labeled BH, is

$$\tilde{p}_{(k)} = \min\Big(1,\ \min_{j \geq k} \frac{m}{j}\, p_{(j)}\Big),$$

and rejecting when $\tilde{p} \leq \alpha$ keeps the expected share of false rejections among all rejections, the false discovery rate, at or below α. The false discovery rate is the less demanding criterion, so a result that fails the BH adjustment fails every familywise adjustment as well. The family is always the entire set of related specifications, the 32 tests of Table 6 and the 132 of Tables 4, A.5 and A.6, so that a specification is not rescued by being evaluated in a family of one.

Sharpe-ratio and forecast-accuracy tests

The Ledoit and Wolf (2008) test compares the Sharpe ratios of two portfolios with return series r_t^A and r_t^B observed over the same months. The statistic is the difference $\widehat{\mathrm{SR}}_A - \widehat{\mathrm{SR}}_B$ divided by a standard error obtained from the delta method applied to the first two moments of the two series, with a heteroskedasticity- and autocorrelation-consistent estimate of their joint variance, and the p-value comes from the studentized circular block bootstrap of the same paper, which is robust to non-normality, autocorrelation and heavy tails. The block size is chosen from $\{1, 3, 6, 10\}$ with the calibration method of Ledoit and Wolf (2008), which selects the block size whose bootstrap confidence interval has the empirical coverage closest to the nominal level in a simulation calibrated to the two return series (500 pilot samples with 199 bootstrap rounds each), and the p-value is then computed with 4,999 bootstrap replications. The test is one-sided in Table 1, where the alternative is that the characteristics

model has the higher Sharpe ratio, and in Figure 15, where the alternative is that the k-variable sustainability model beats the univariate greenness-score benchmark, and two-sided in Tables 2 and 3, where neither direction is privileged.

The Diebold and Mariano (1995) test compares the forecast accuracy of two models A and B. Because many stocks respond to the same shock in a given month, the stock-month loss differentials are not independent, so we first average within each month,

$$d_t = \frac{1}{N_t} \sum_{i=1}^{N_t} \left[\left(y_{i,t} - \hat{y}_{i,t}^{A} \right)^2 - \left(y_{i,t} - \hat{y}_{i,t}^{B} \right)^2 \right],$$

where N_t is the number of stocks with predictions from both models in month t, and then test whether the mean of the monthly series d_t is zero with a Newey–West t-statistic. In Figure IA.39 the alternative is that the k-variable sustainability model has the lower loss, so the test is one-sided.